Lecture 3: Convolutional Neural Networks

Today:

- Expectations of visual recognition systems (10%)
- A Conv layer, stride, padding etc. (25%)
- Conv2D (20%)
- Transposed Conv (5%)
- Implementation and backprop (20%)
- What is encoded in feature maps? (10%)
- Max-pooling, convnet and conv variants (10%)






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Monitor

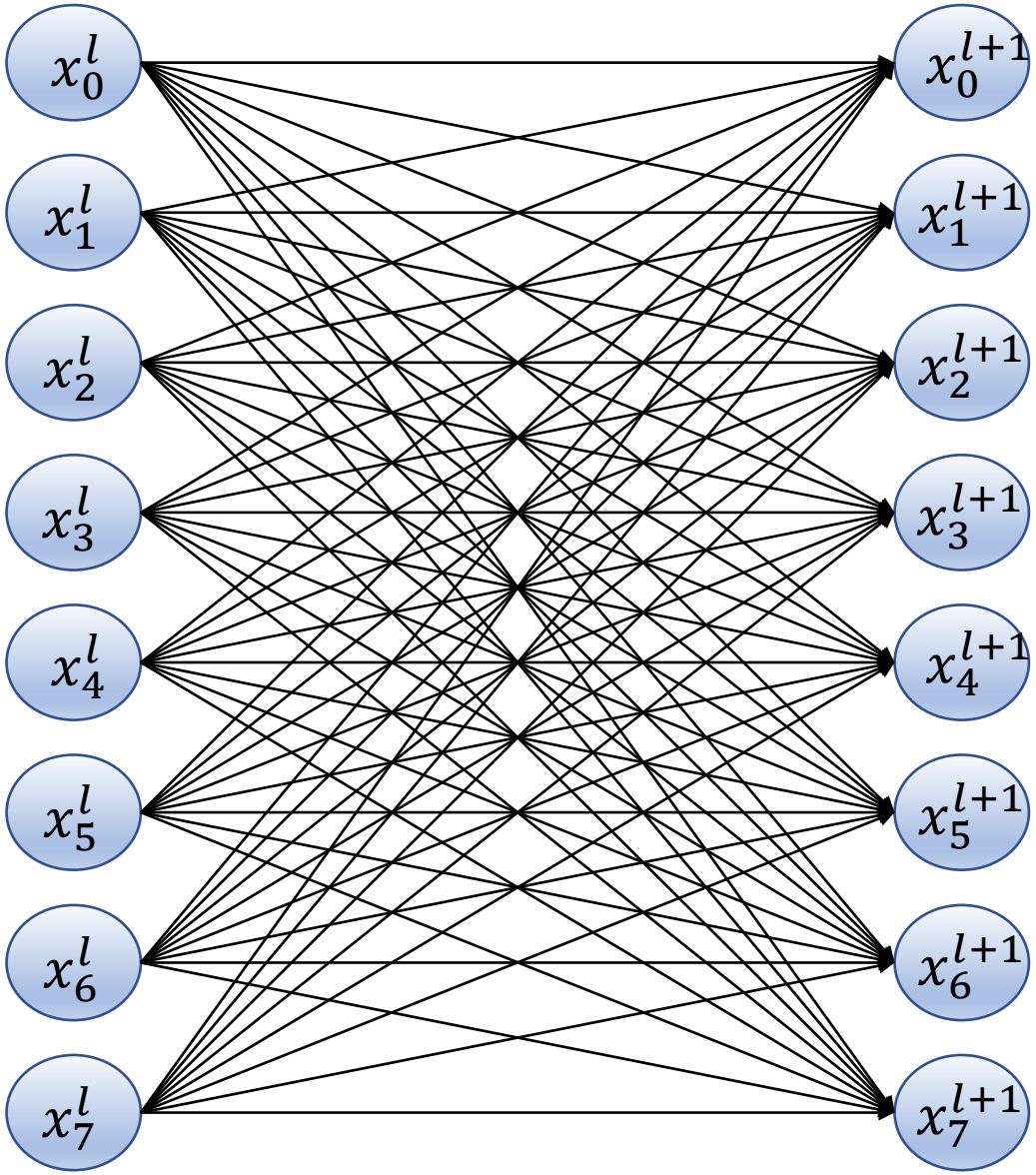

Monitor


Monitor

Expectations of visual recognition network

1. Maintain 2D structure logic
2. Shift invariant (actually, equivariant)
3. Consider local correlations first
4. Hierarchically growing field of view
5. Hierarchically progressing complexity
6. Reasonable amount of params

Fully connected layer

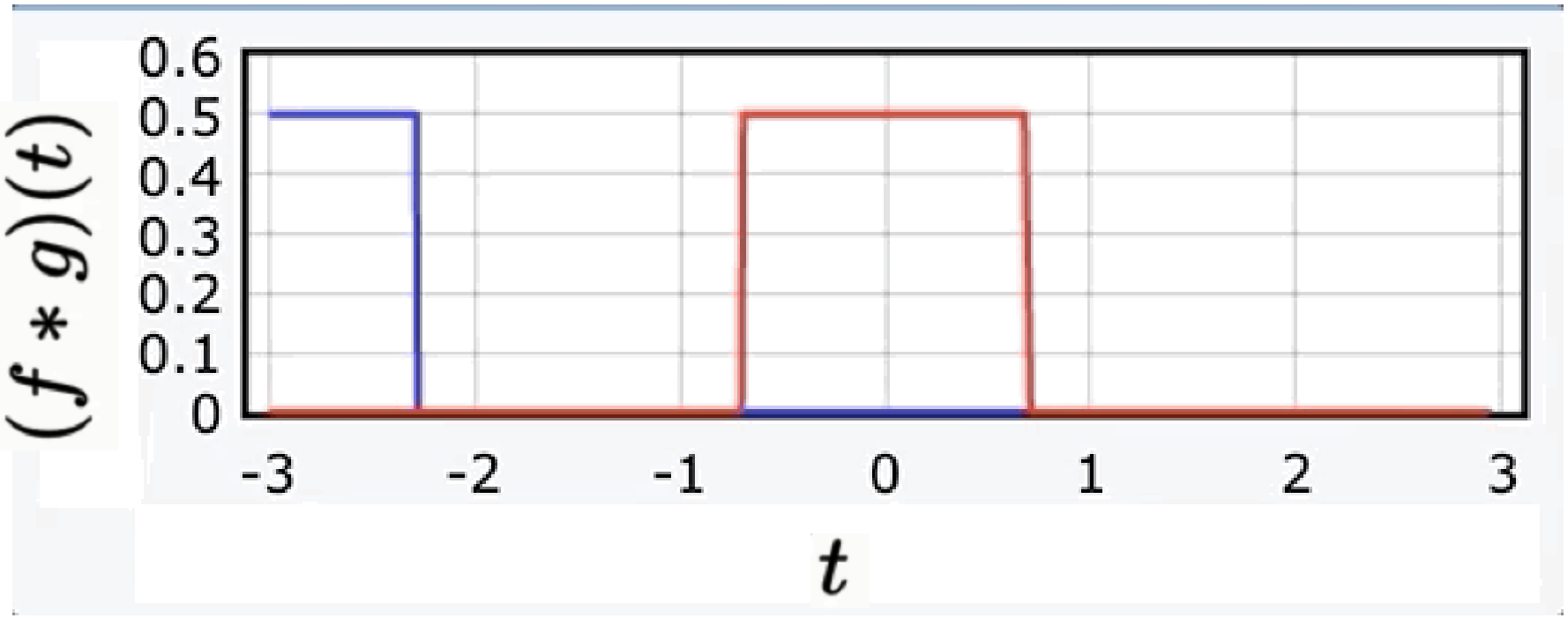


$$\begin{bmatrix} w_{00}^l & w_{01}^l & & \\ w_{10}^l & w_{11}^l & \cdots & \\ \vdots & & \ddots & \\ & & & \end{bmatrix}$$

1. ✗ Maintain 2D structure logic
2. ✗ Shift invariant (actually, equivariant)
3. ✗ Consider only local correlations
4. ✗ Hierarchically growing field of view
5. ✓ Hierarchically progressing complexity
6. ✗ Reasonable amount of params

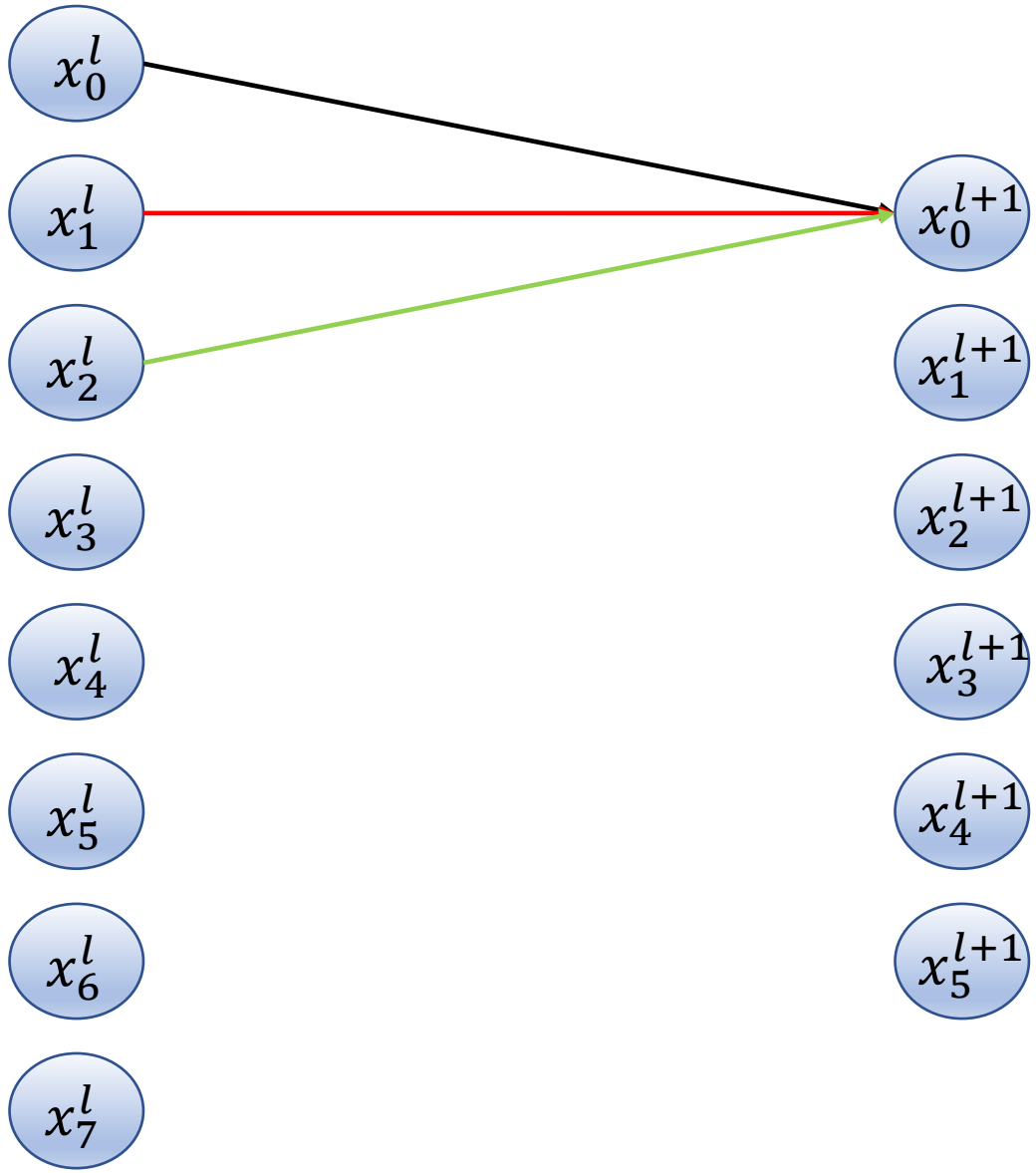
Convolution





$$(f * g)(t) \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} f(\tau)\, g(t-\tau)\, d\tau$$

Convolution layer

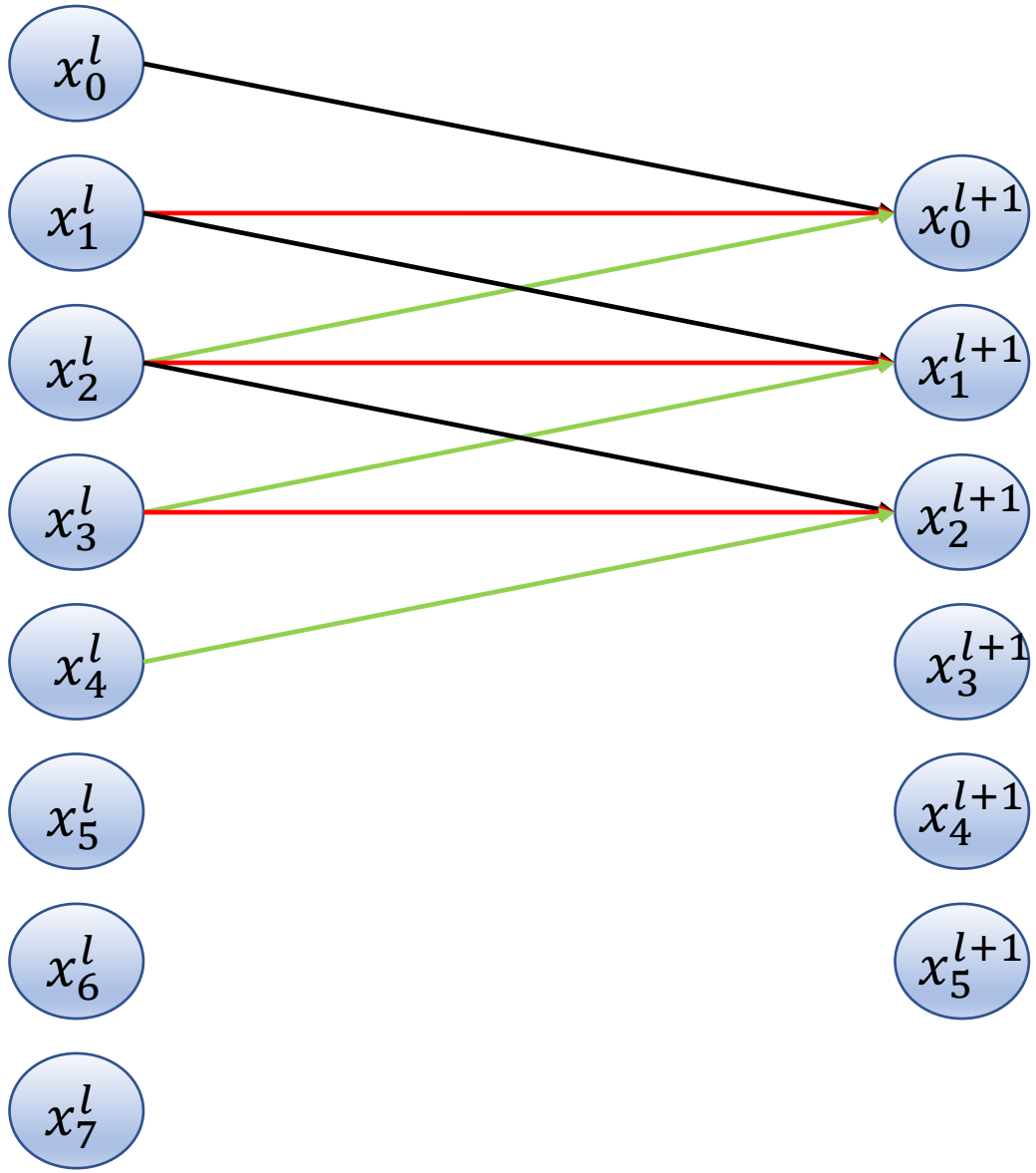


a

b

c

Convolution layer

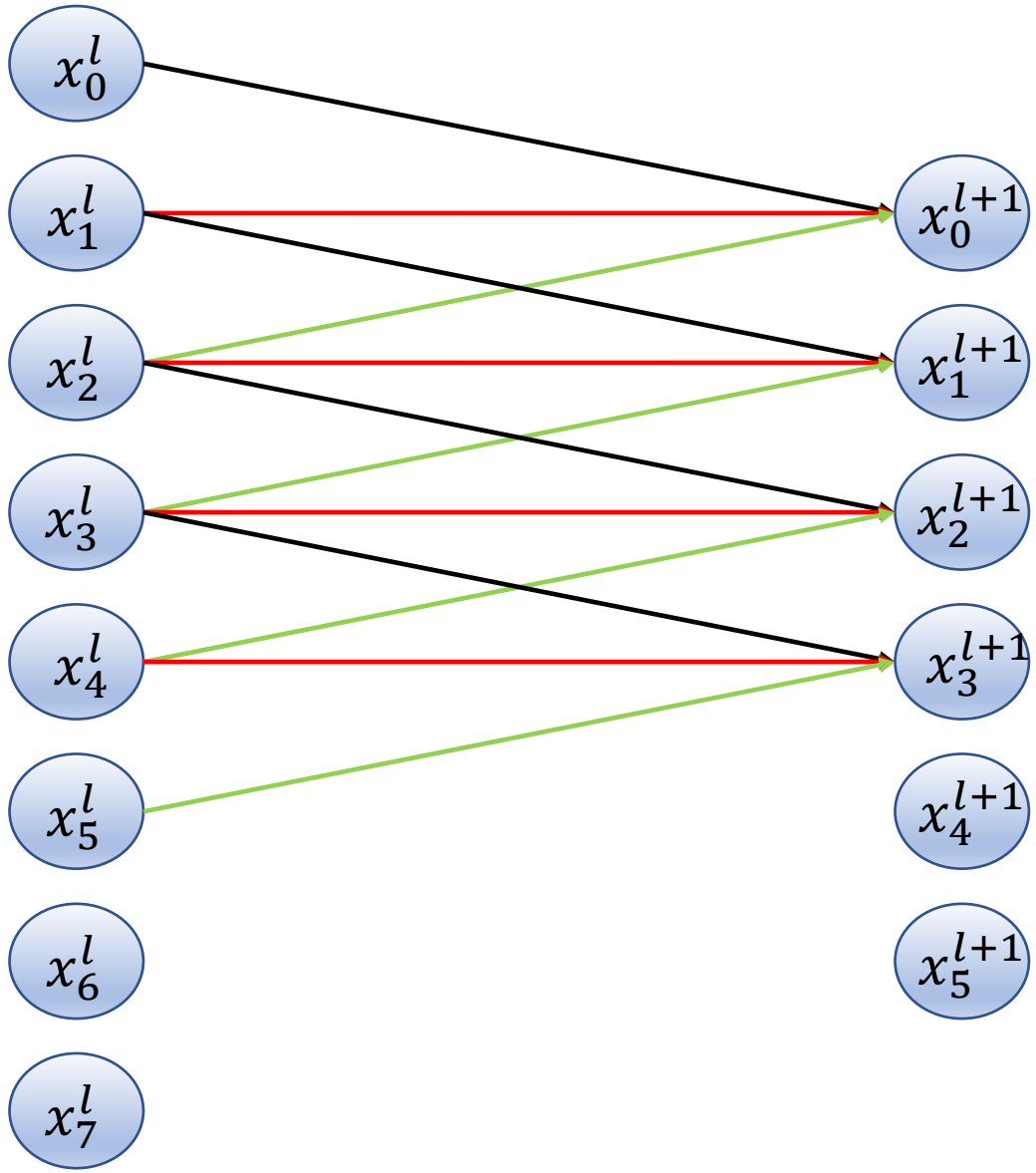


a

b

c

Convolution layer

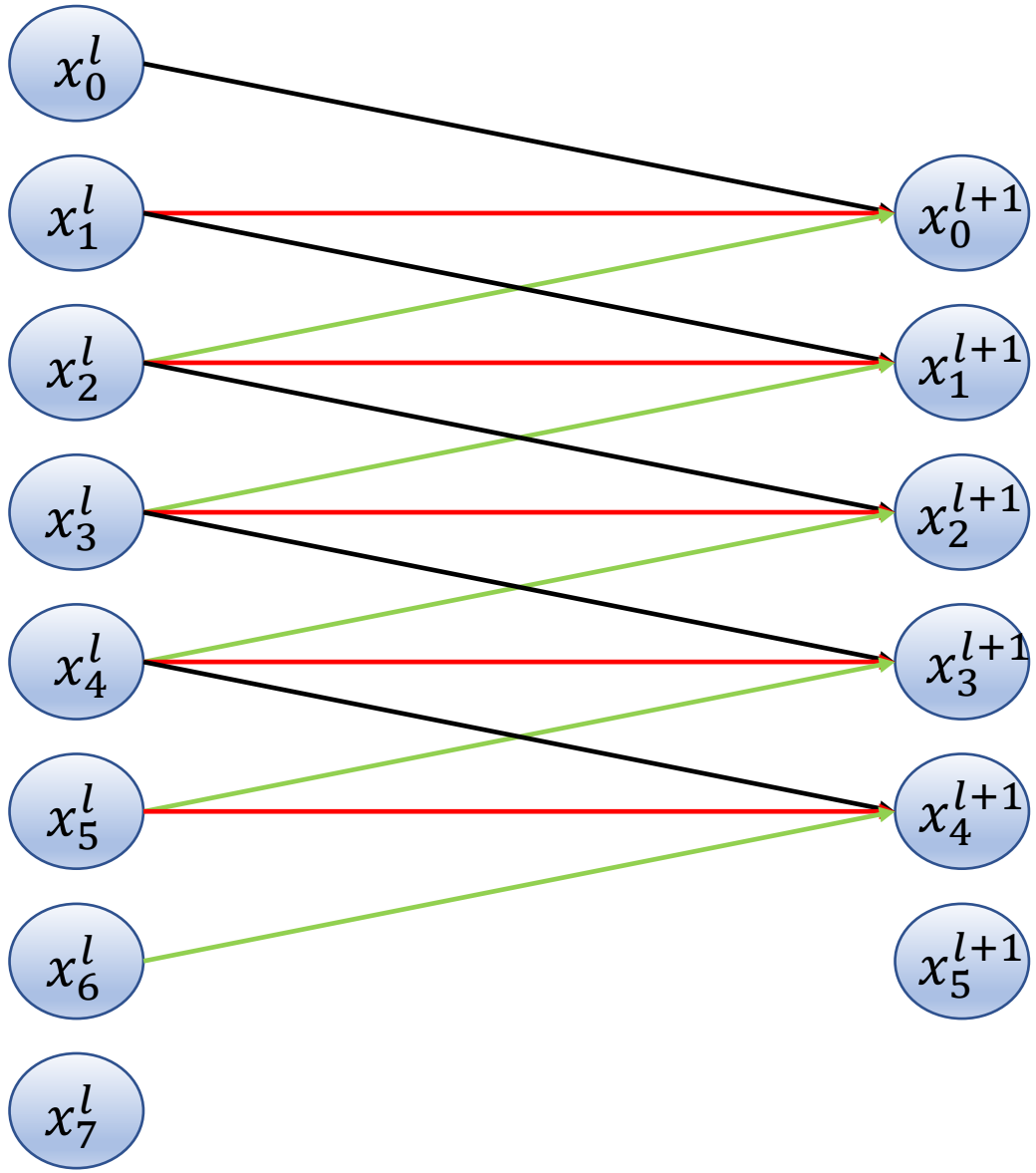


a

b

c

Convolution layer

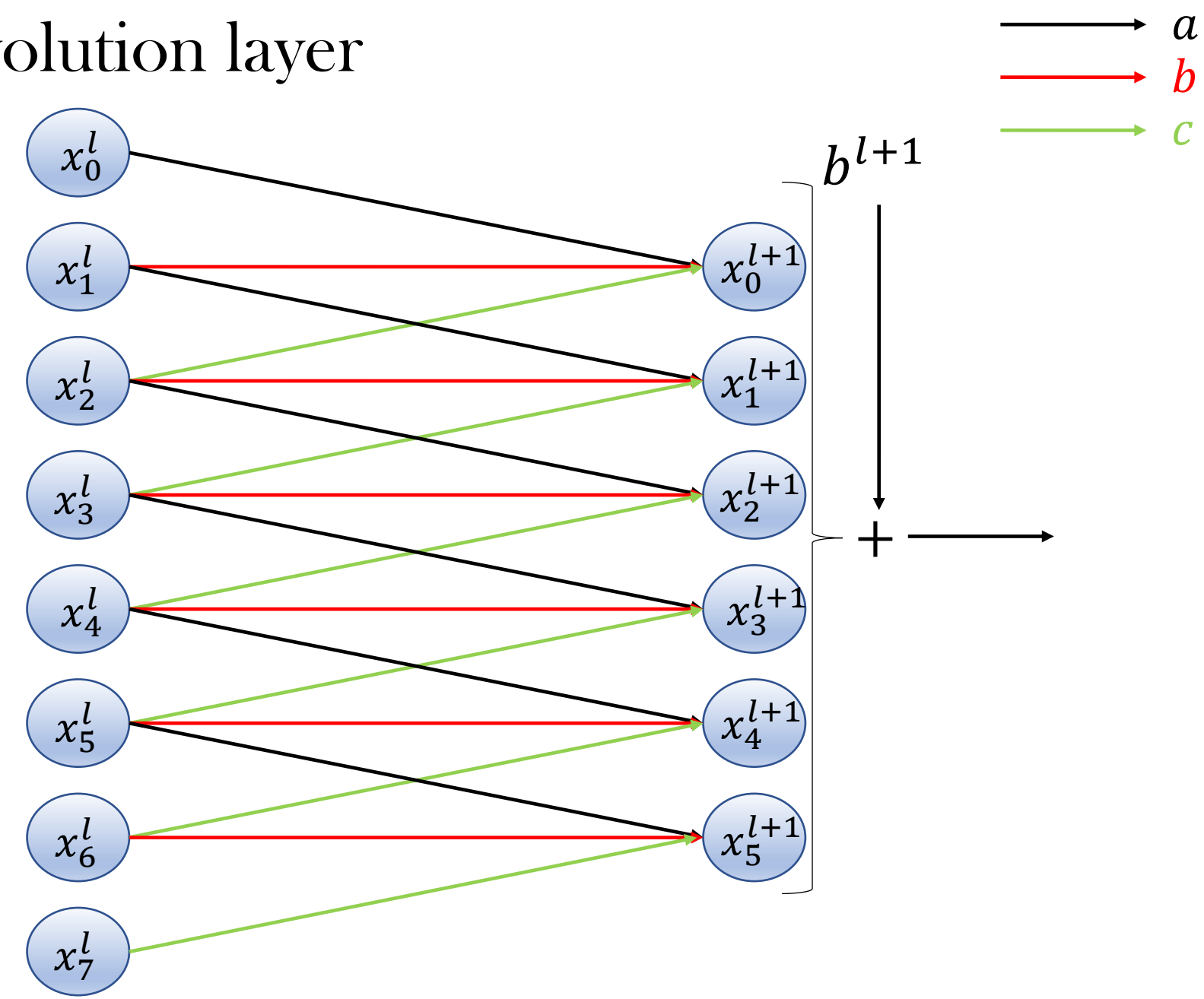


a

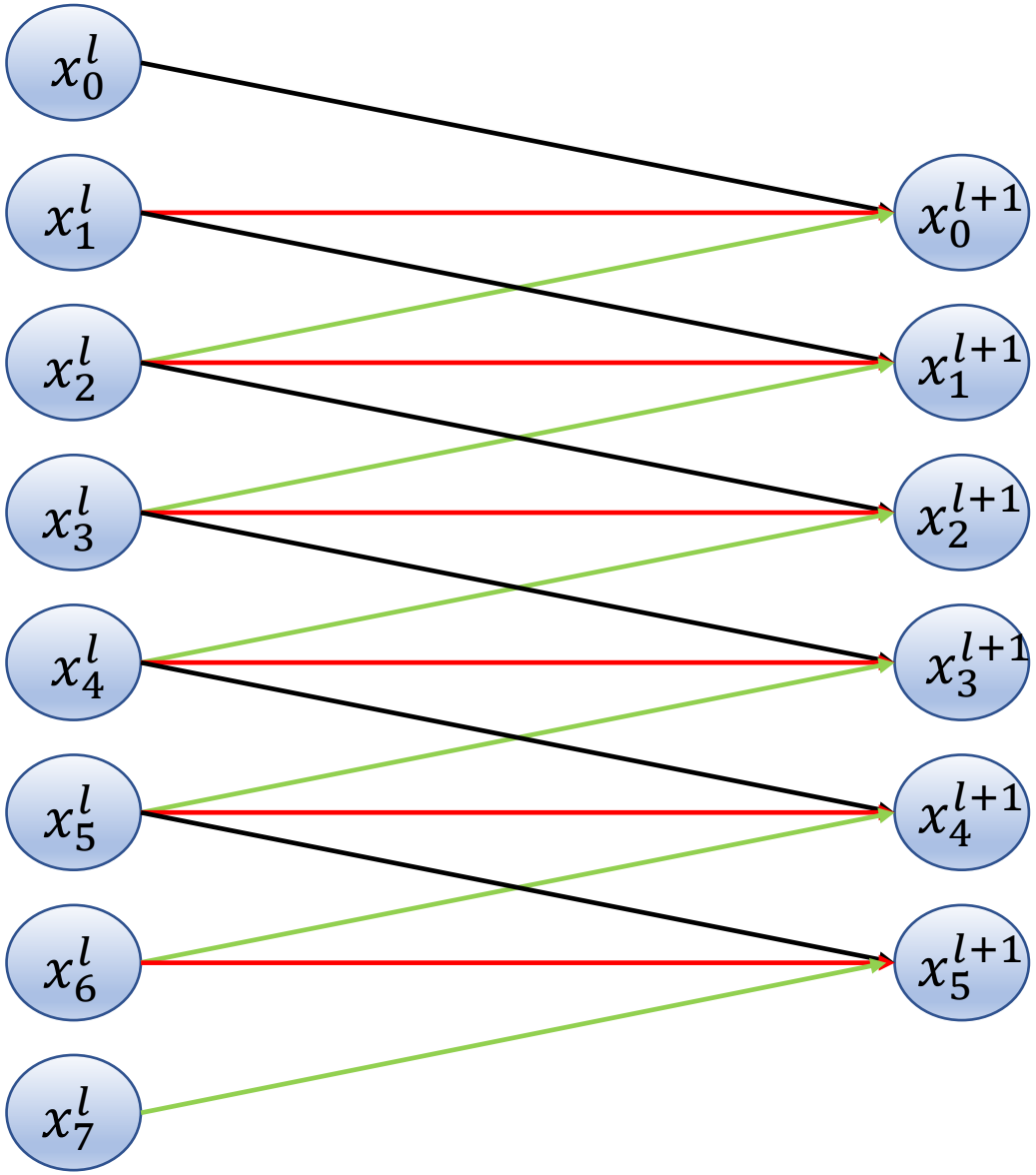
b

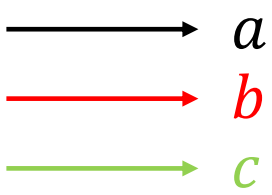
c

Convolution layer



Convolution layer



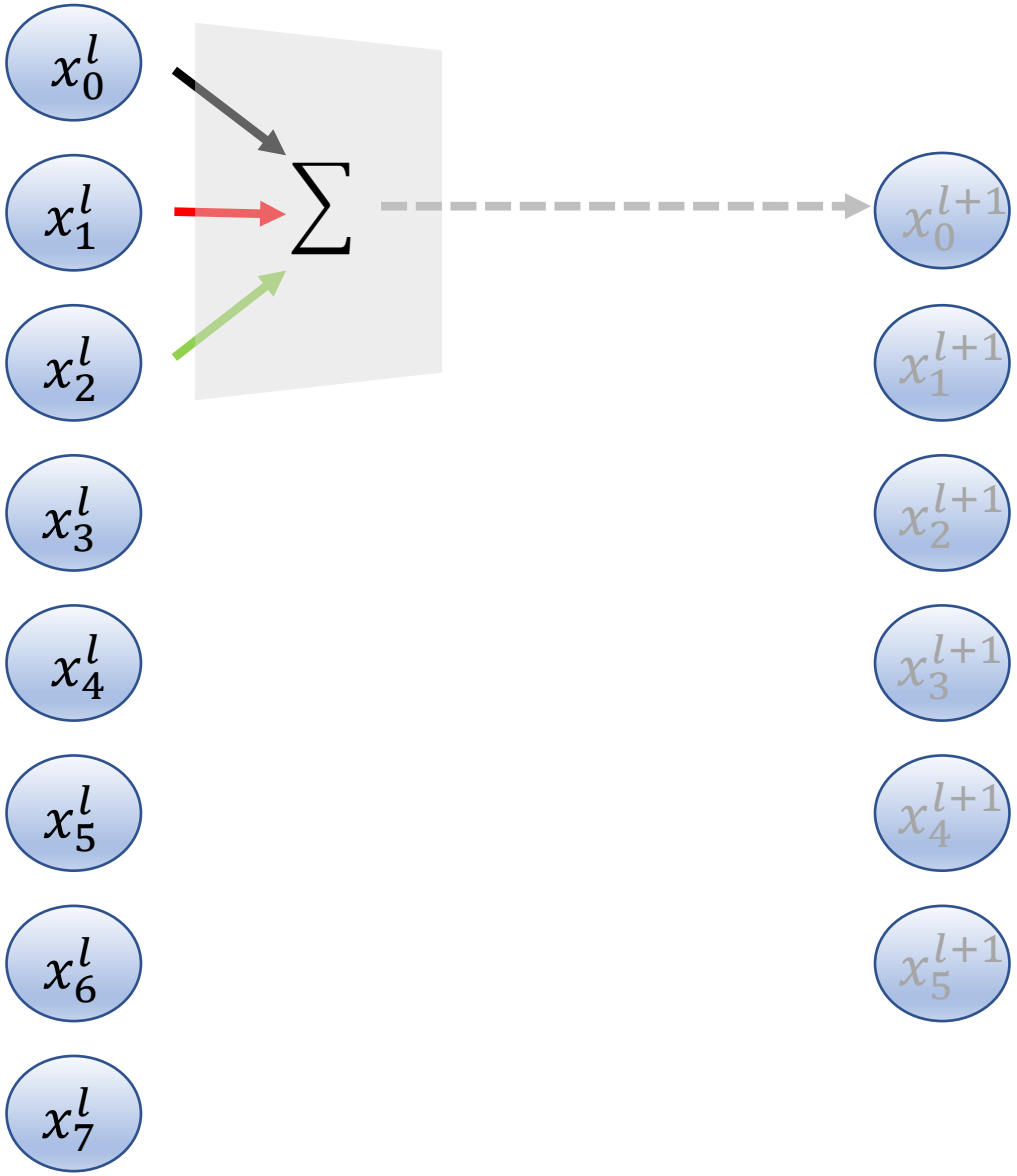


$$\begin{bmatrix} a & b & c & 0 & 0 & 0 & 0 & 0 \\ 0 & a & b & c & 0 & 0 & 0 & 0 \\ 0 & 0 & a & b & c & 0 & 0 & 0 \\ 0 & 0 & 0 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & 0 & a & b & c & 0 \\ 0 & 0 & 0 & 0 & 0 & a & b & c \end{bmatrix}$$

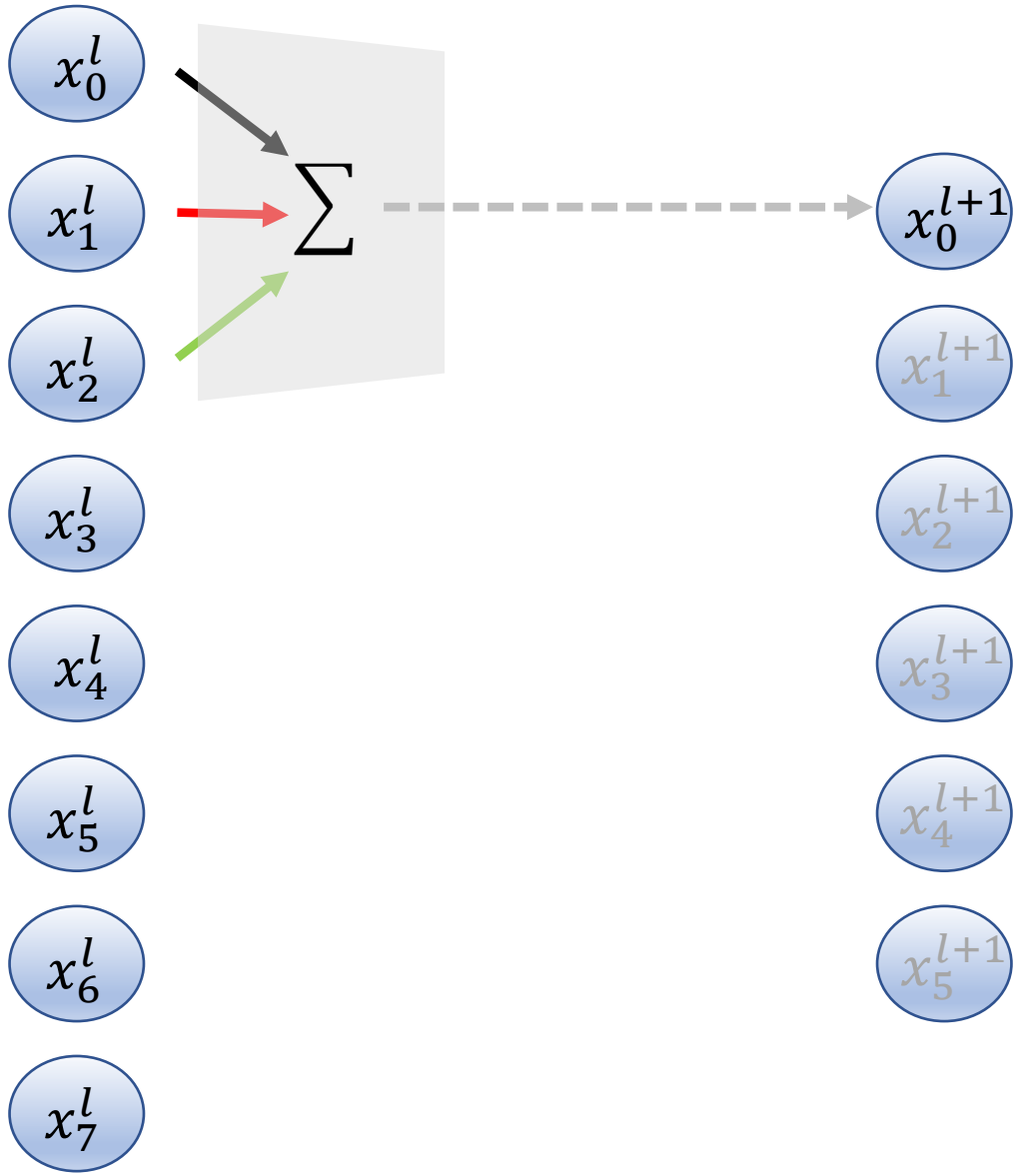
Special case of Linear Layer

Toeplitz matrix

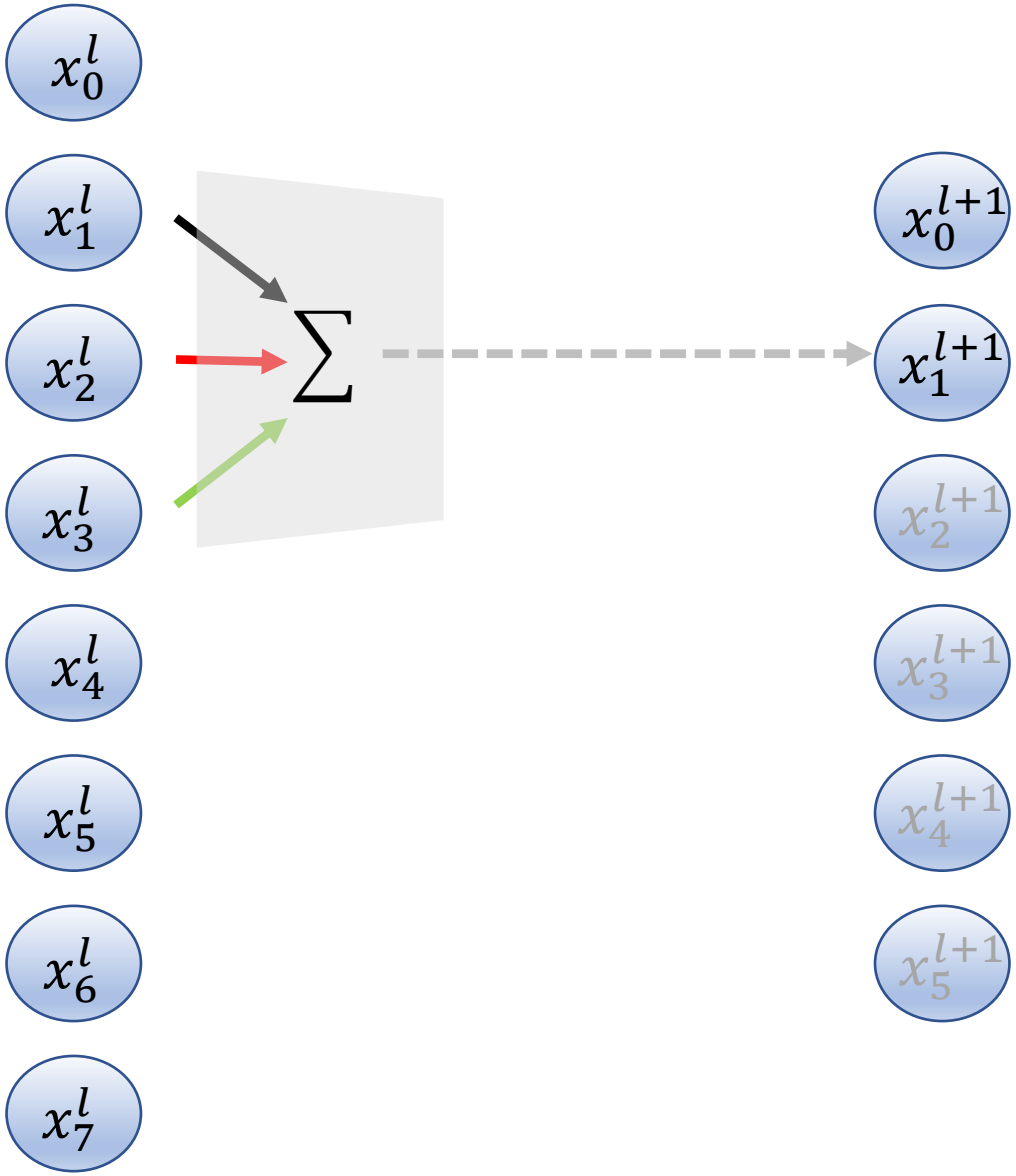
Convolution Filter



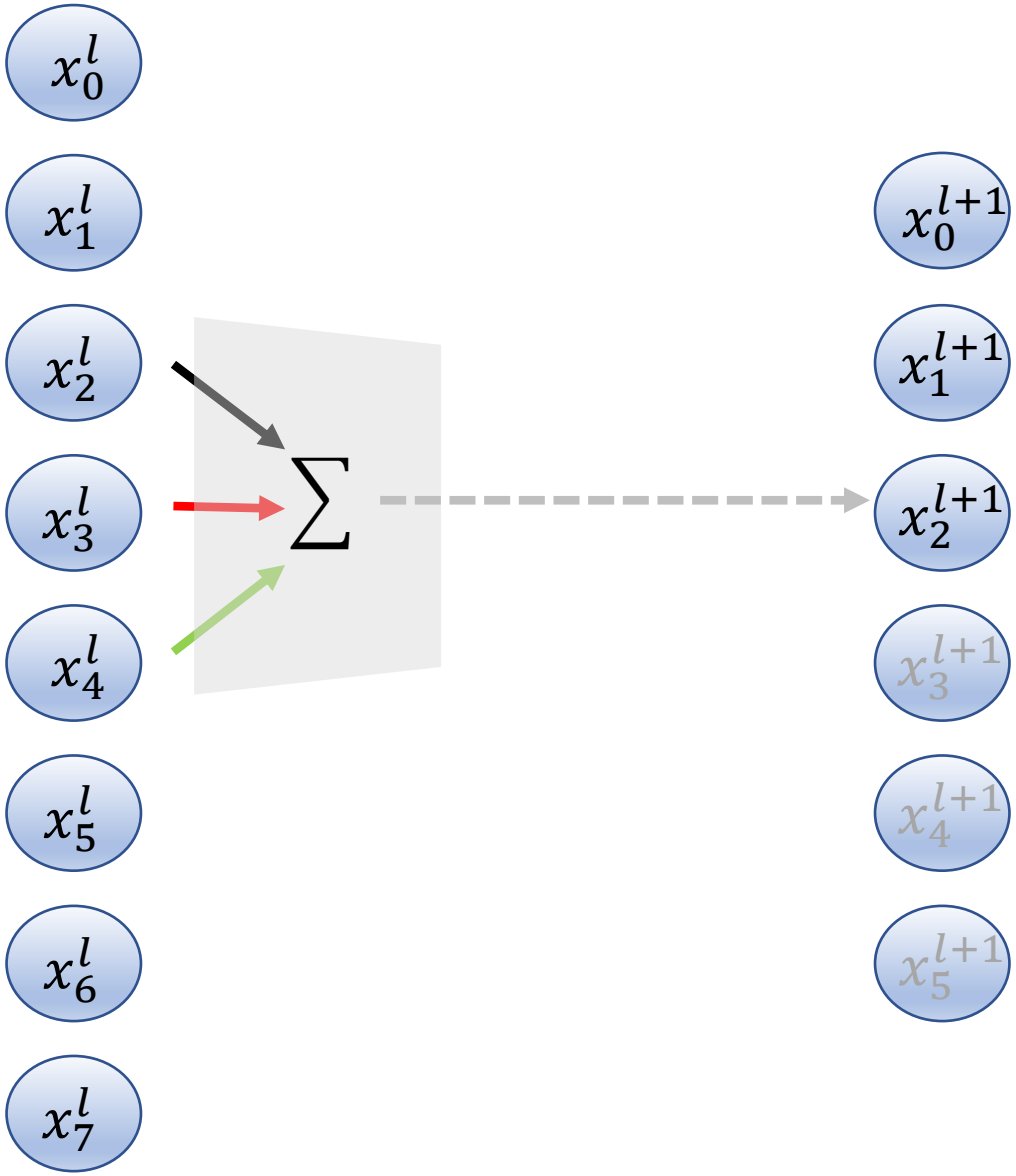
Convolution Filter



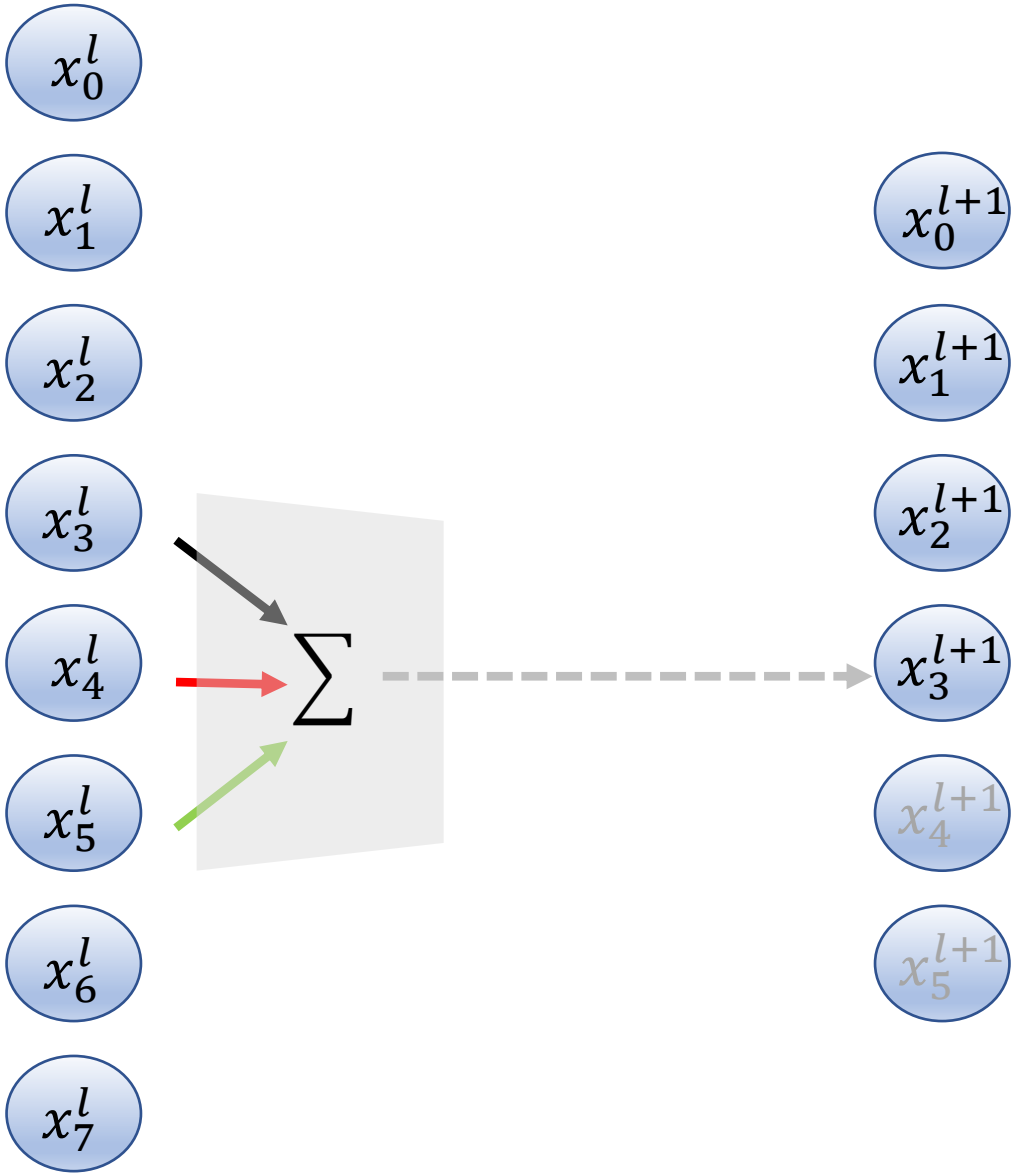
Convolution Filter



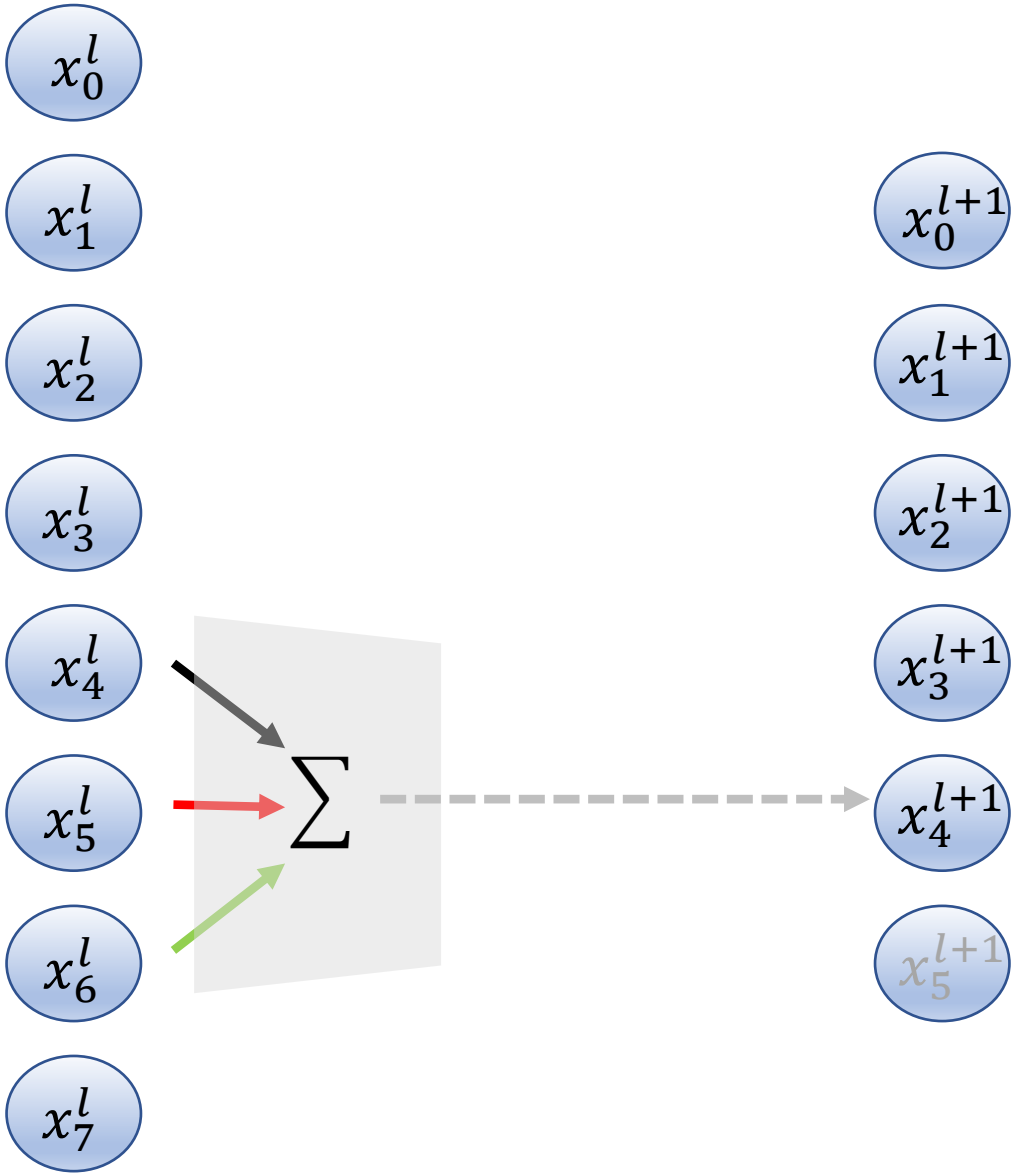
Convolution Filter



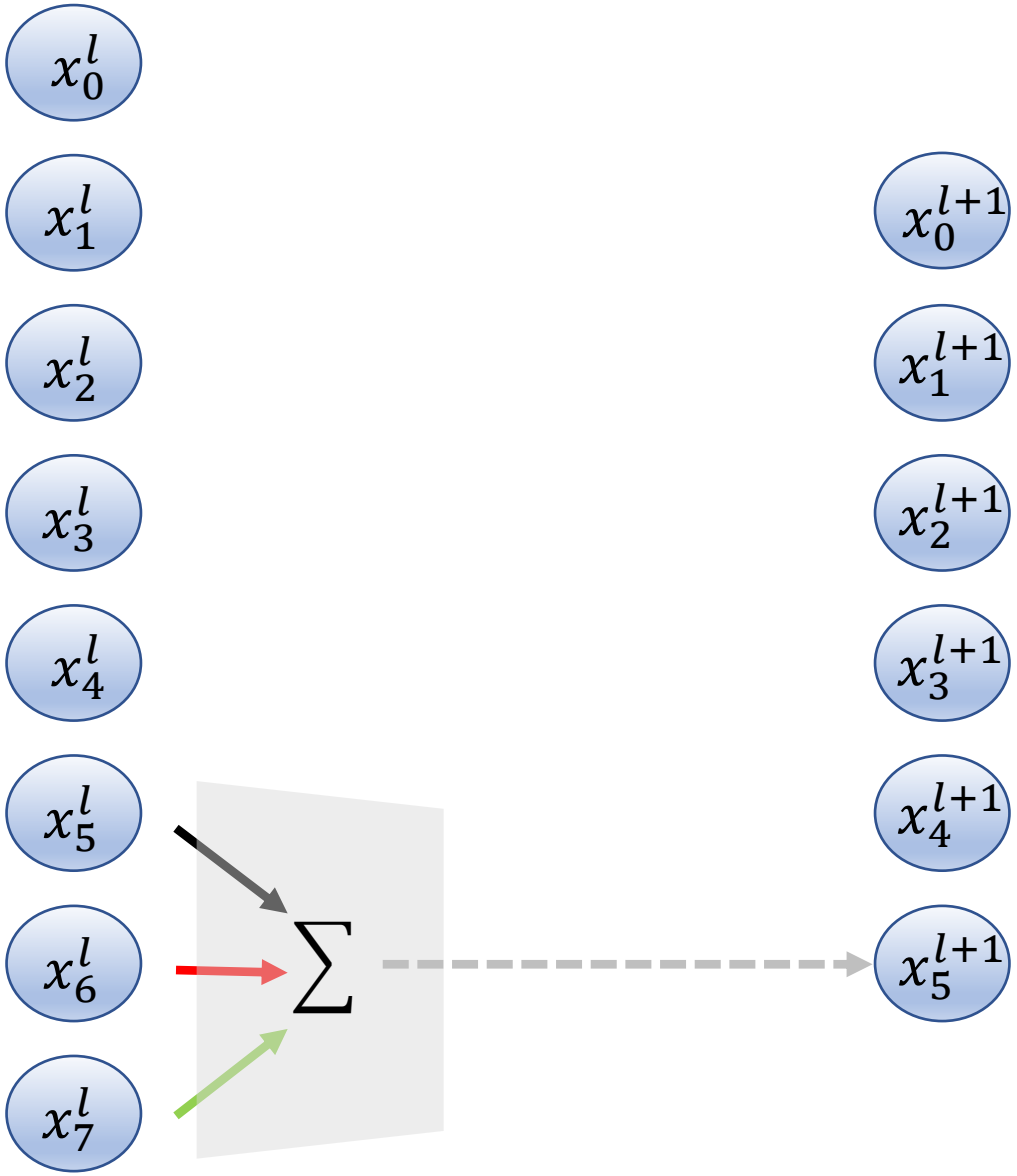
Convolution Filter



Convolution Filter



Convolution Filter

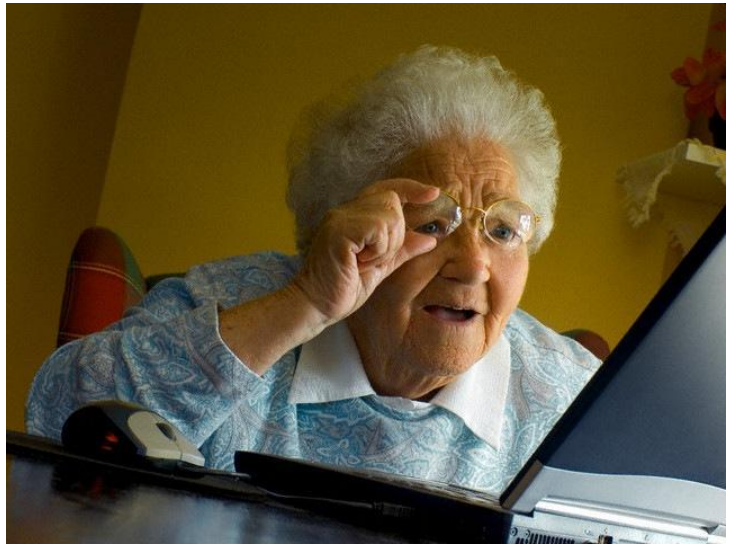


Q: Given input size N and filter size K, find output size.

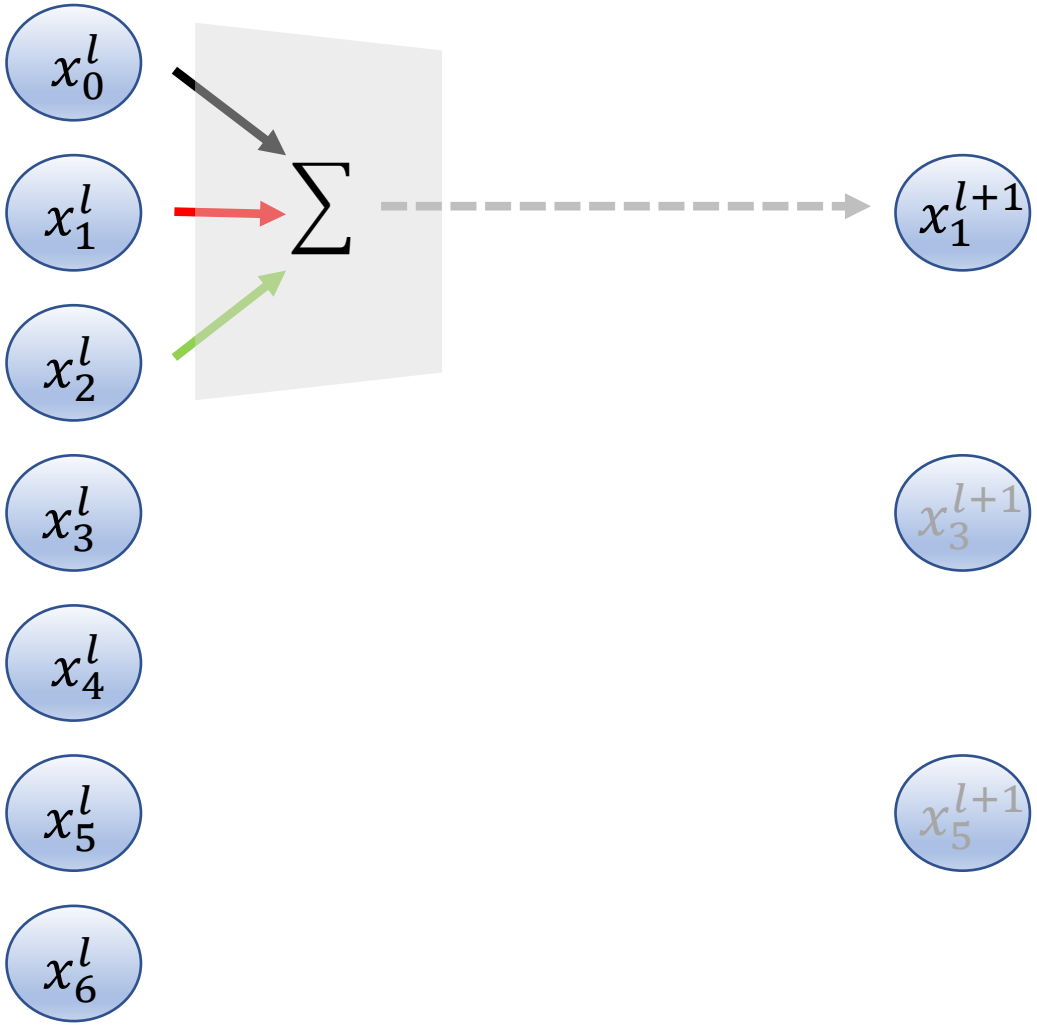
A: $N - K + 1$

Q: is this a convolution?

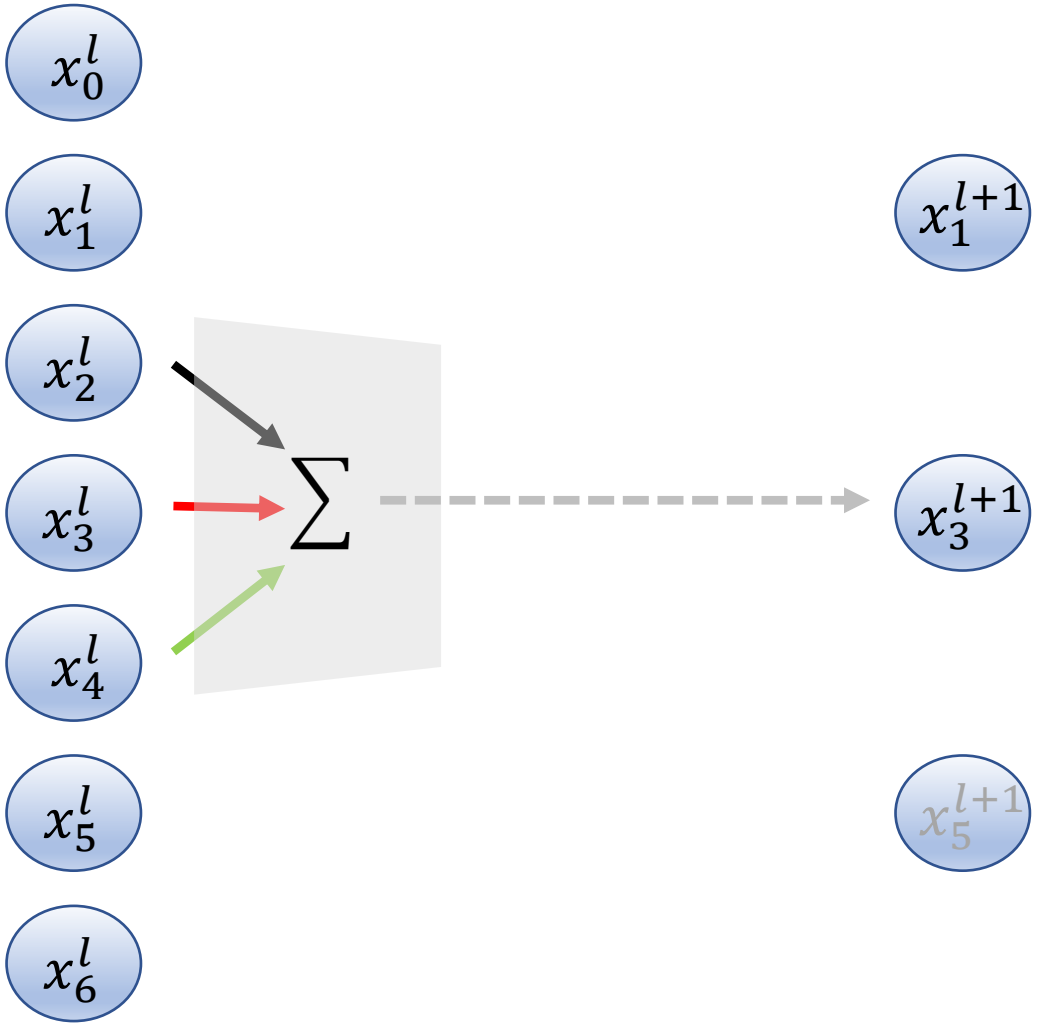
A: Yes, but with the flipped filter.
This is cross-correlation.



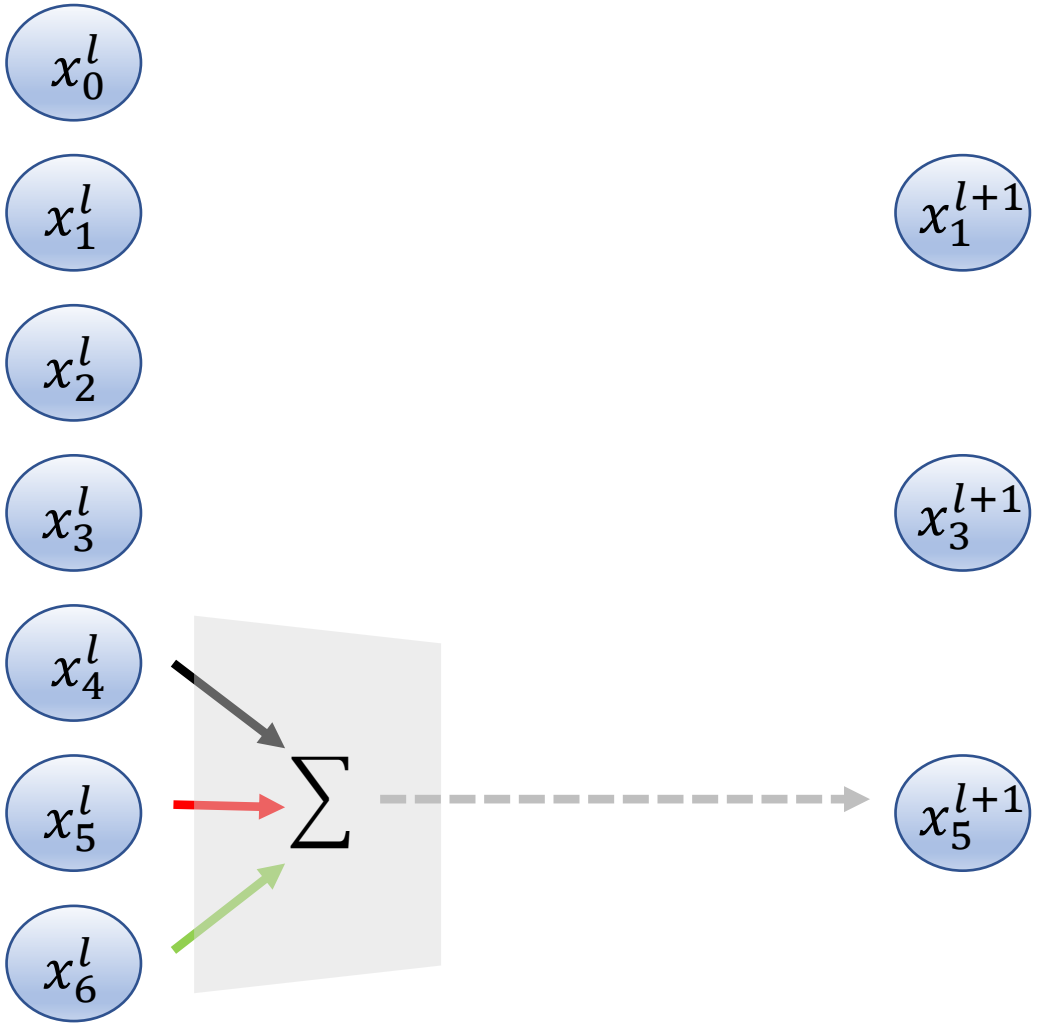
Stride



Stride

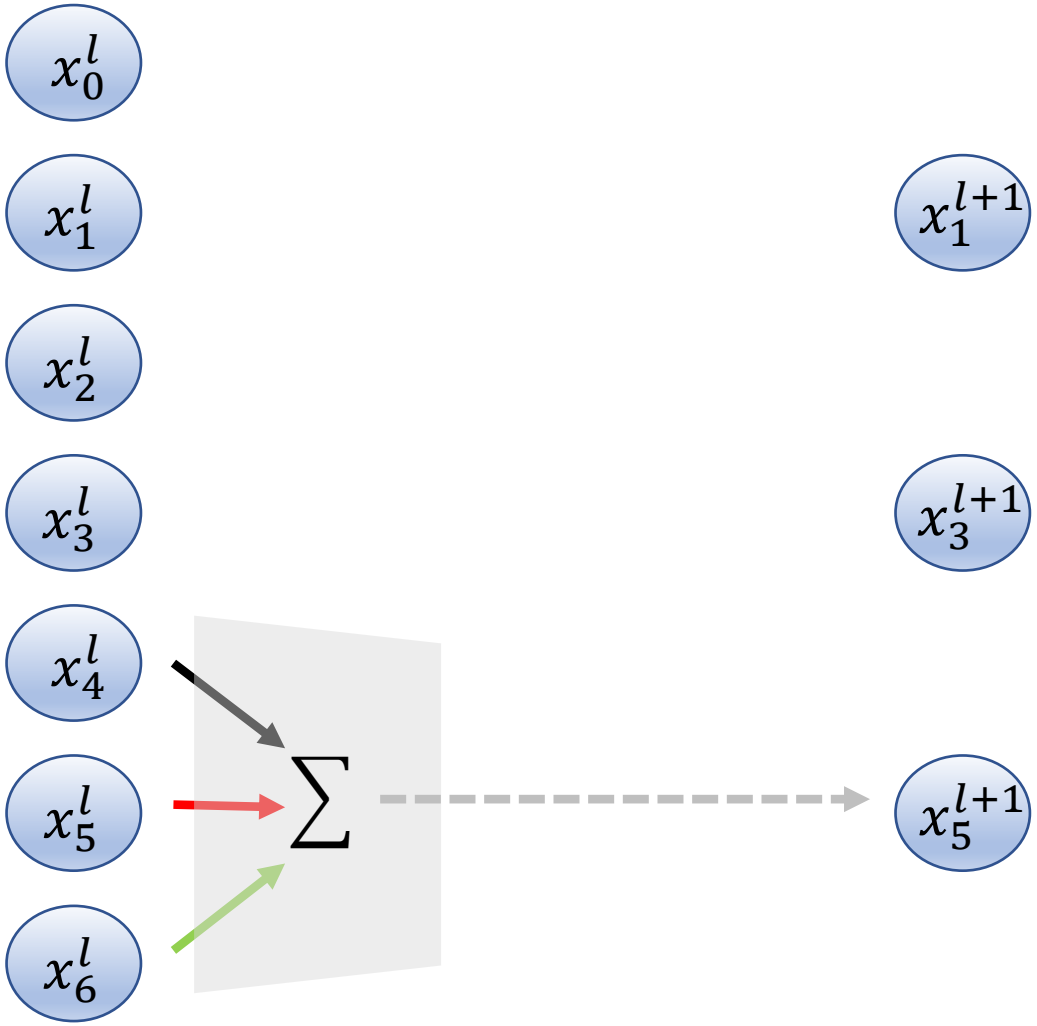


Stride



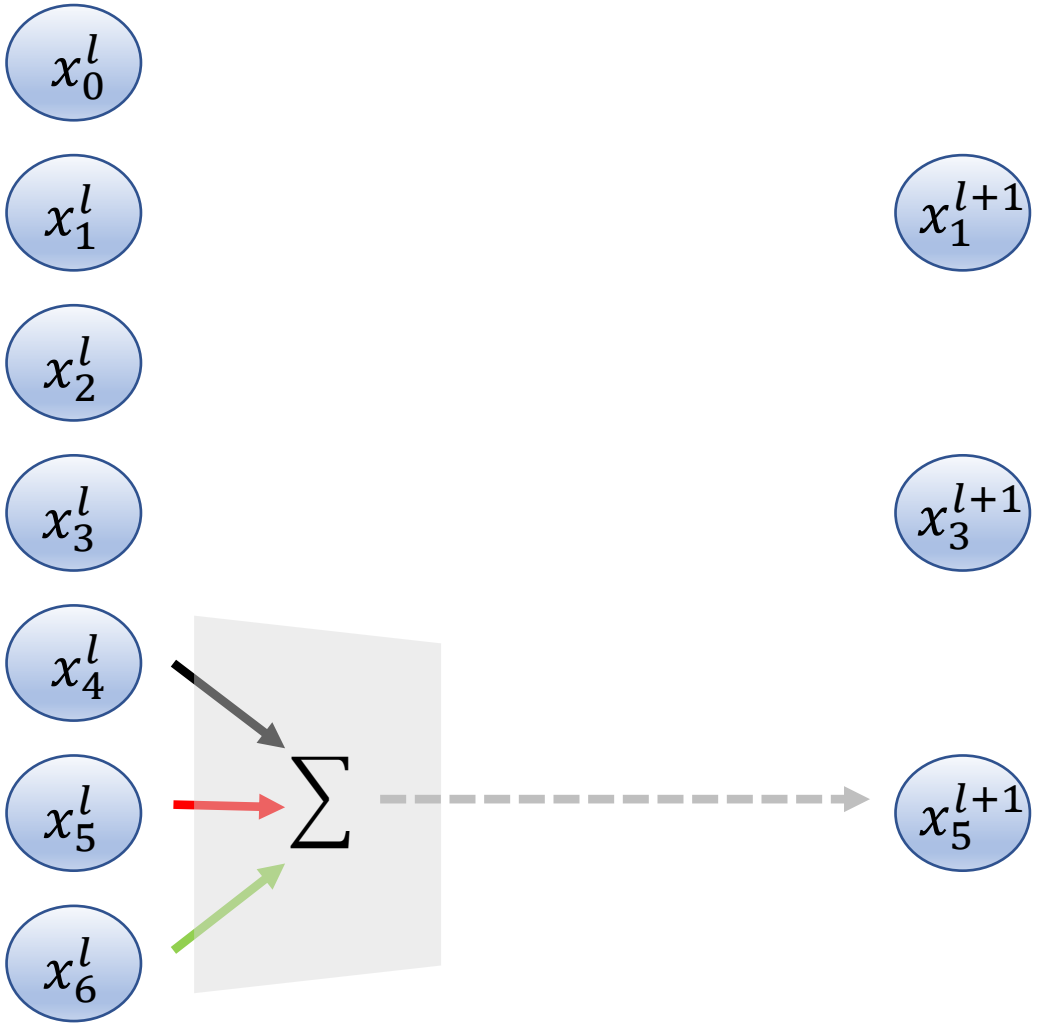
$$\begin{bmatrix} a & b & c & 0 & 0 & 0 & 0 \\ 0 & a & b & c & 0 & 0 & 0 \\ 0 & 0 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & a & b & c & 0 \\ 0 & 0 & 0 & 0 & a & b & c \\ 0 & 0 & 0 & 0 & 0 & a & b \end{bmatrix}$$

Stride



$$\begin{bmatrix} a & b & c & 0 & 0 & 0 & 0 \\ 0 & a & b & c & 0 & 0 & 0 \\ 0 & 0 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & a & b & c & 0 \\ 0 & 0 & 0 & 0 & a & b & c \\ 0 & 0 & 0 & 0 & 0 & a & b \end{bmatrix}$$

Stride

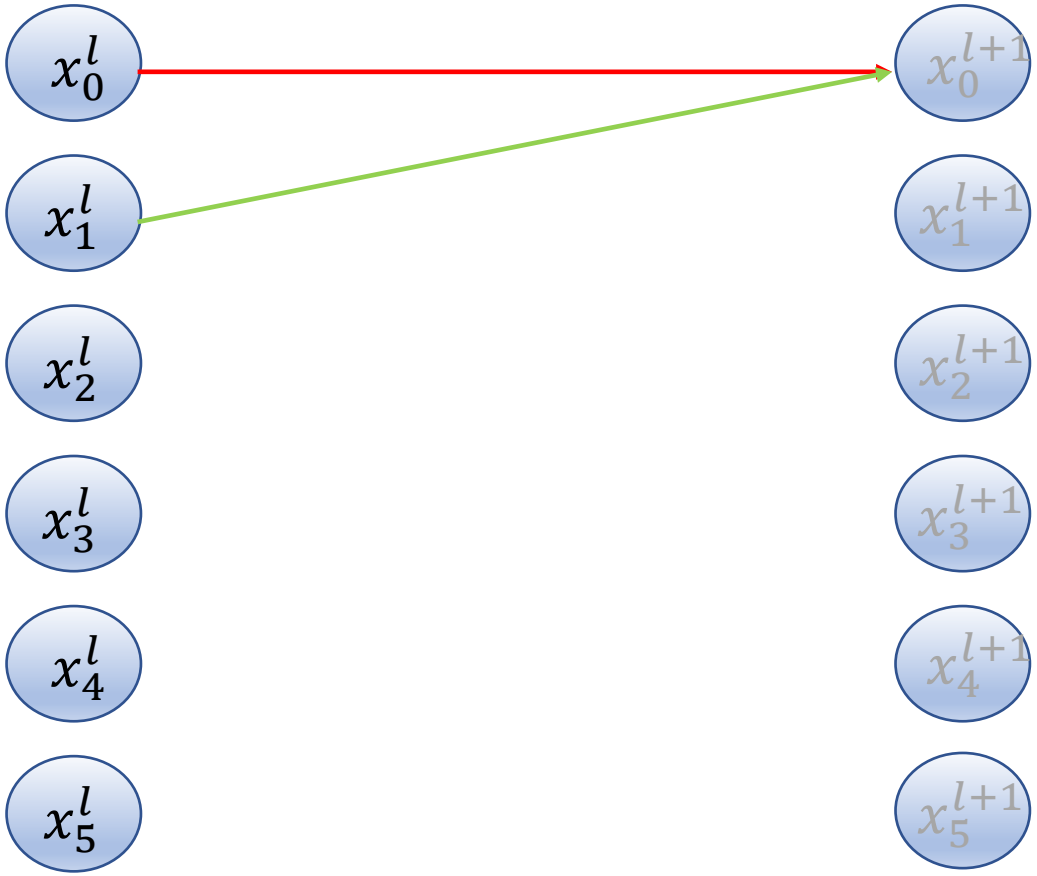


$$\begin{bmatrix} \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} & 0 & 0 & 0 & 0 \\ 0 & 0 & \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} & 0 & 0 \\ 0 & 0 & 0 & 0 & \boldsymbol{a} & \boldsymbol{b} & \boldsymbol{c} \end{bmatrix}$$

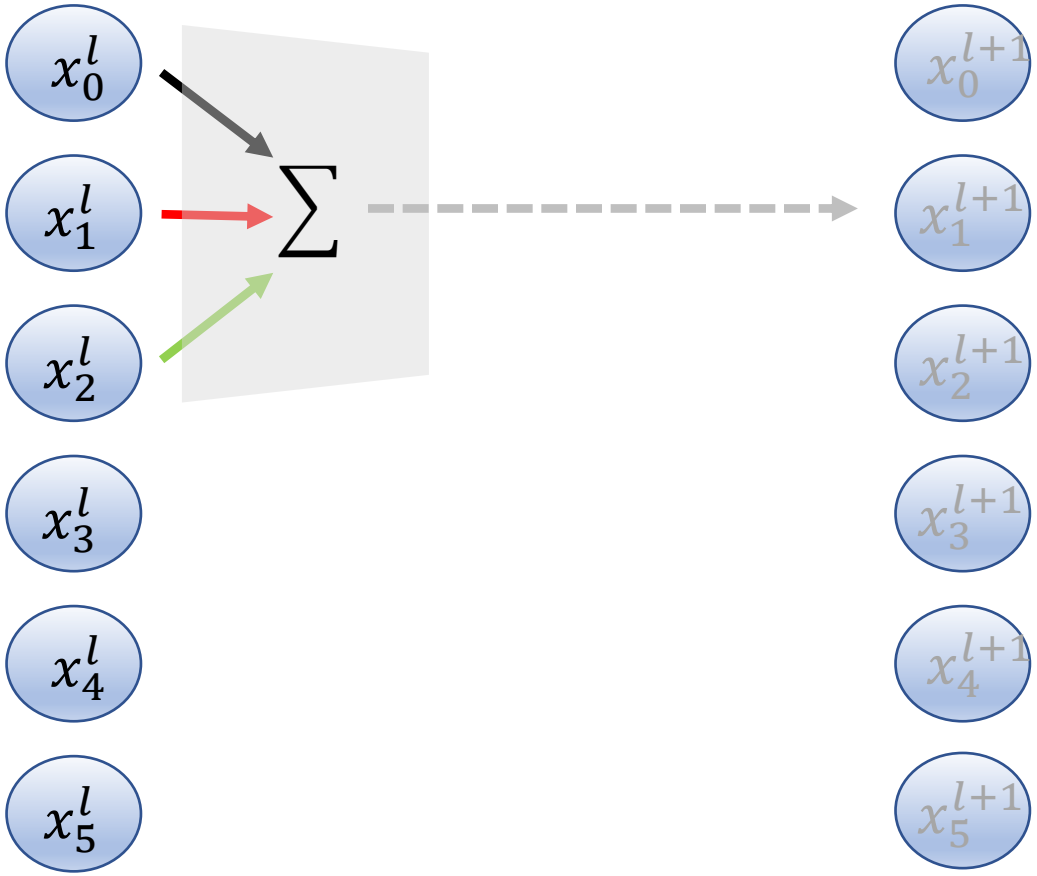
Q: Given input size N, filter size K and stride S , find output size.

A: $\left\lfloor \frac{N-K}{S} + 1 \right\rfloor$

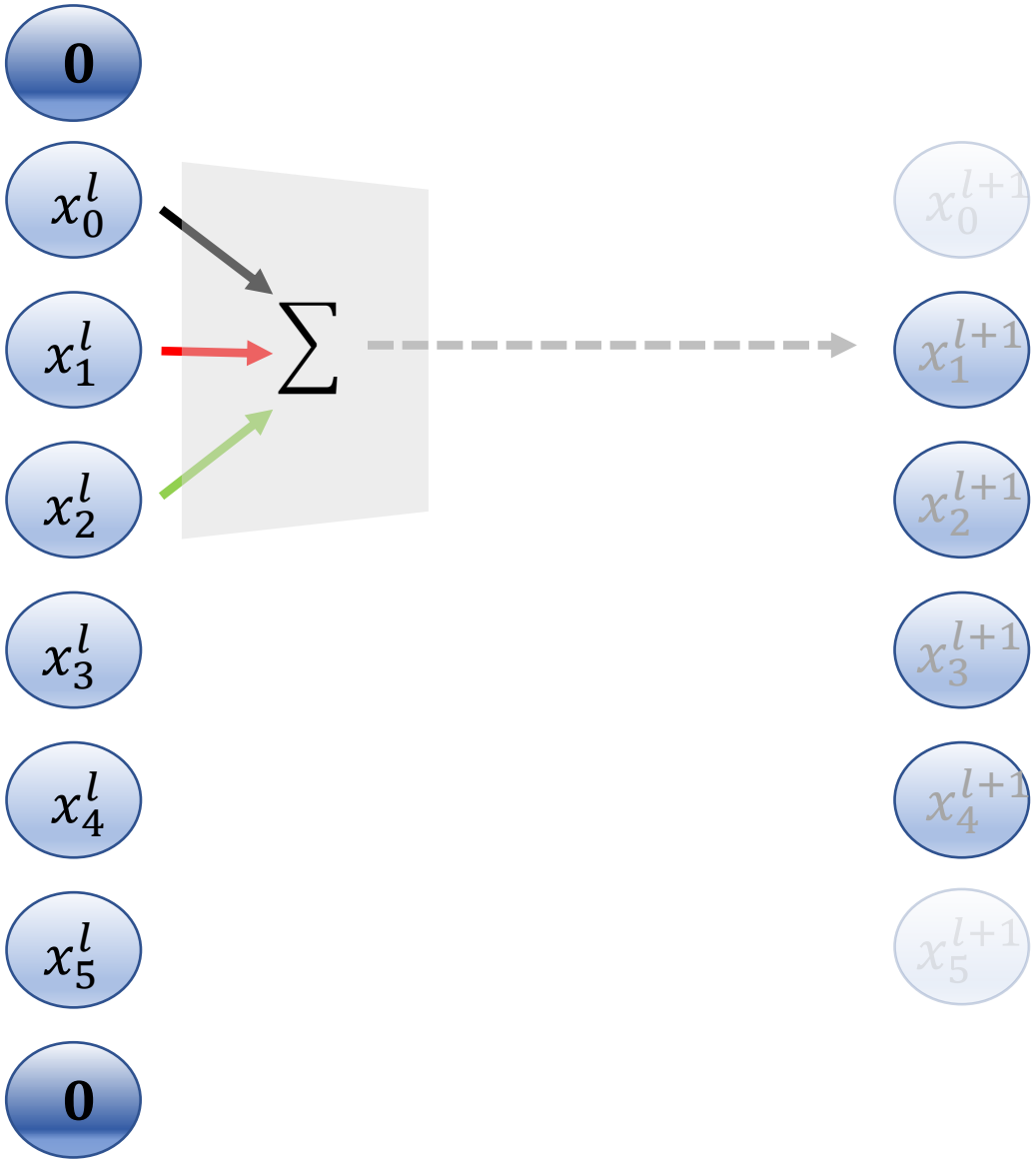
Padding



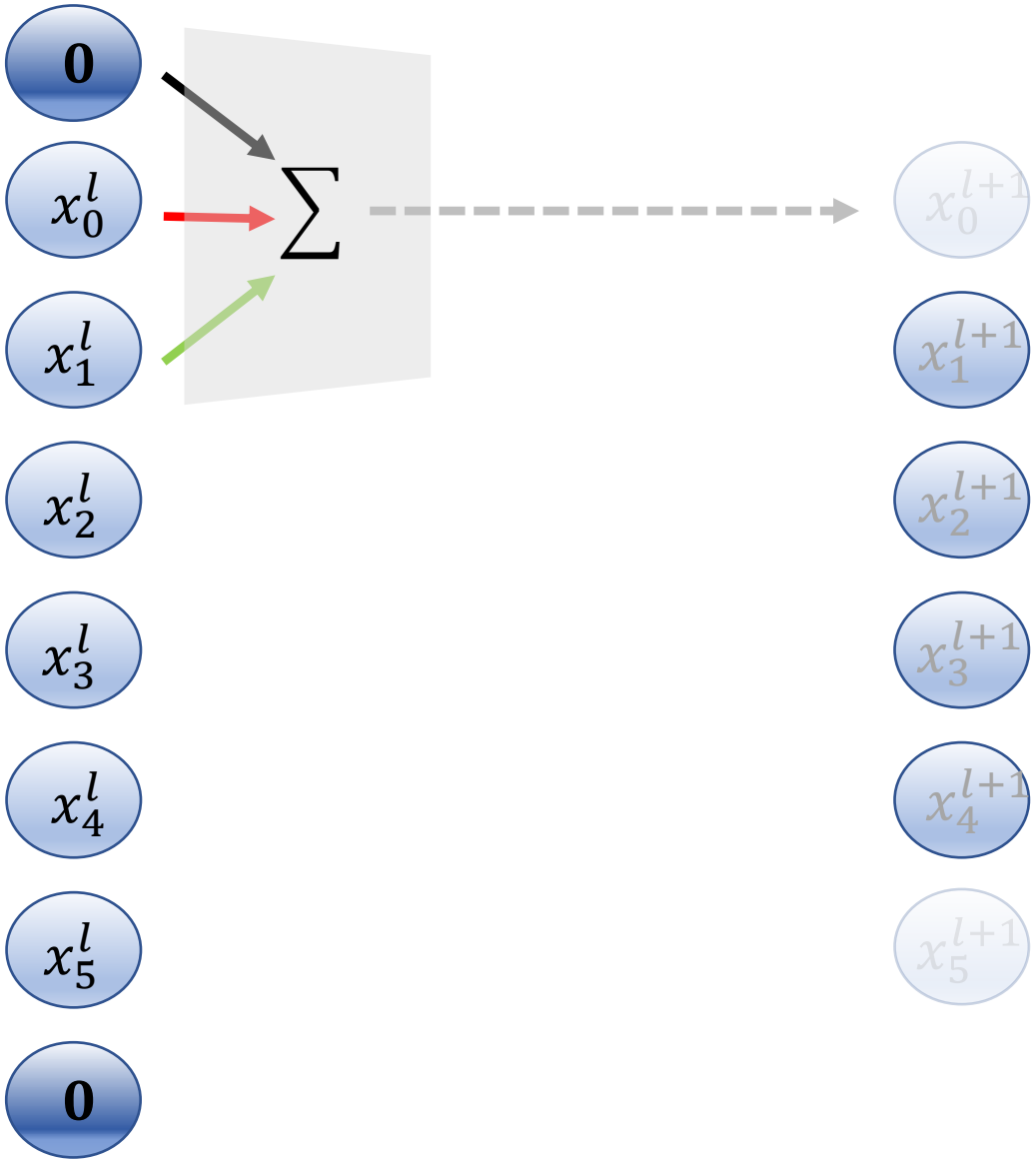
Padding



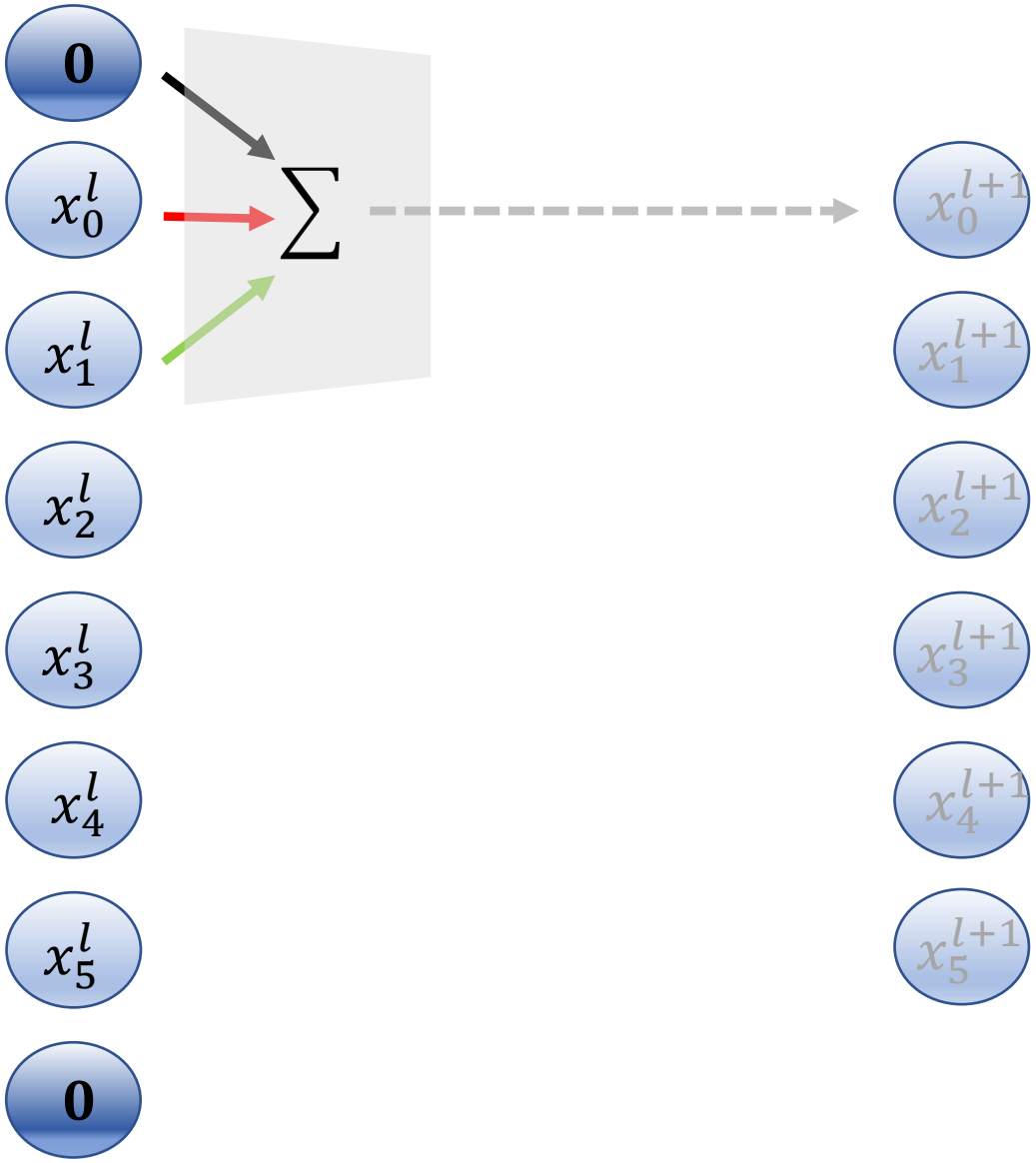
Padding



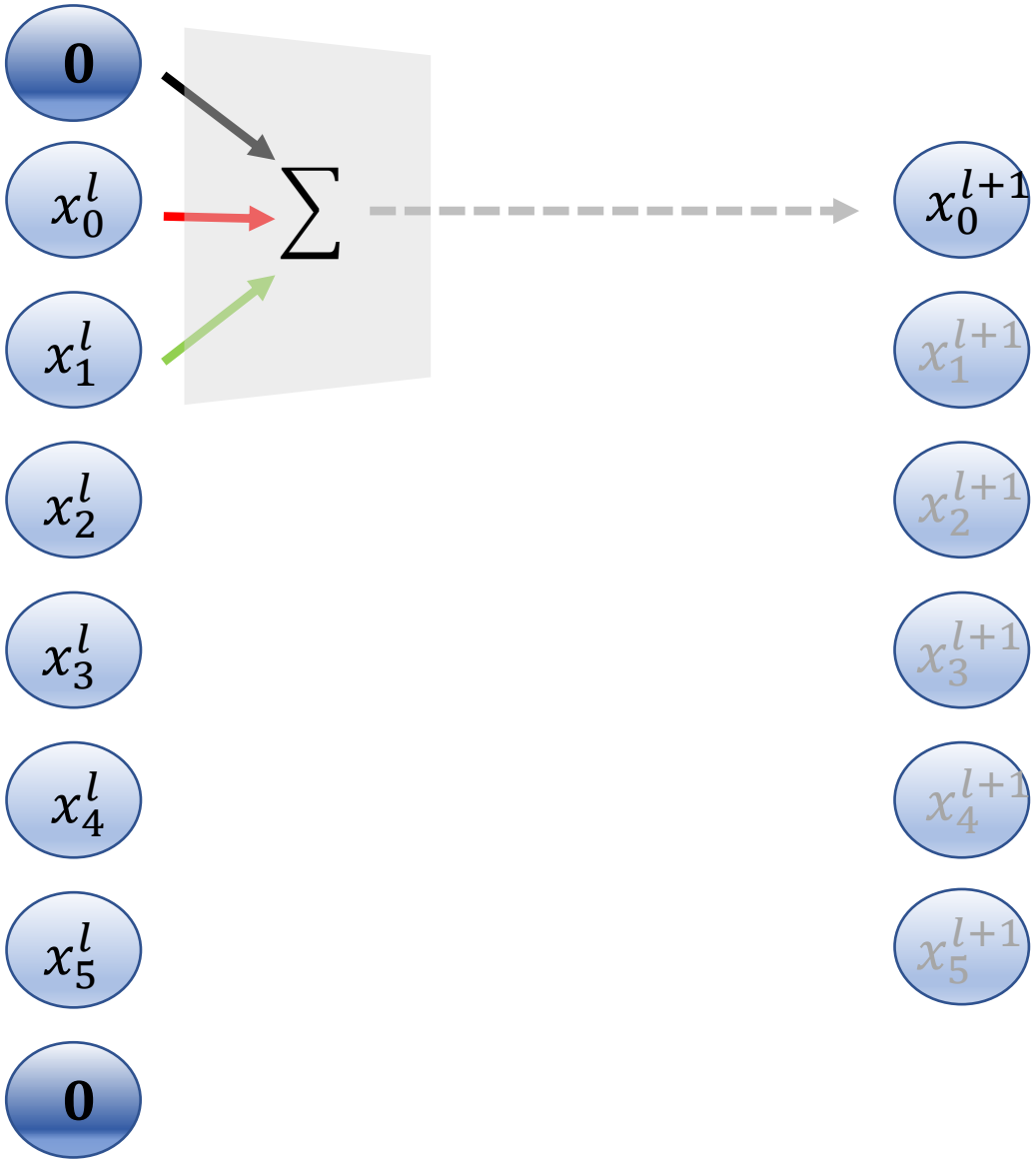
Padding



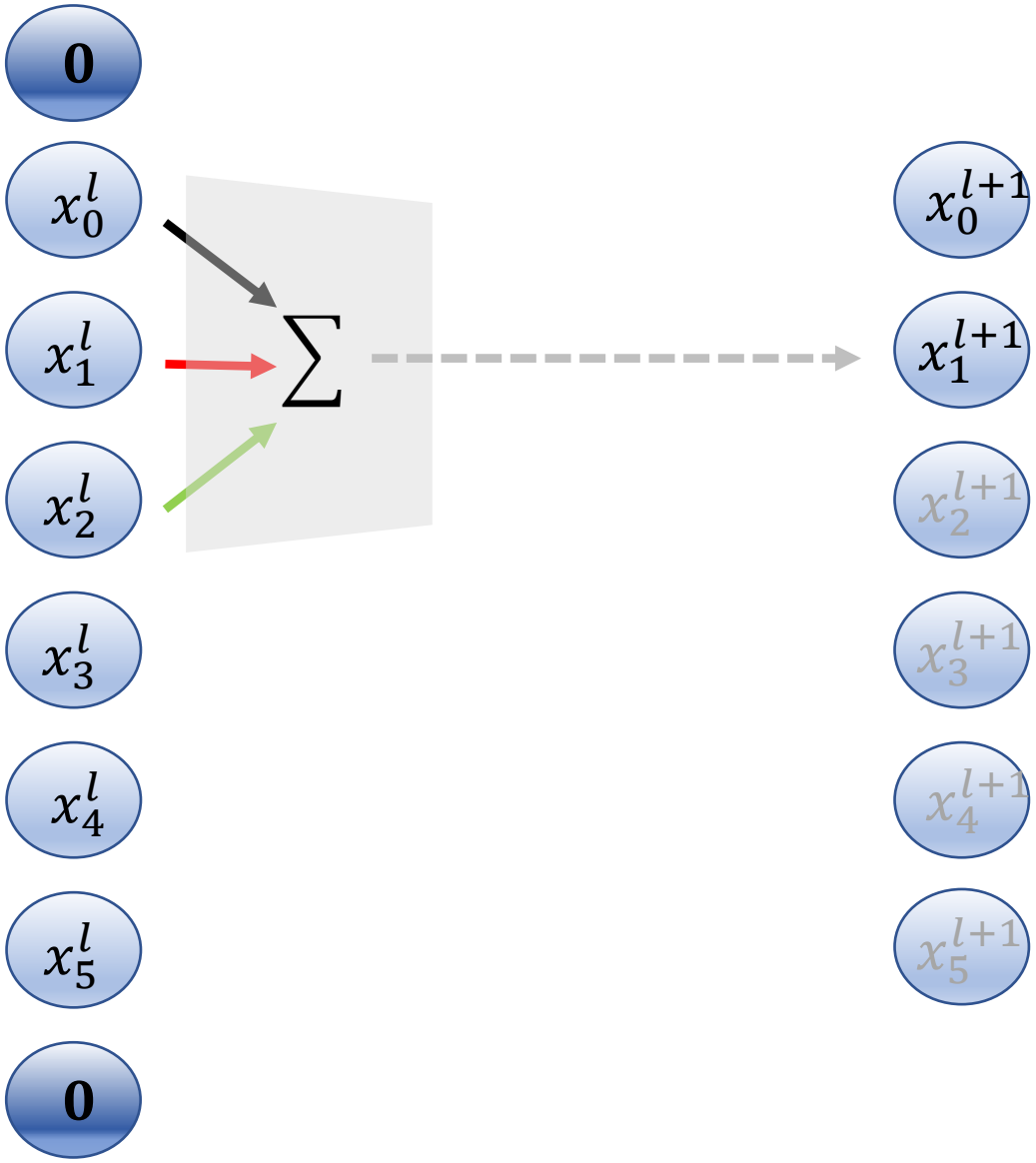
Padding



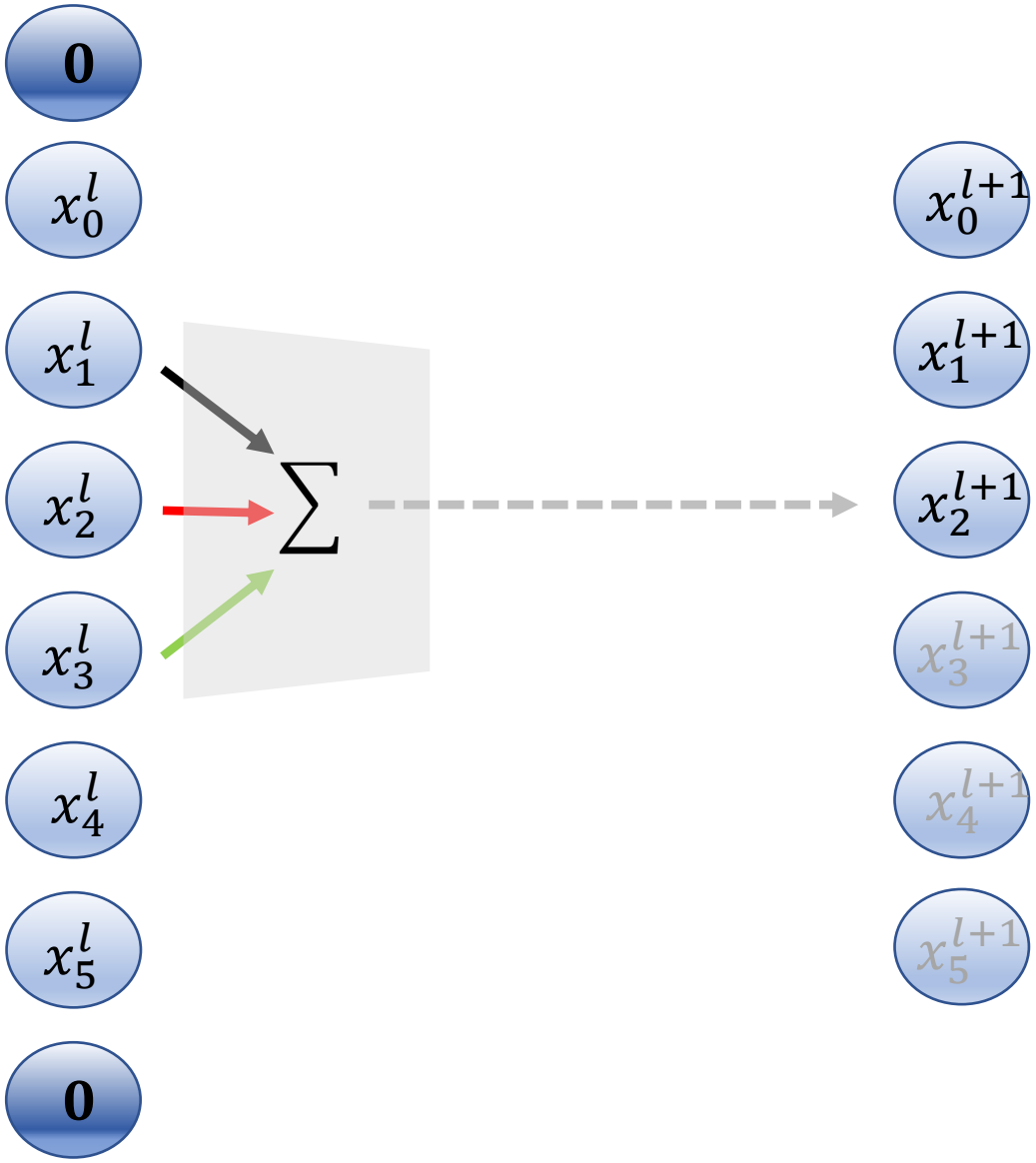
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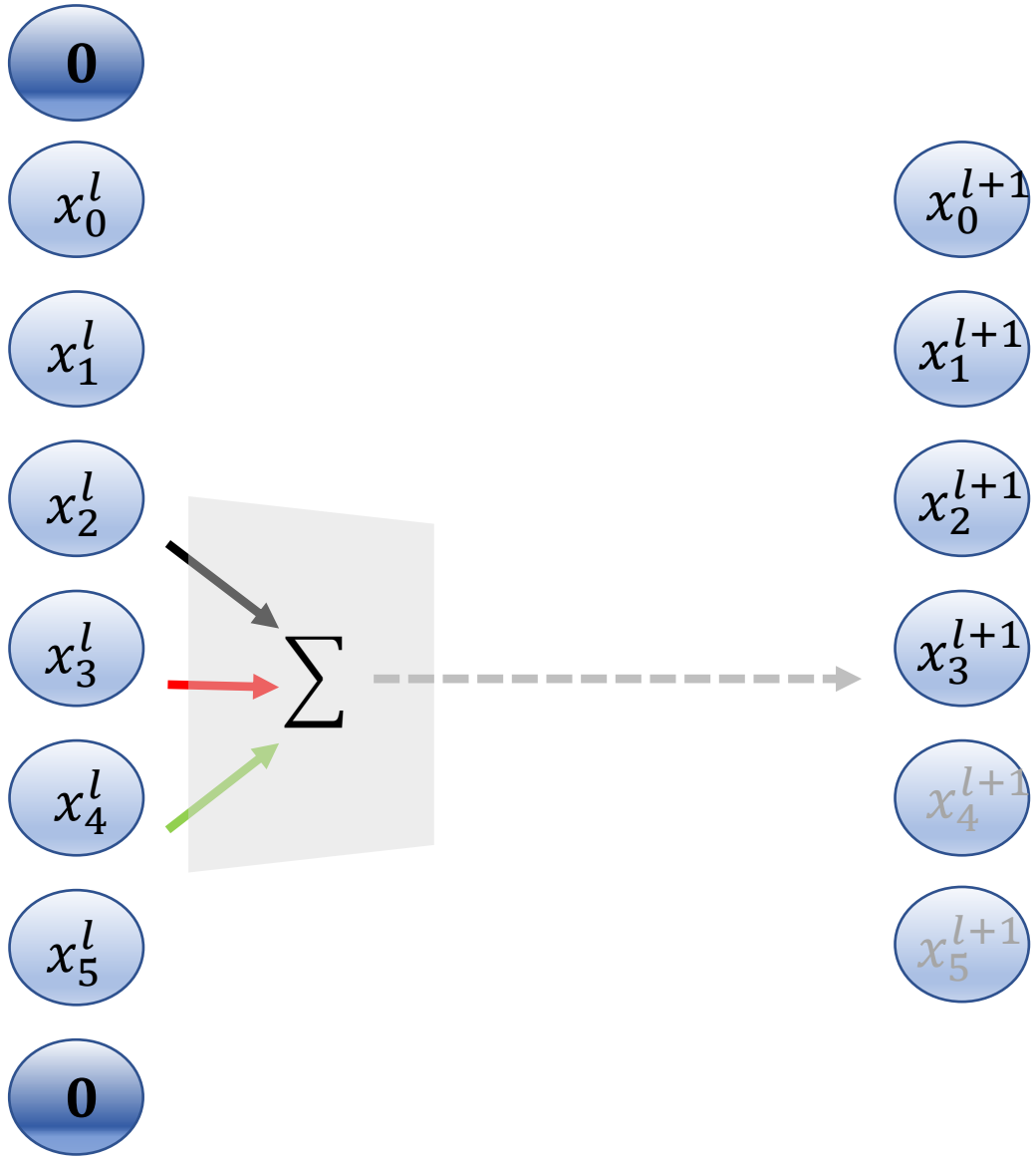
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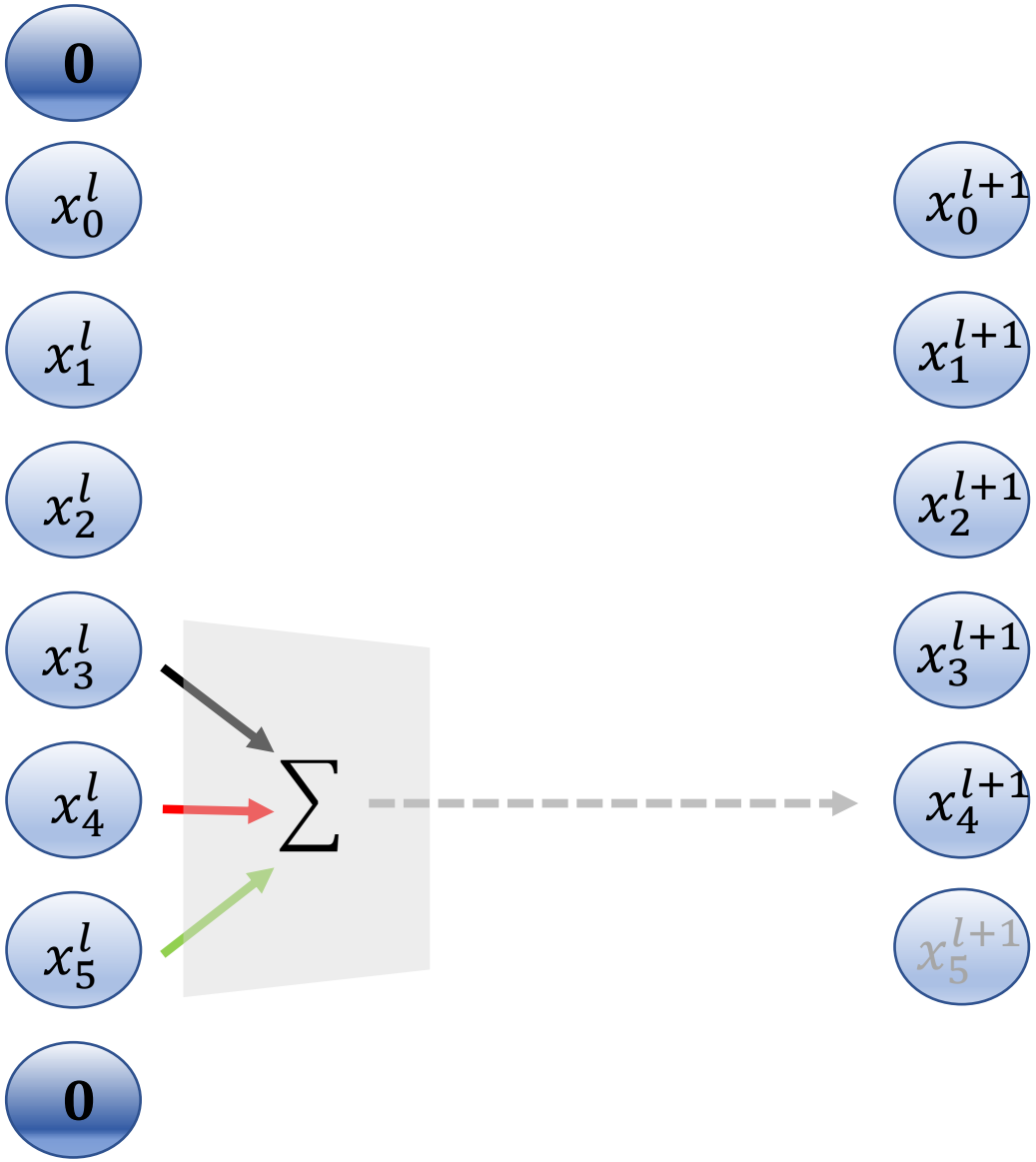
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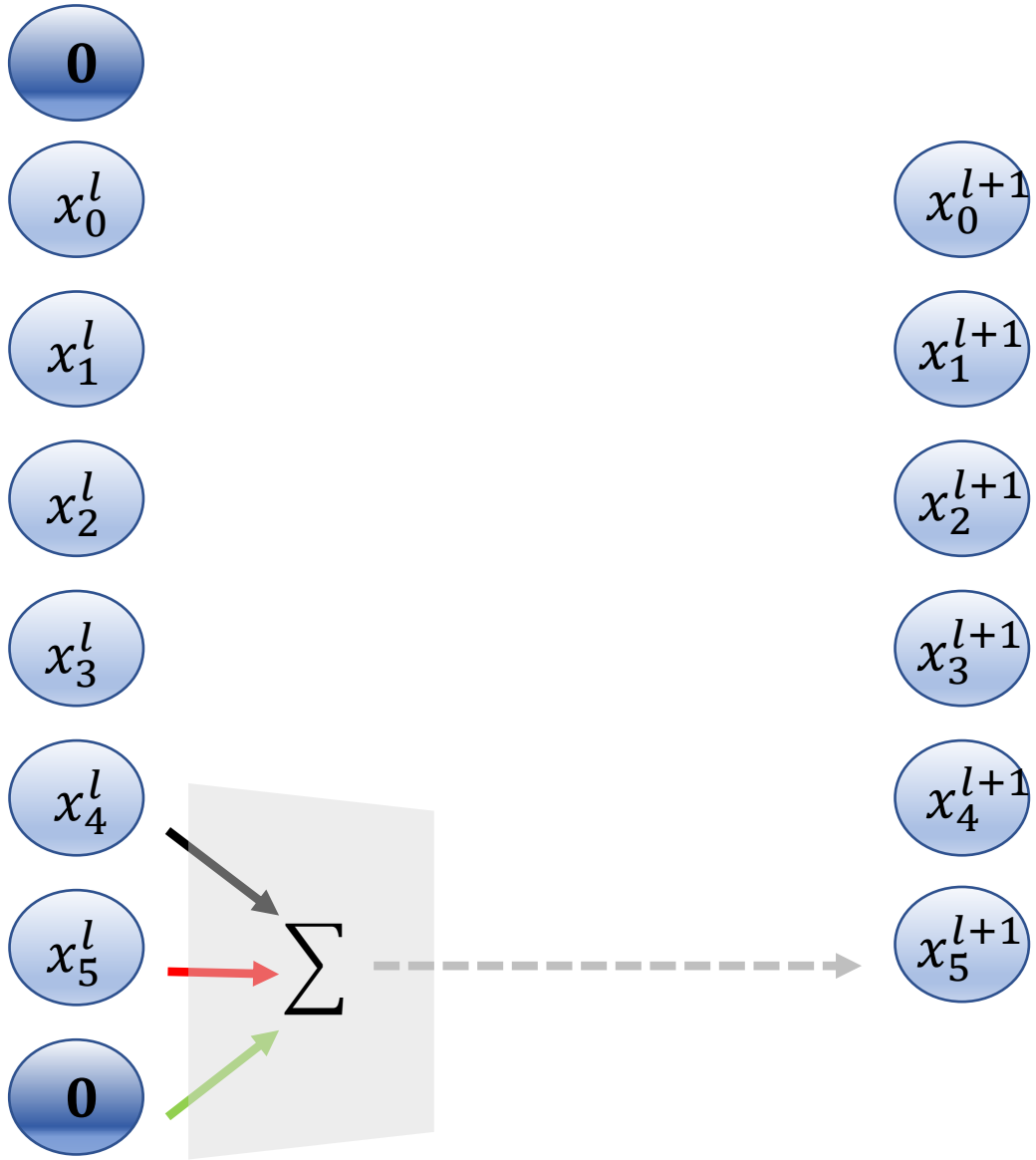
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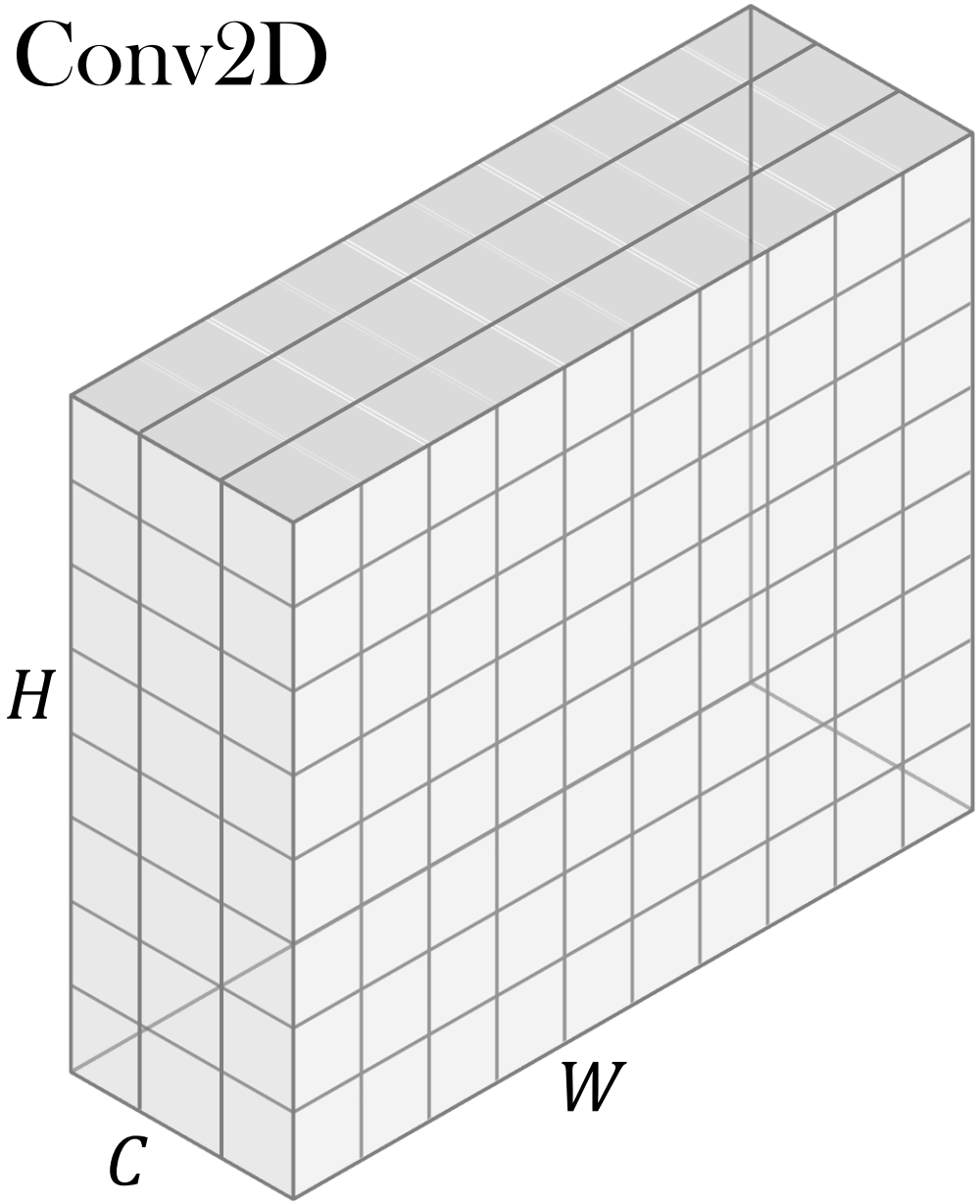
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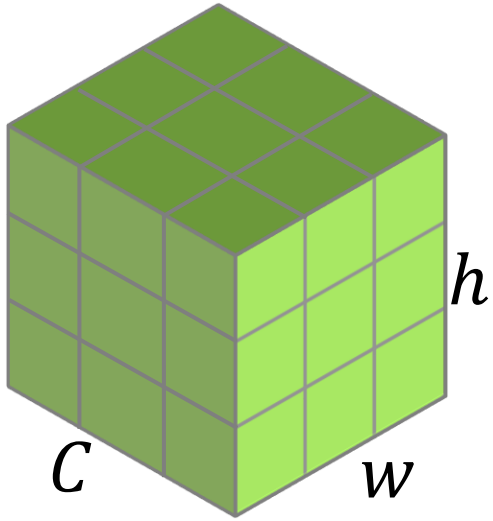
Padding



Conv2D

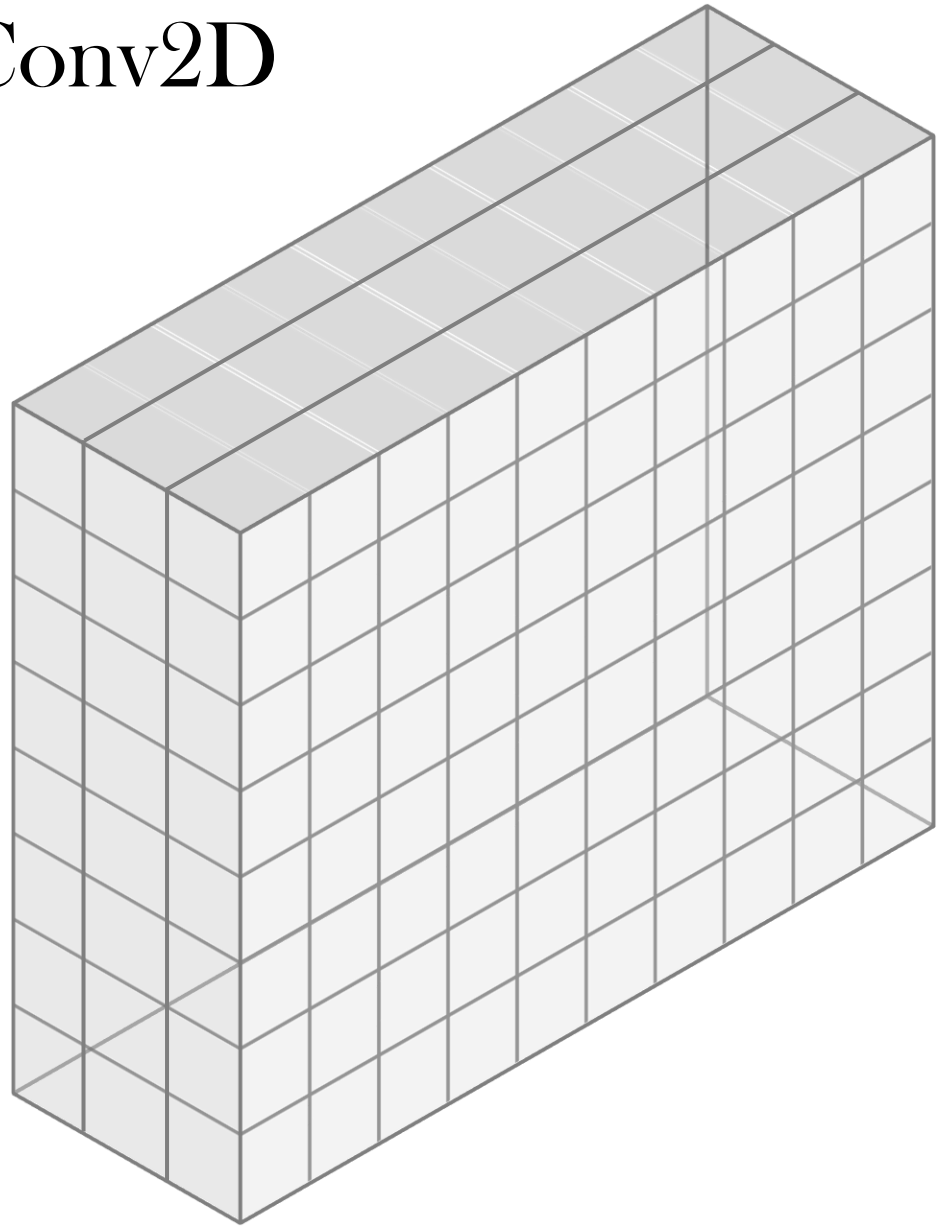


N, C, H, W



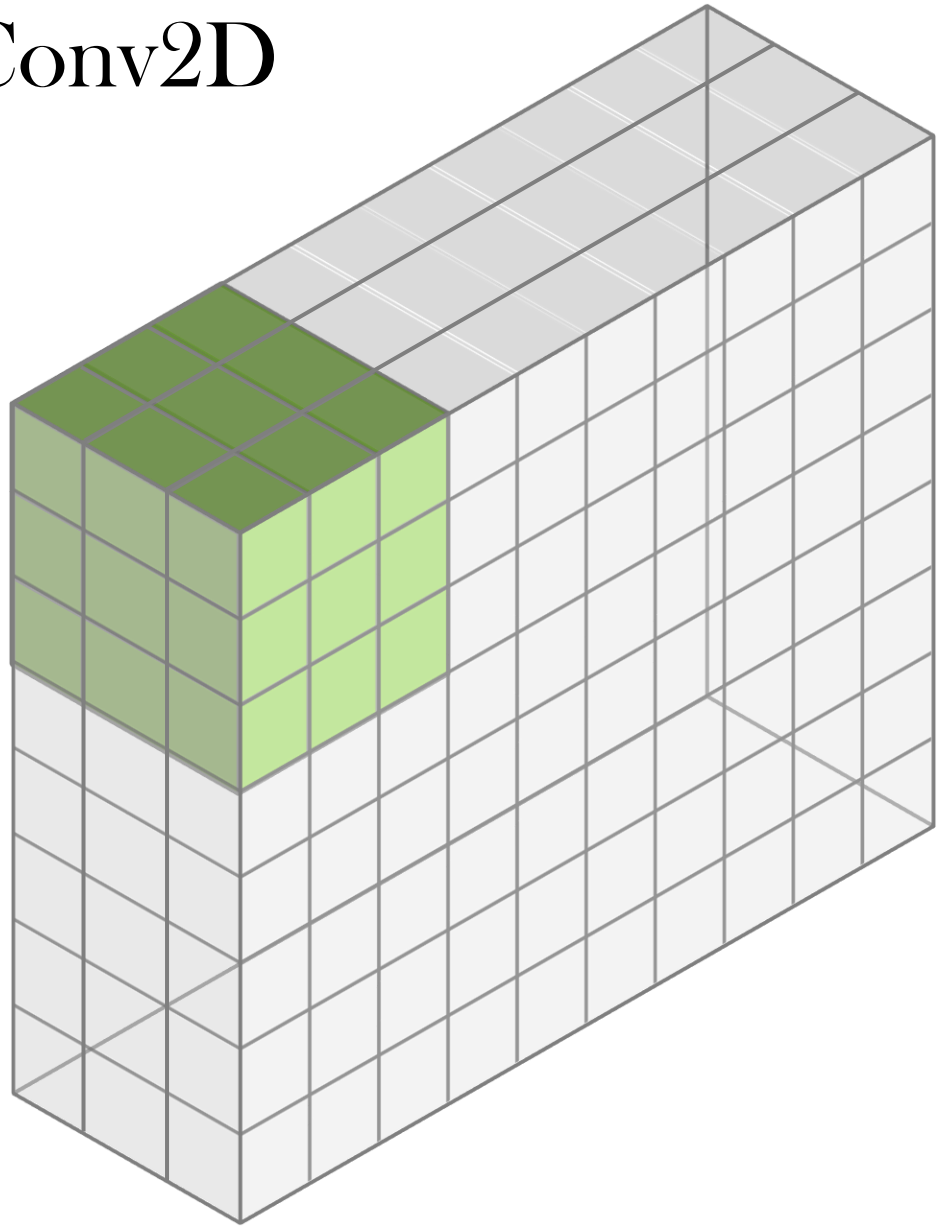
C, h, w

Conv2D

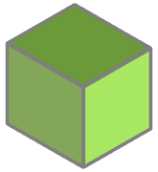


N, C, H, W

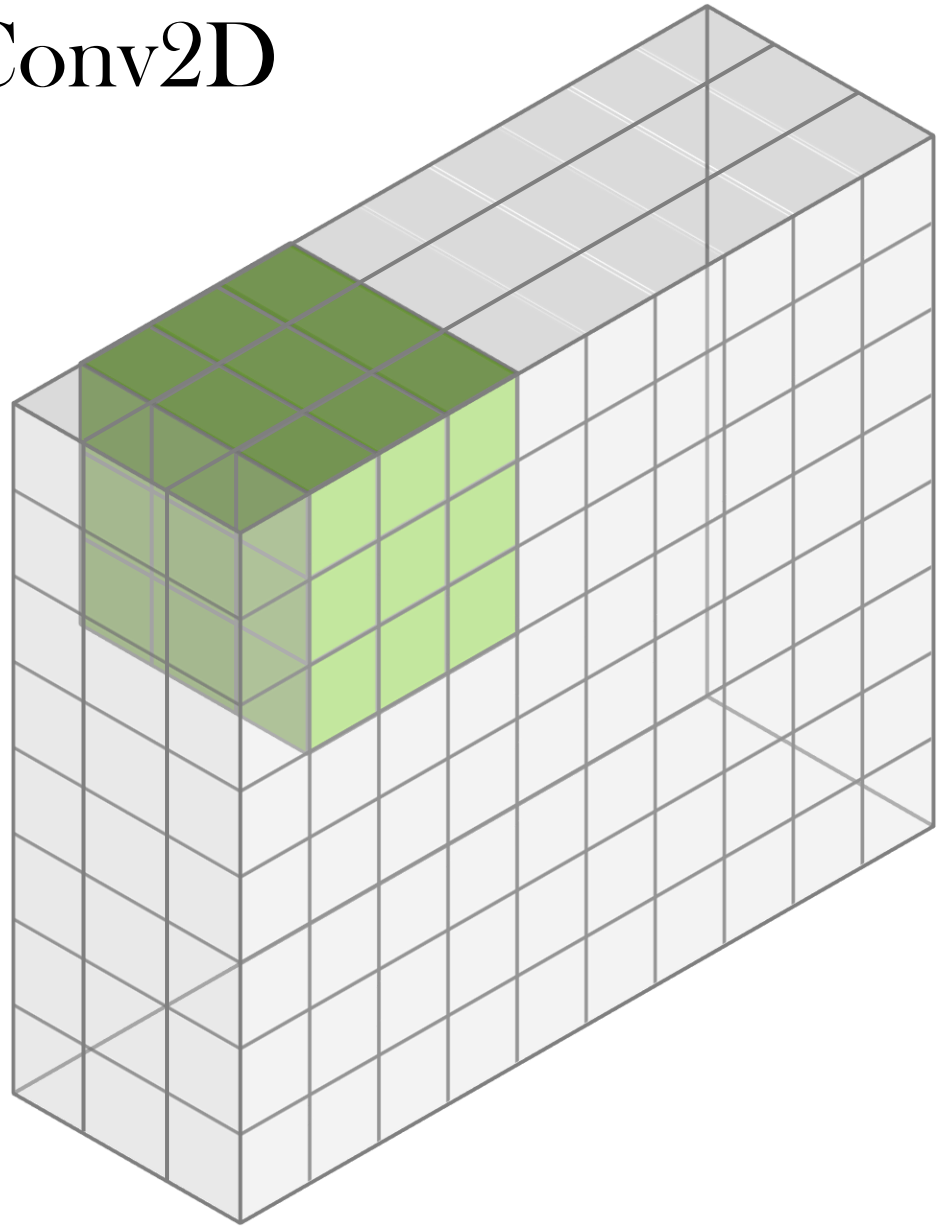
Conv2D

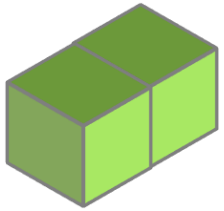


N, C, H, W



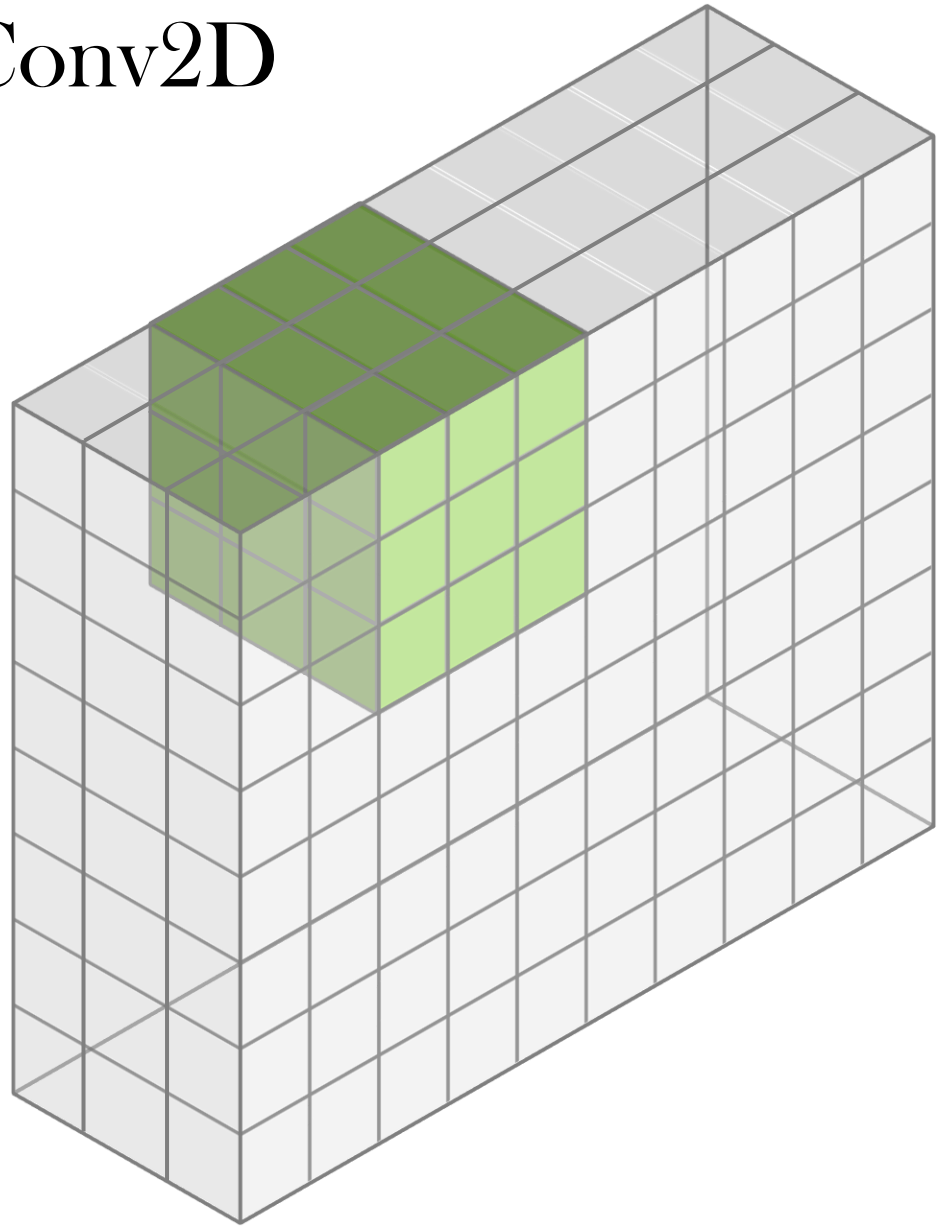
Conv2D



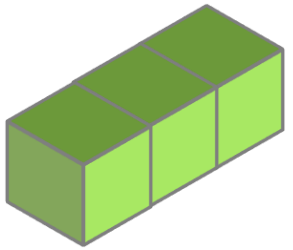


N, C, H, W

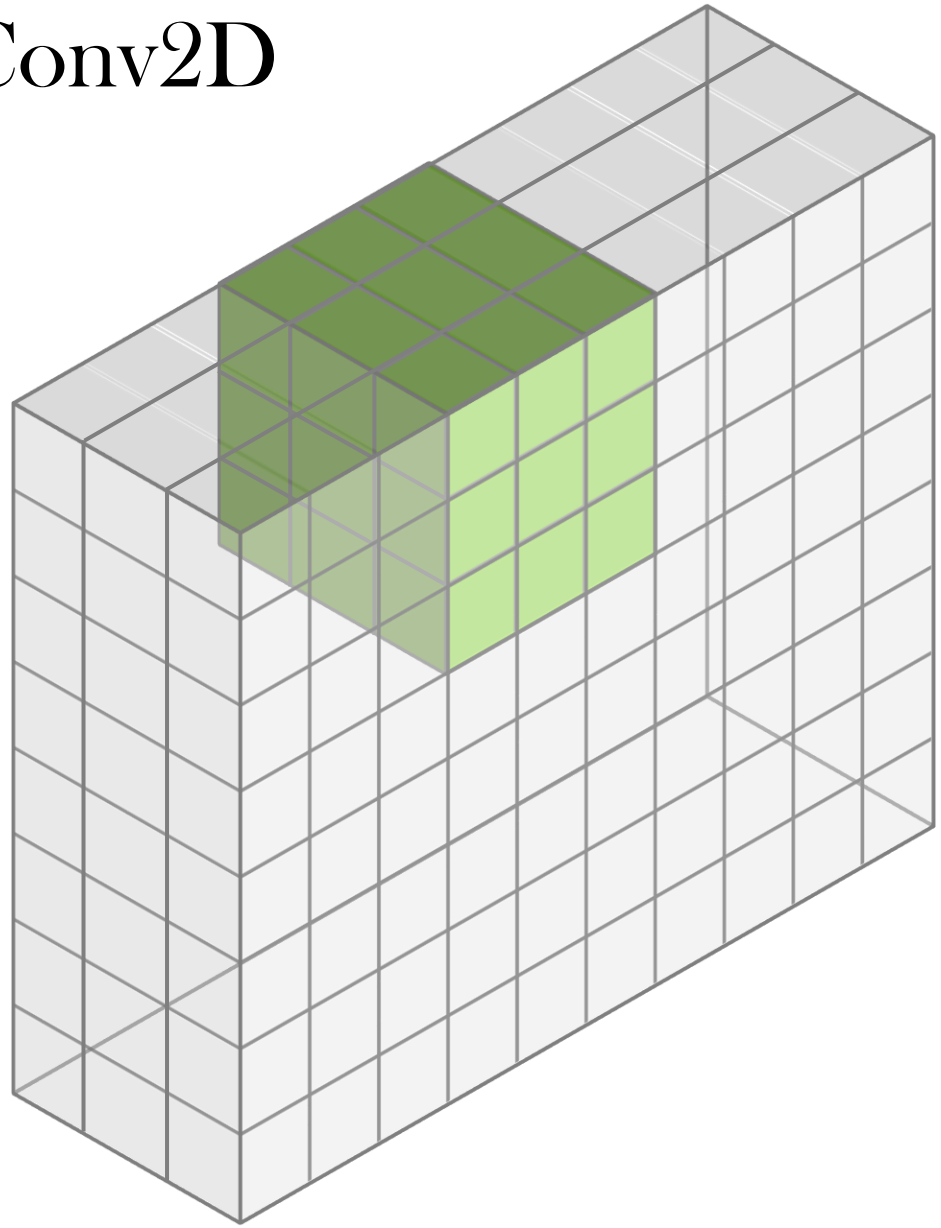
Conv2D

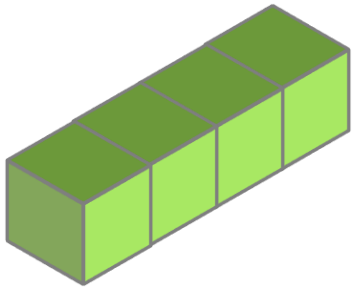


N, C, H, W



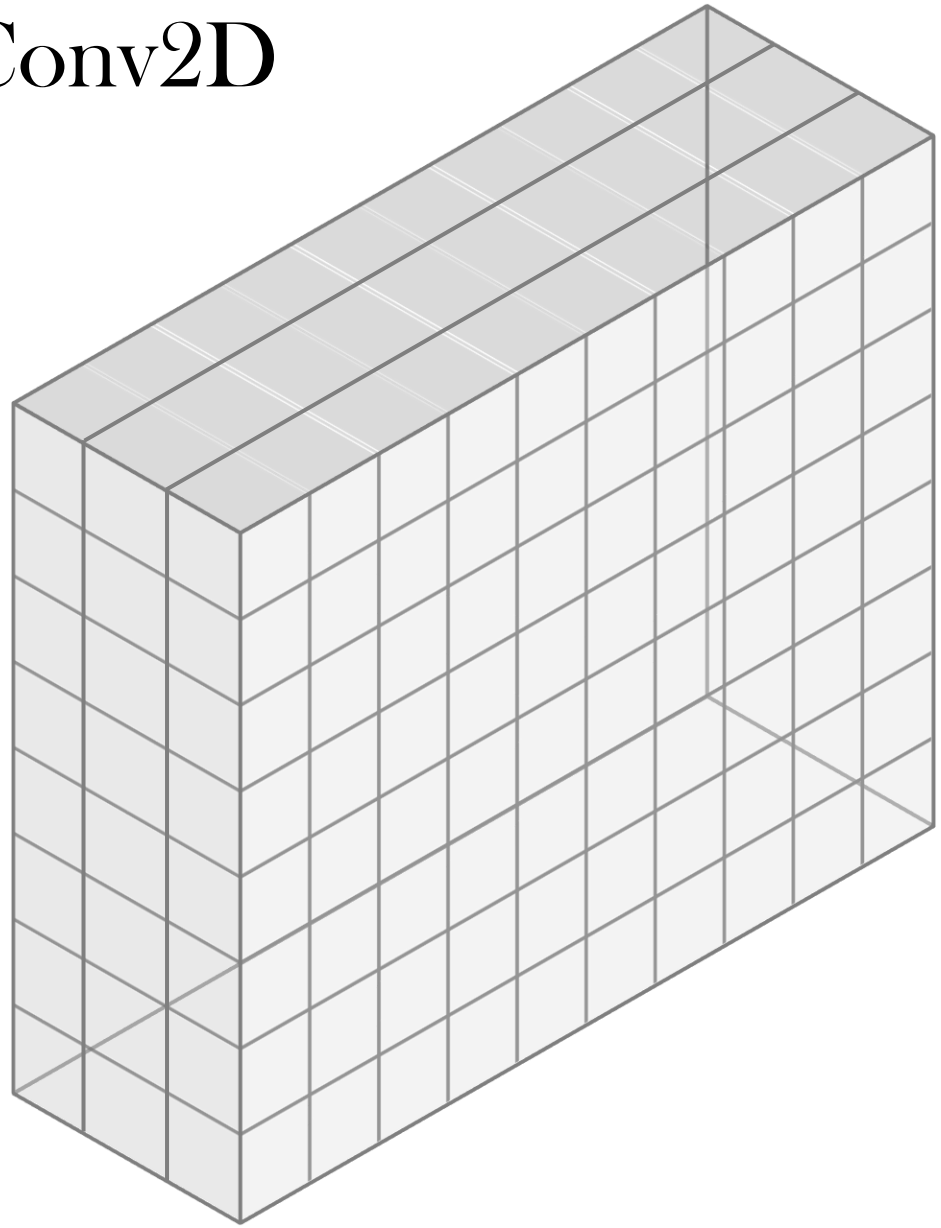
Conv2D



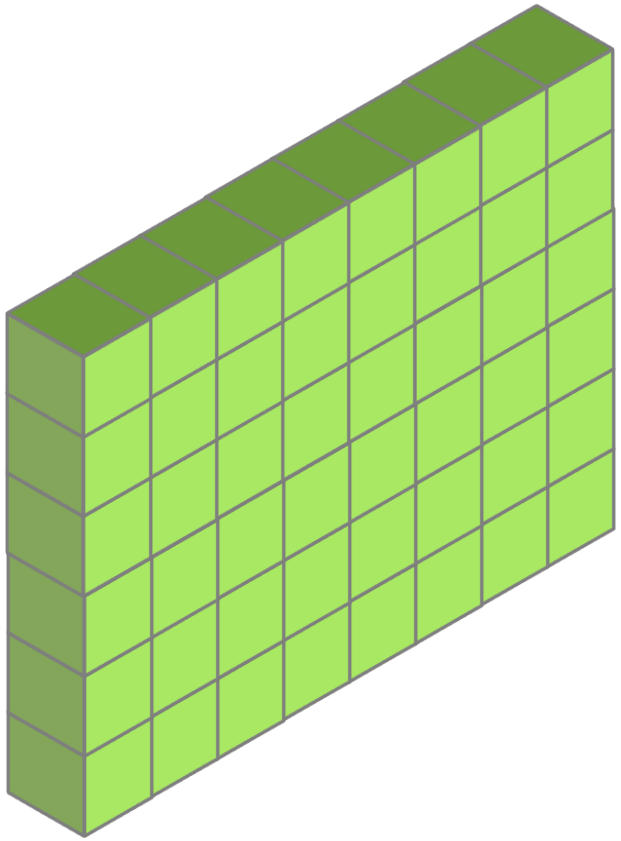


N, C, H, W

Conv2D

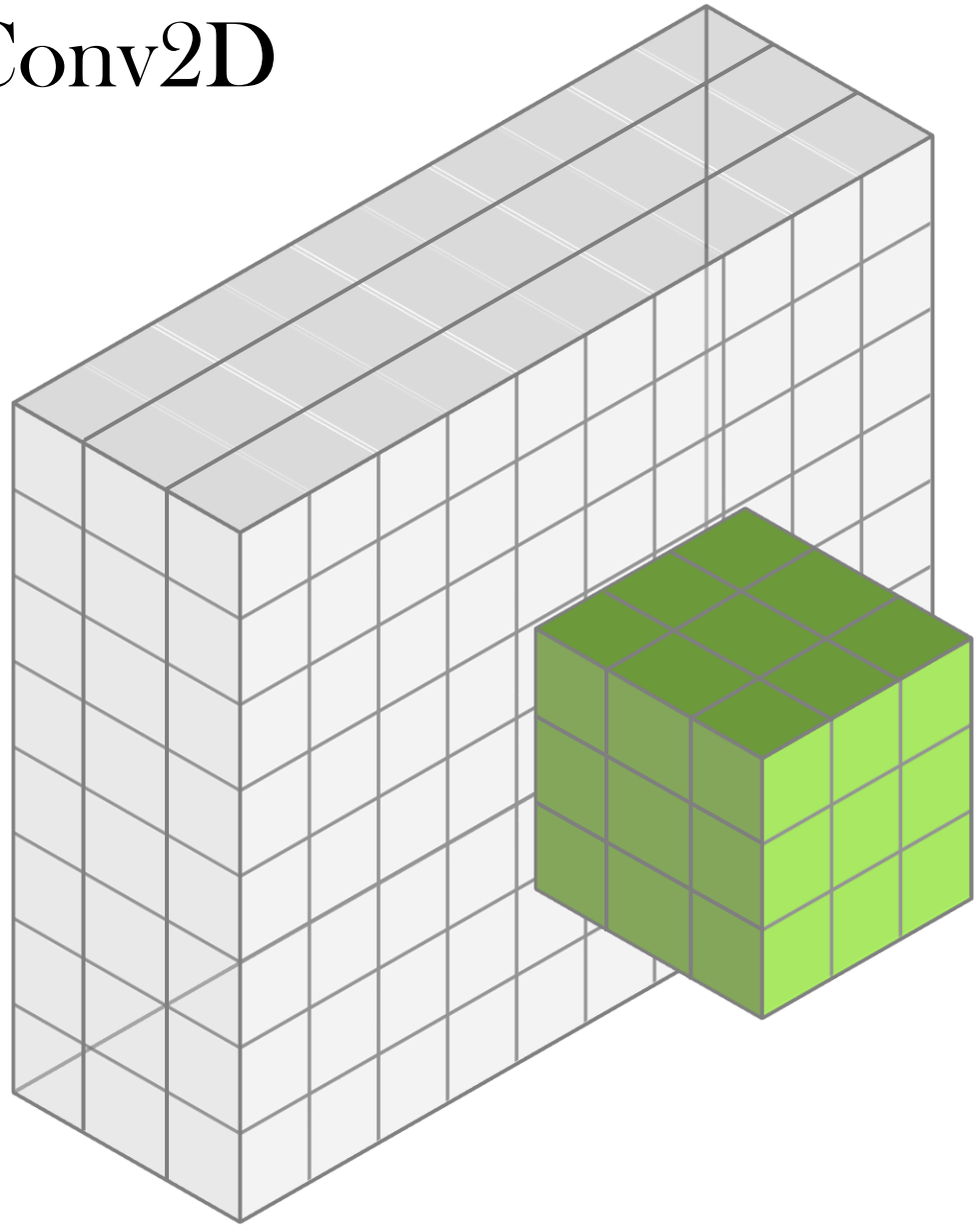


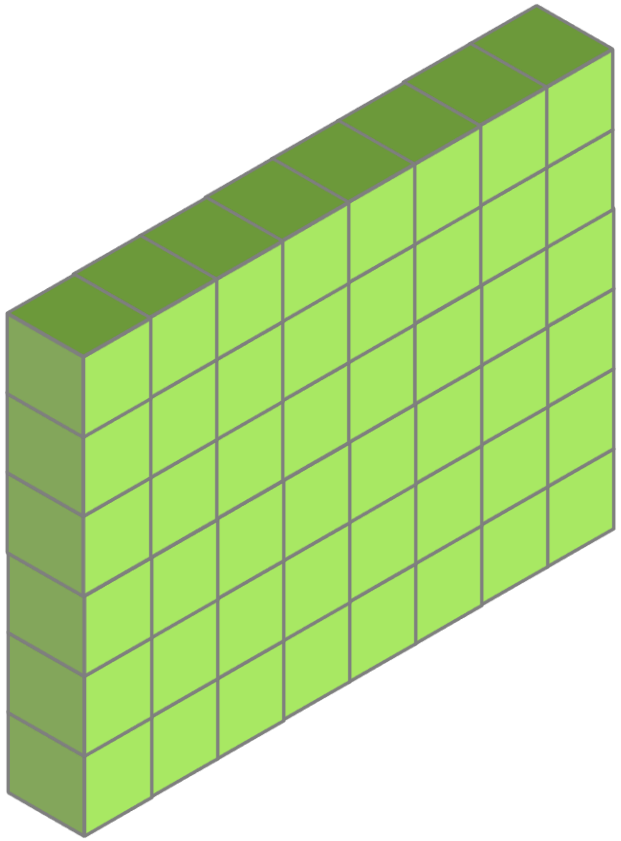
N, C, H, W



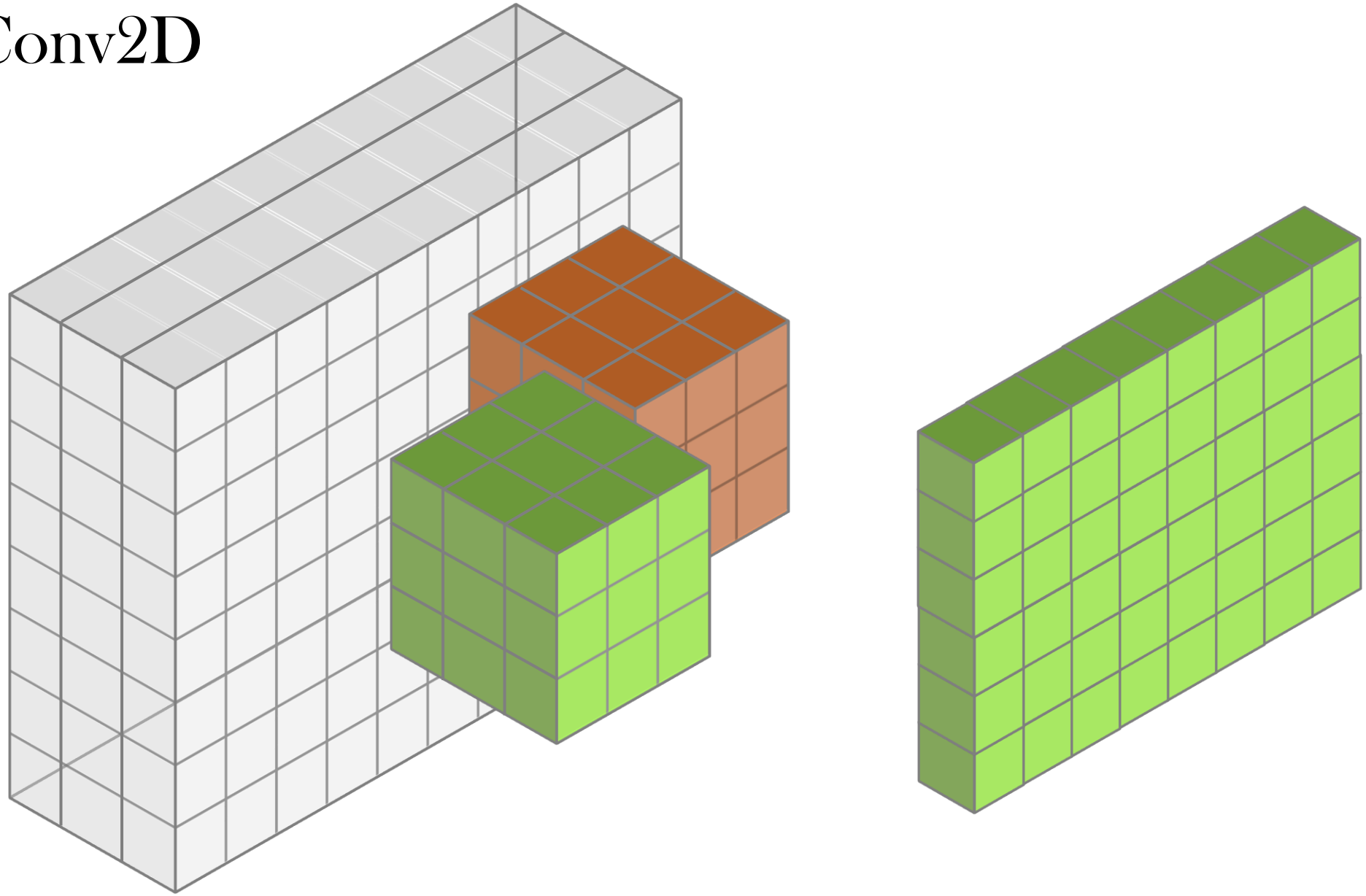
$[N, 1, \widetilde{H}, \widetilde{W}]$

Conv2D

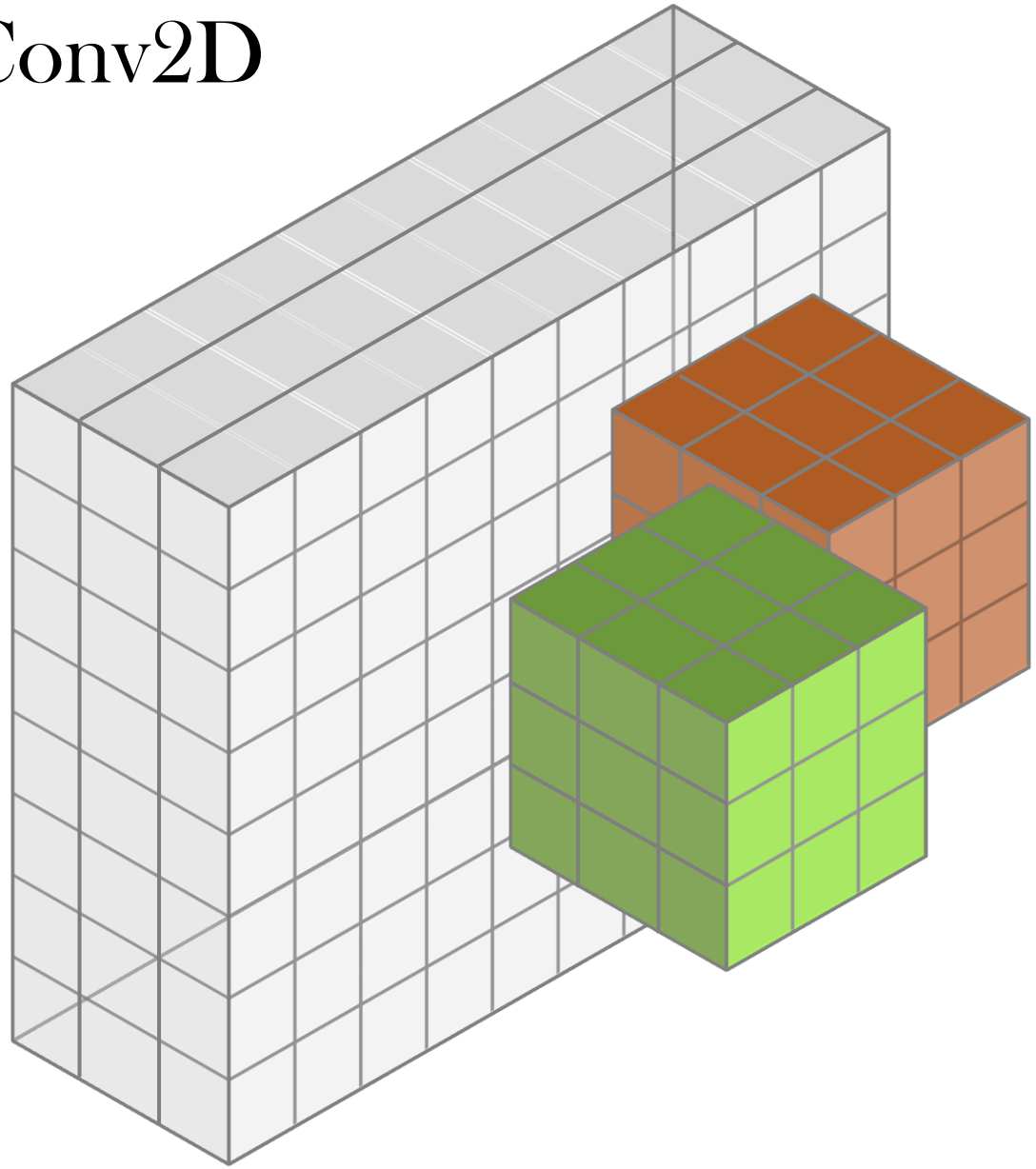


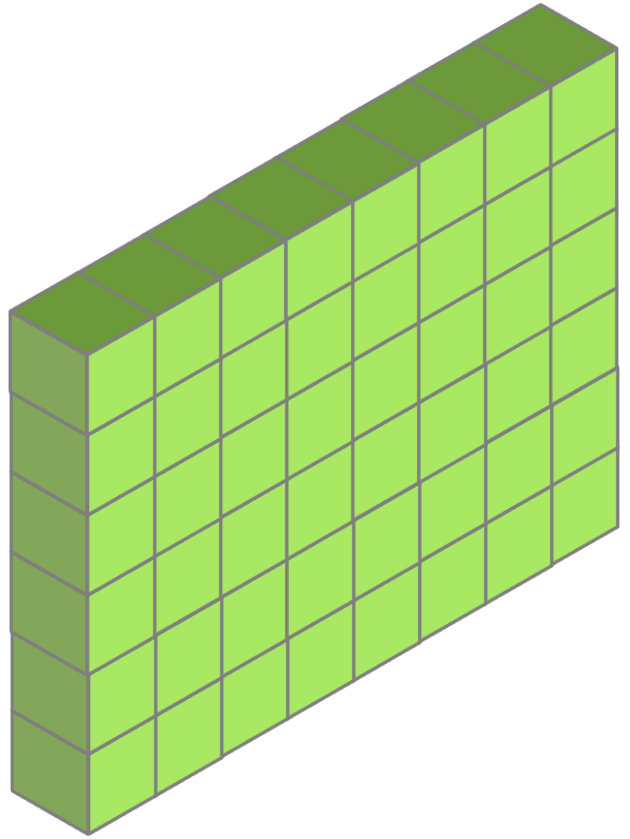


Conv2D

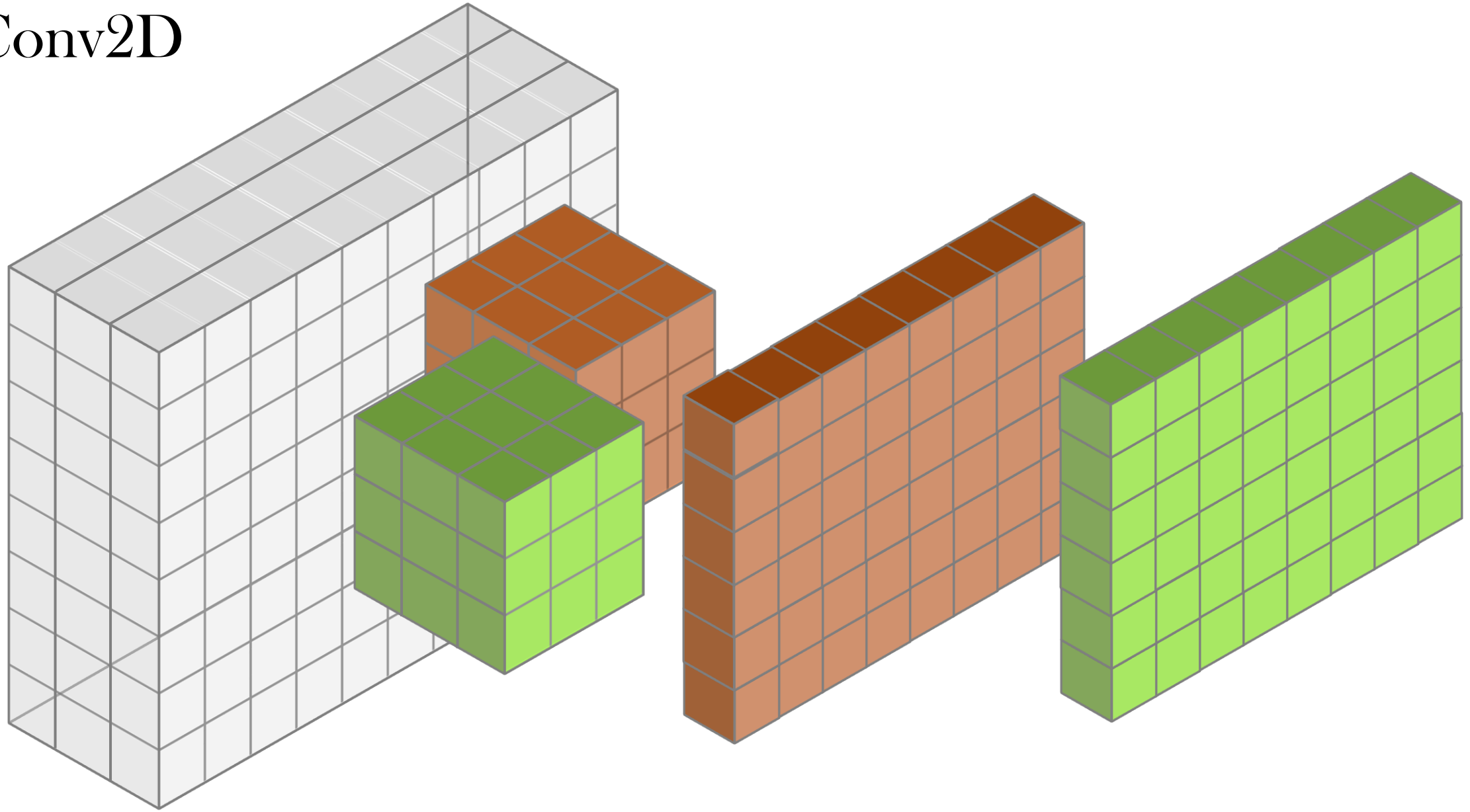


Conv2D

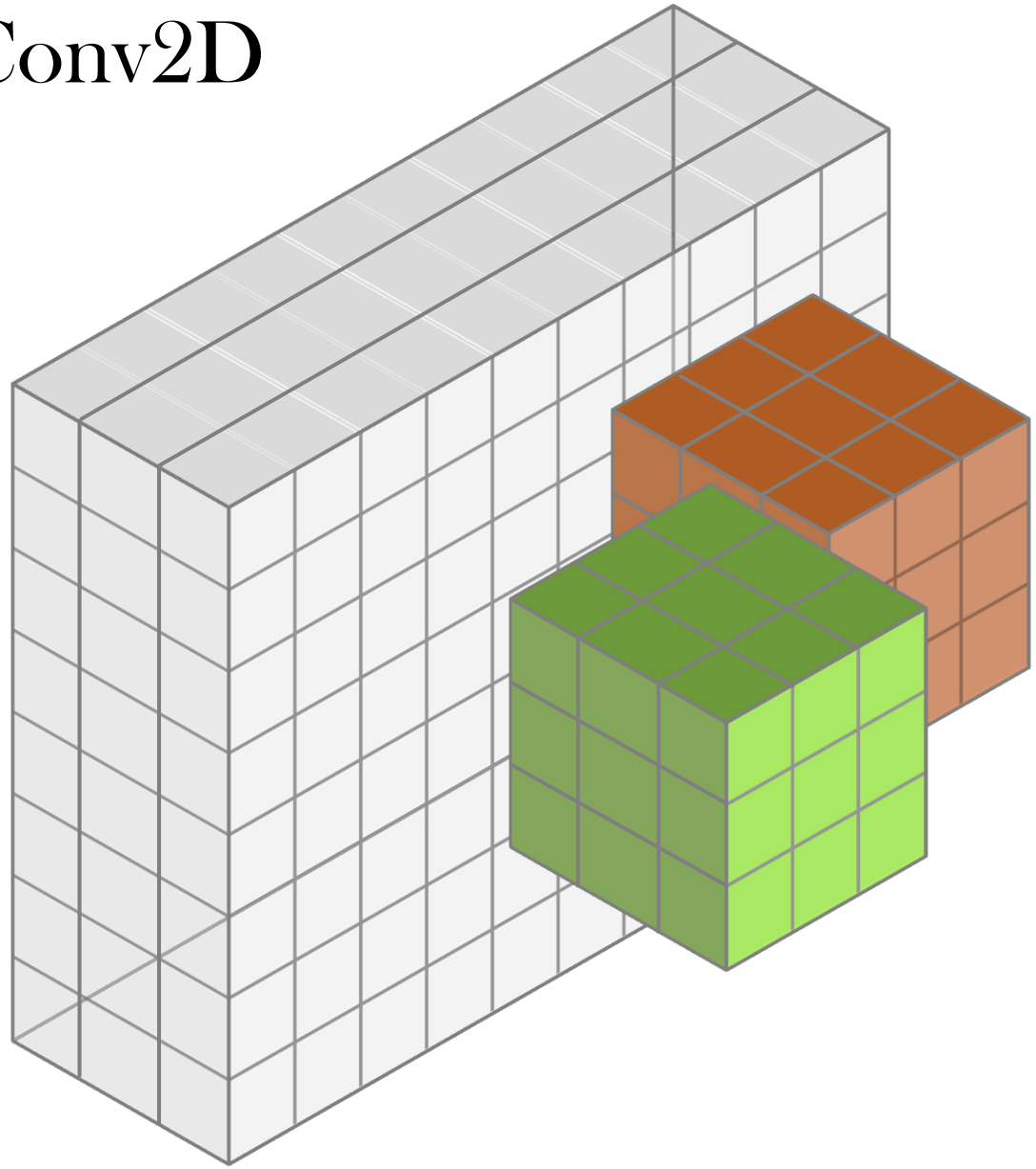


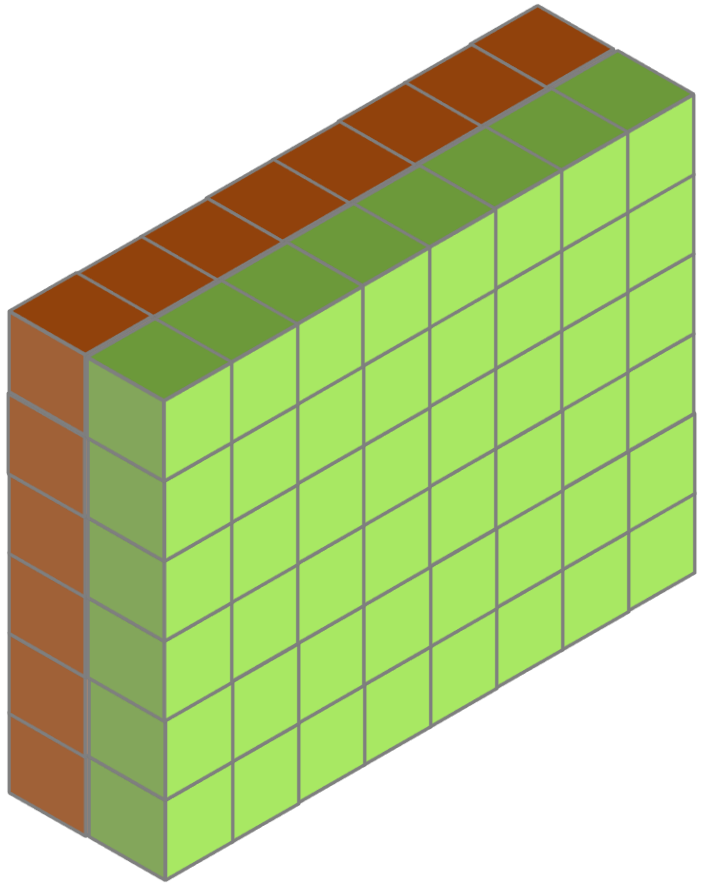


Conv2D

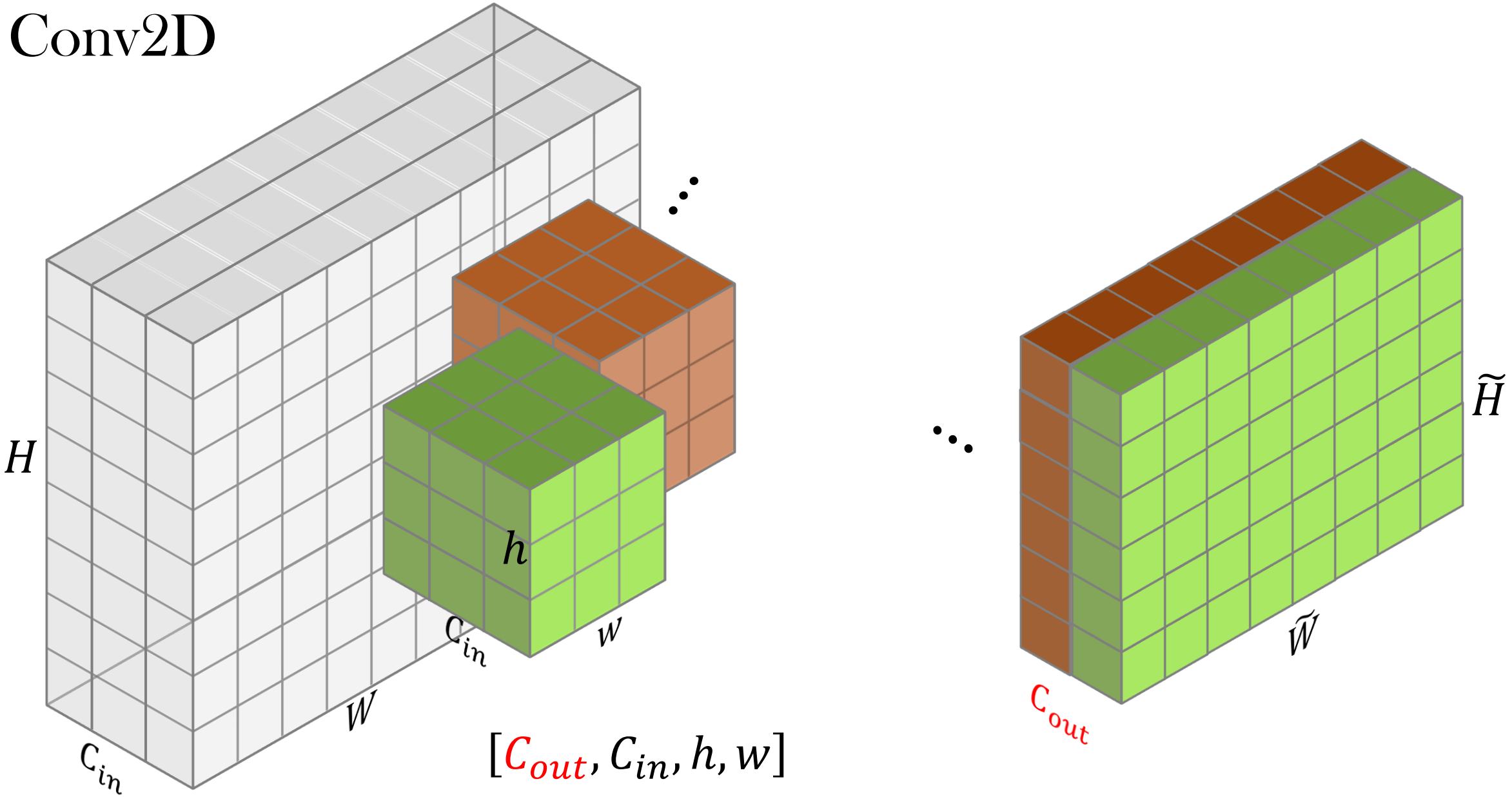


Conv2D





Conv2D



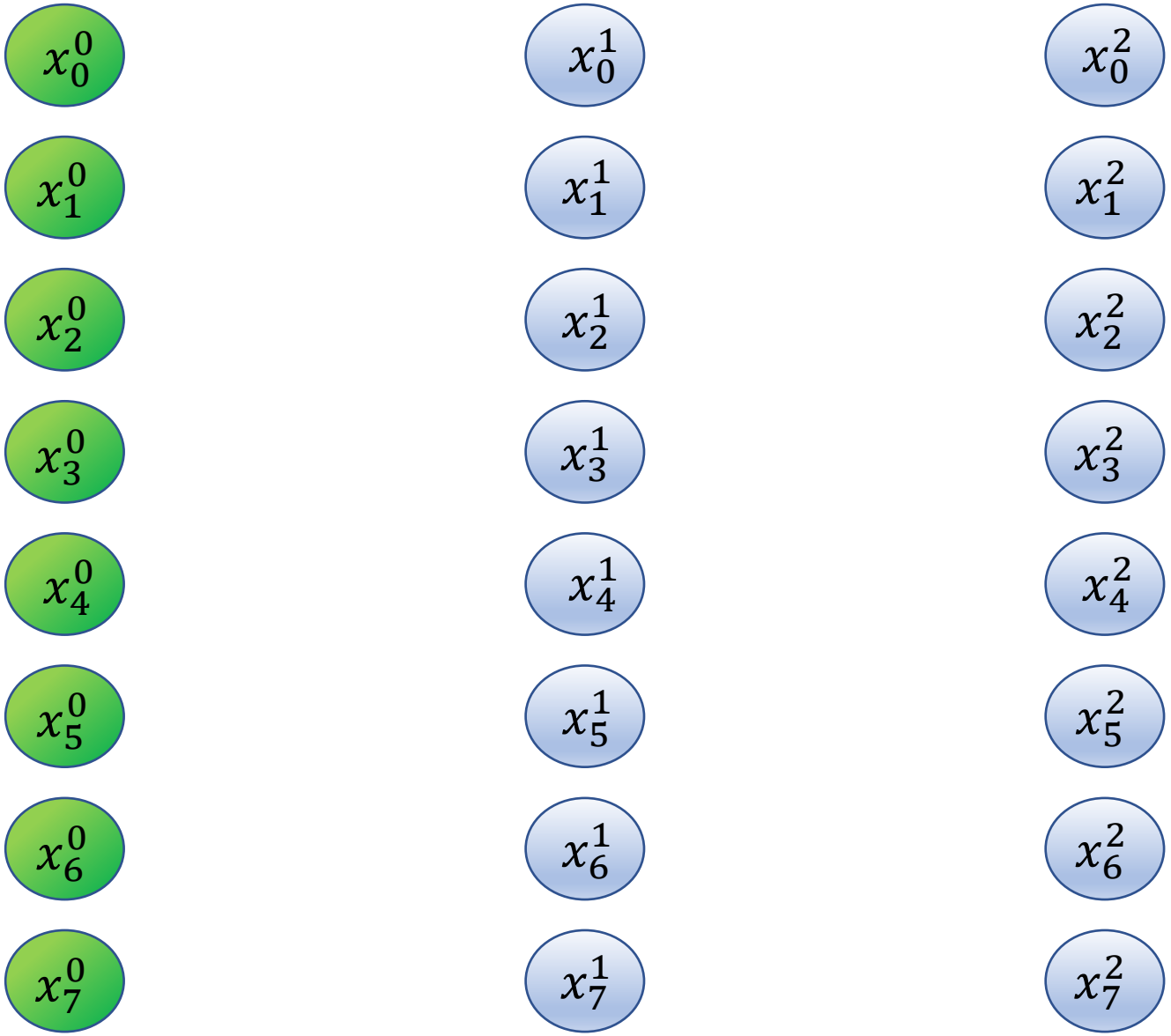
$[C_{out}, C_{in}, h, w]$

$[N, C_{in}, H, W]$

$[N, C_{out}, \widetilde{H}, \widetilde{W}]$

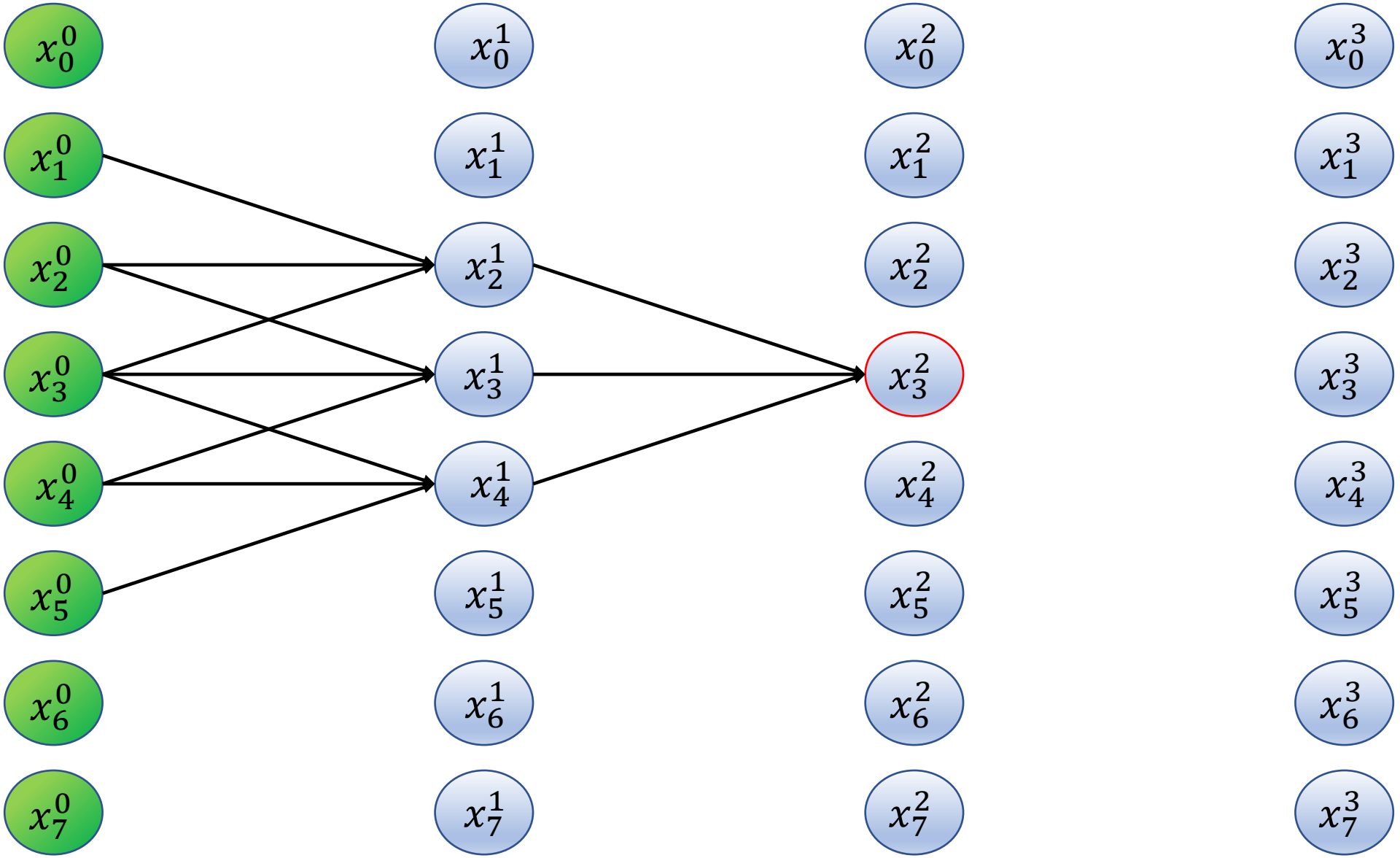
Receptive Field

$[Filter\ size: 3]$



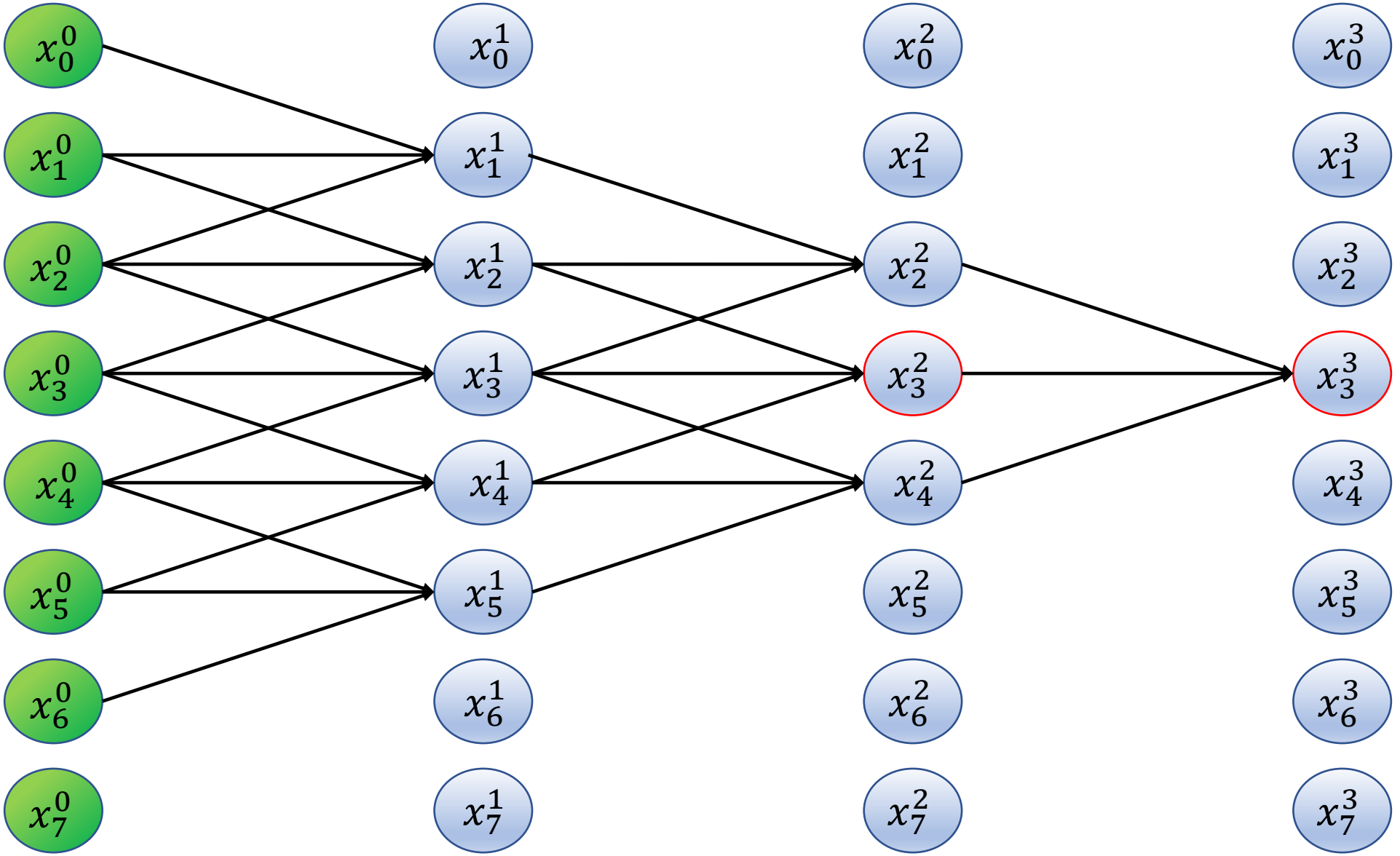
Receptive Field

$[Filter\ size: 3]$



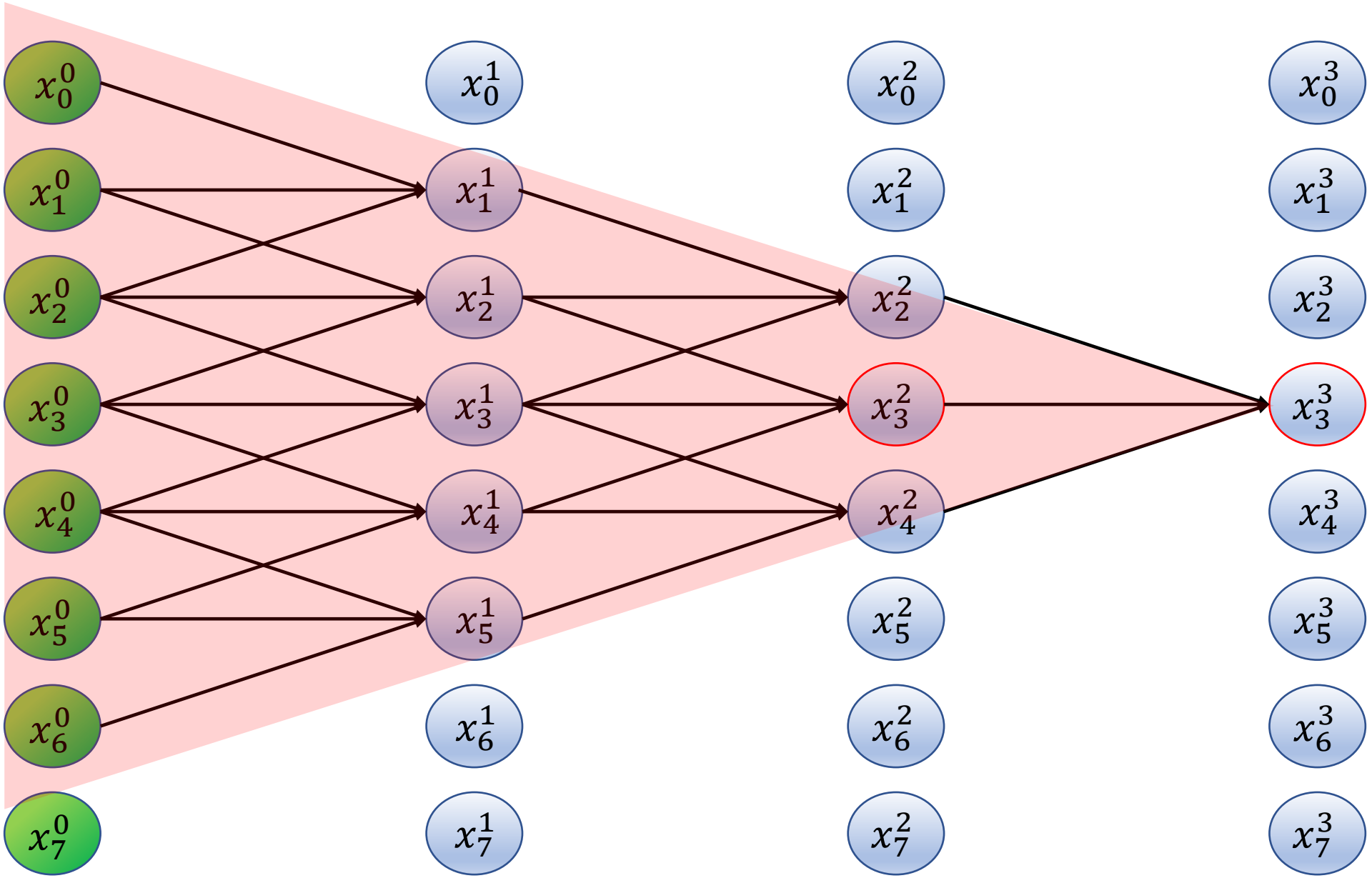
Receptive Field

$[Filter\ size: 3]$

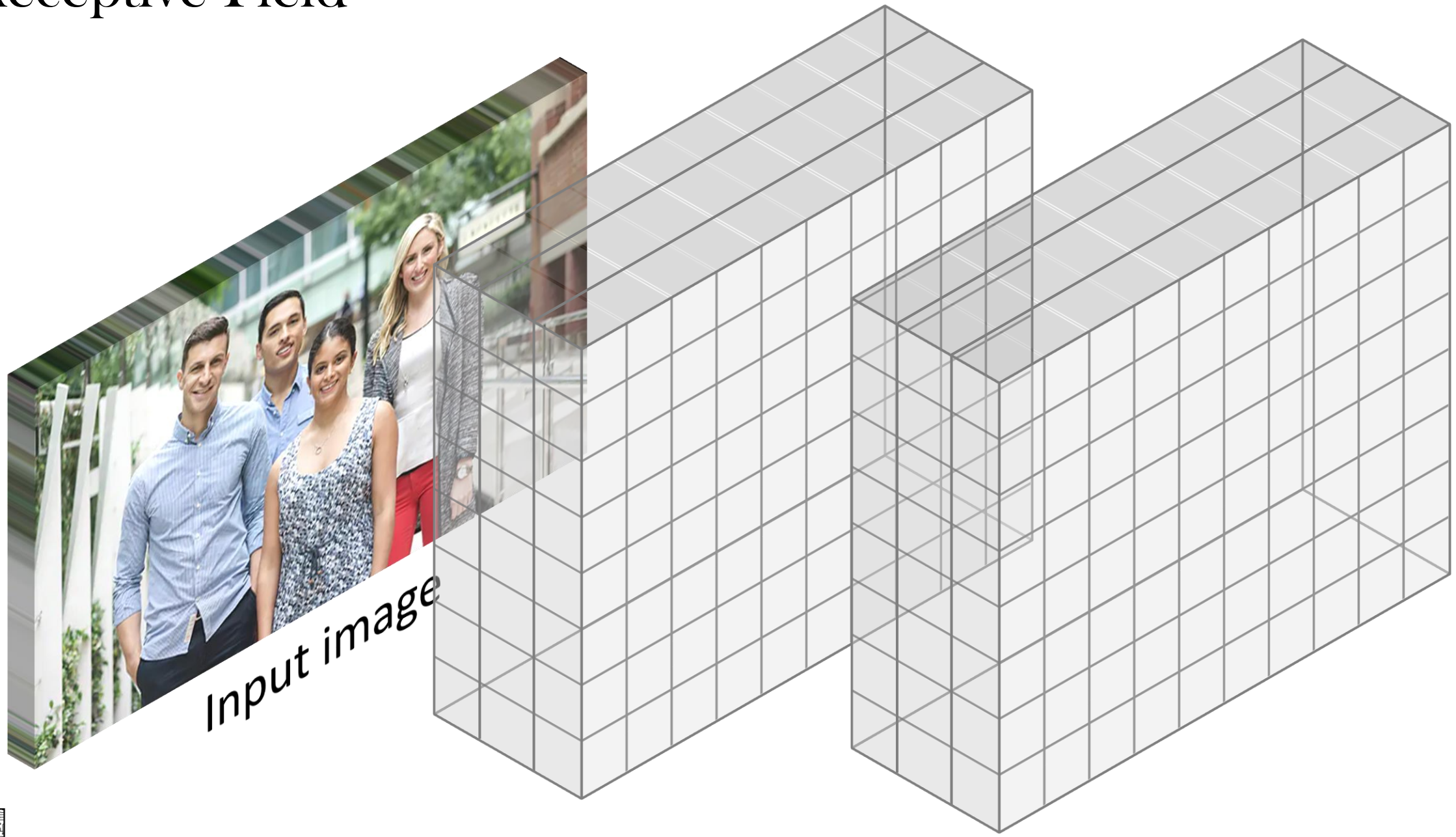


Receptive Field

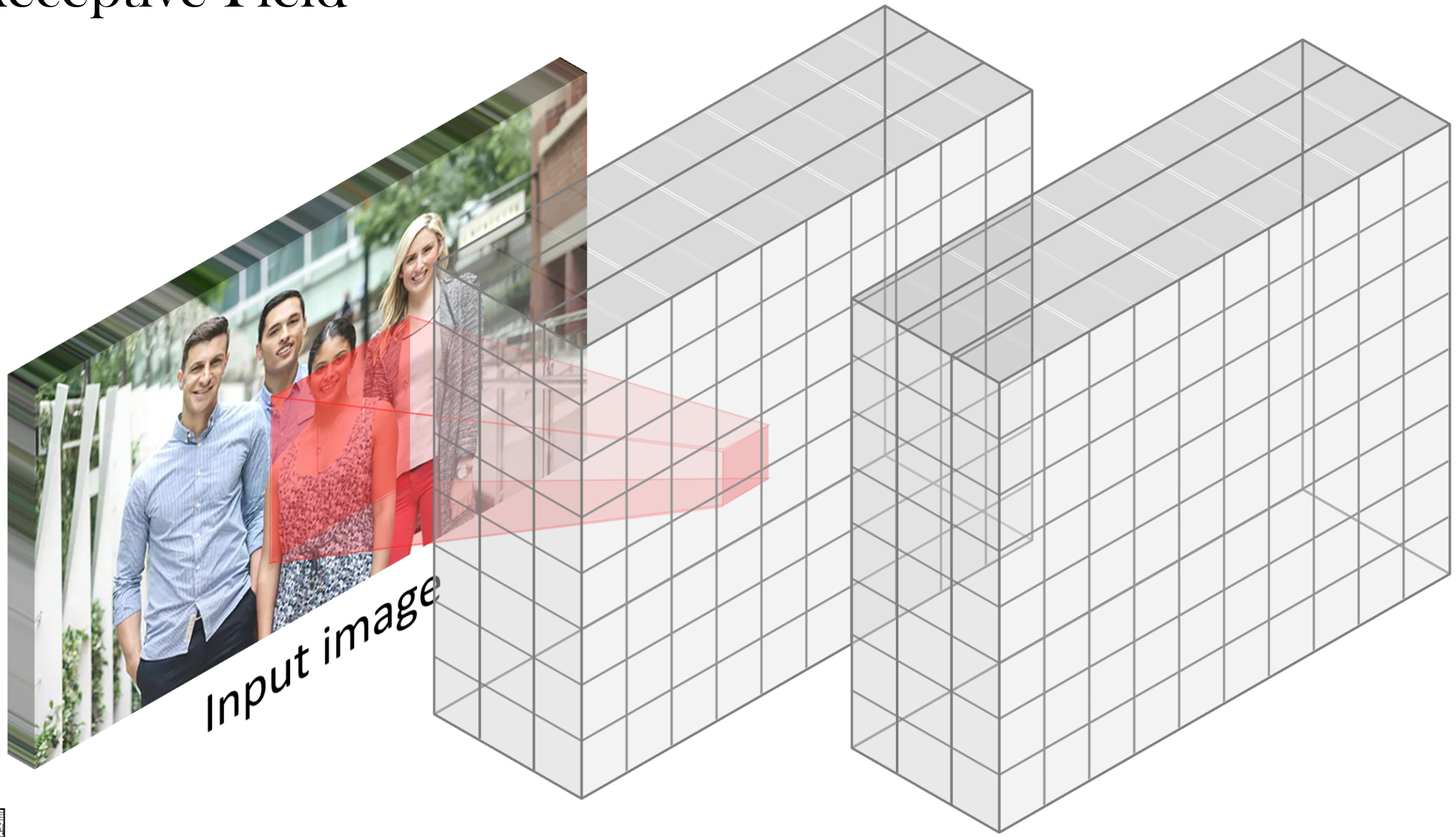
$[Filter\ size: 3]$



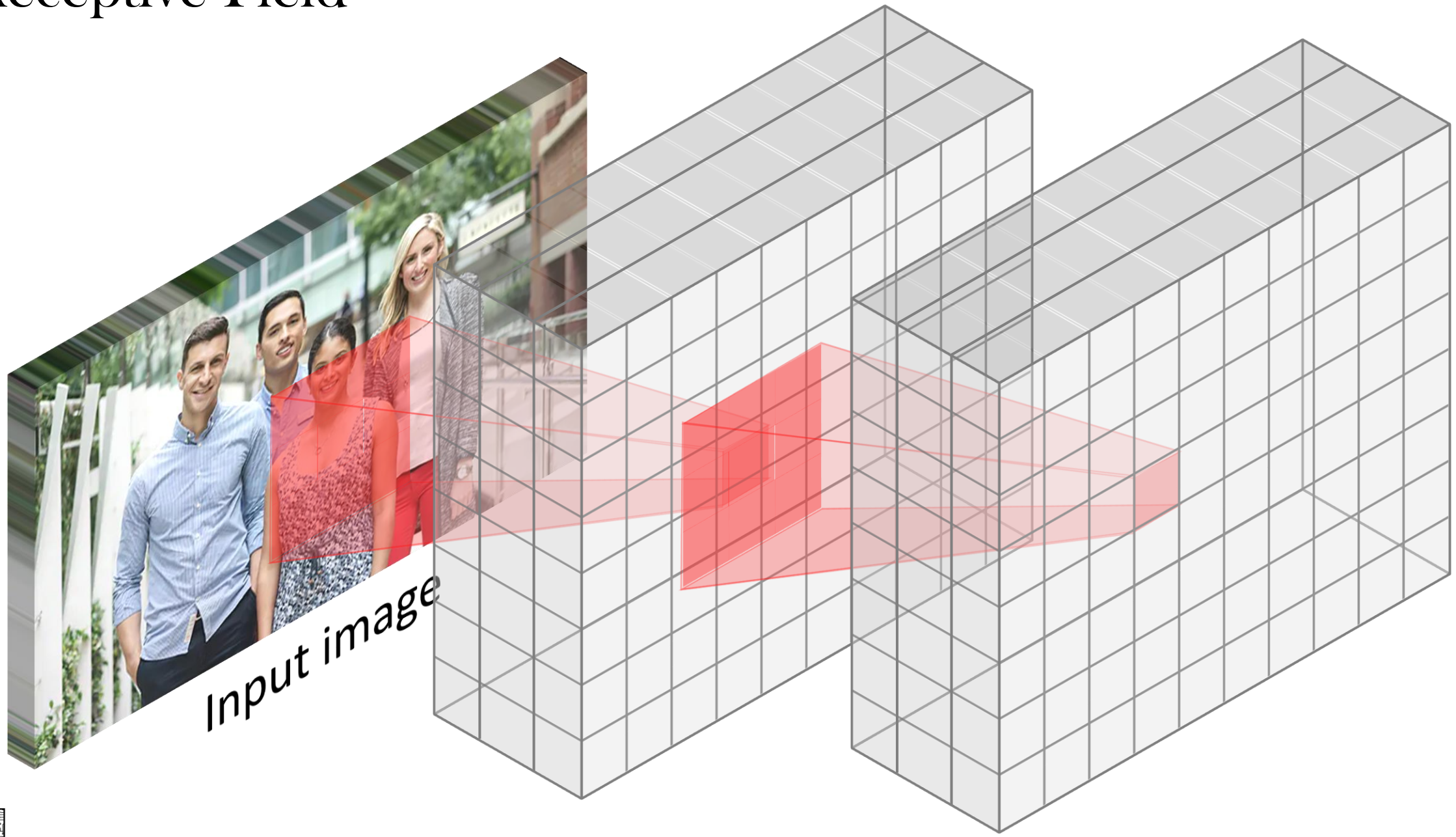
Receptive Field



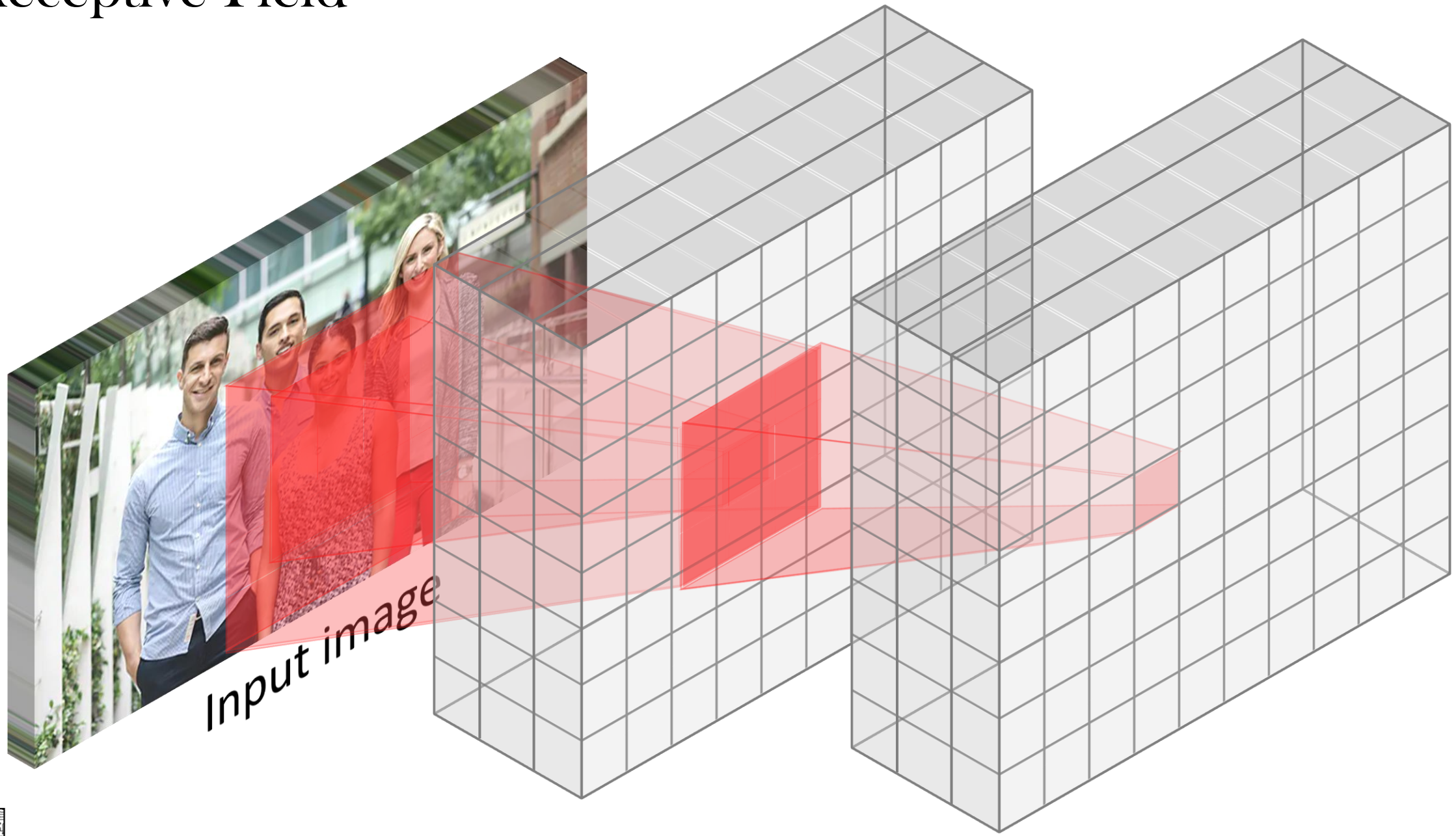
Receptive Field



Receptive Field



Receptive Field

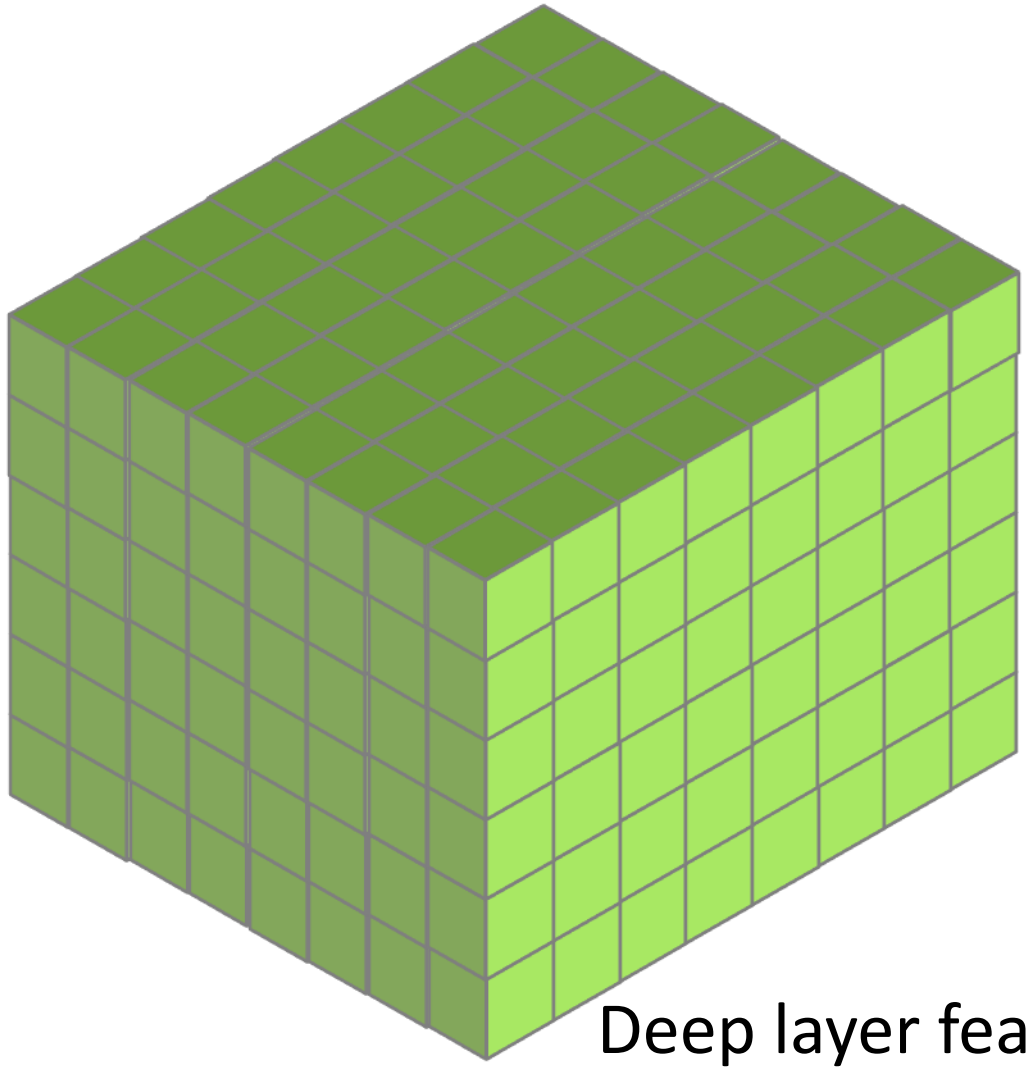


Convs rock!

1. ✓ Maintain 2D structure logic
2. ✓ Shift invariant (actually, equivariant)
3. ✓ Consider local correlations first
4. ✓ Hierarchically growing field of view
5. ✓ Hierarchically progressing complexity
6. ✓ Reasonable amount of params

Two important intuitions about feature maps

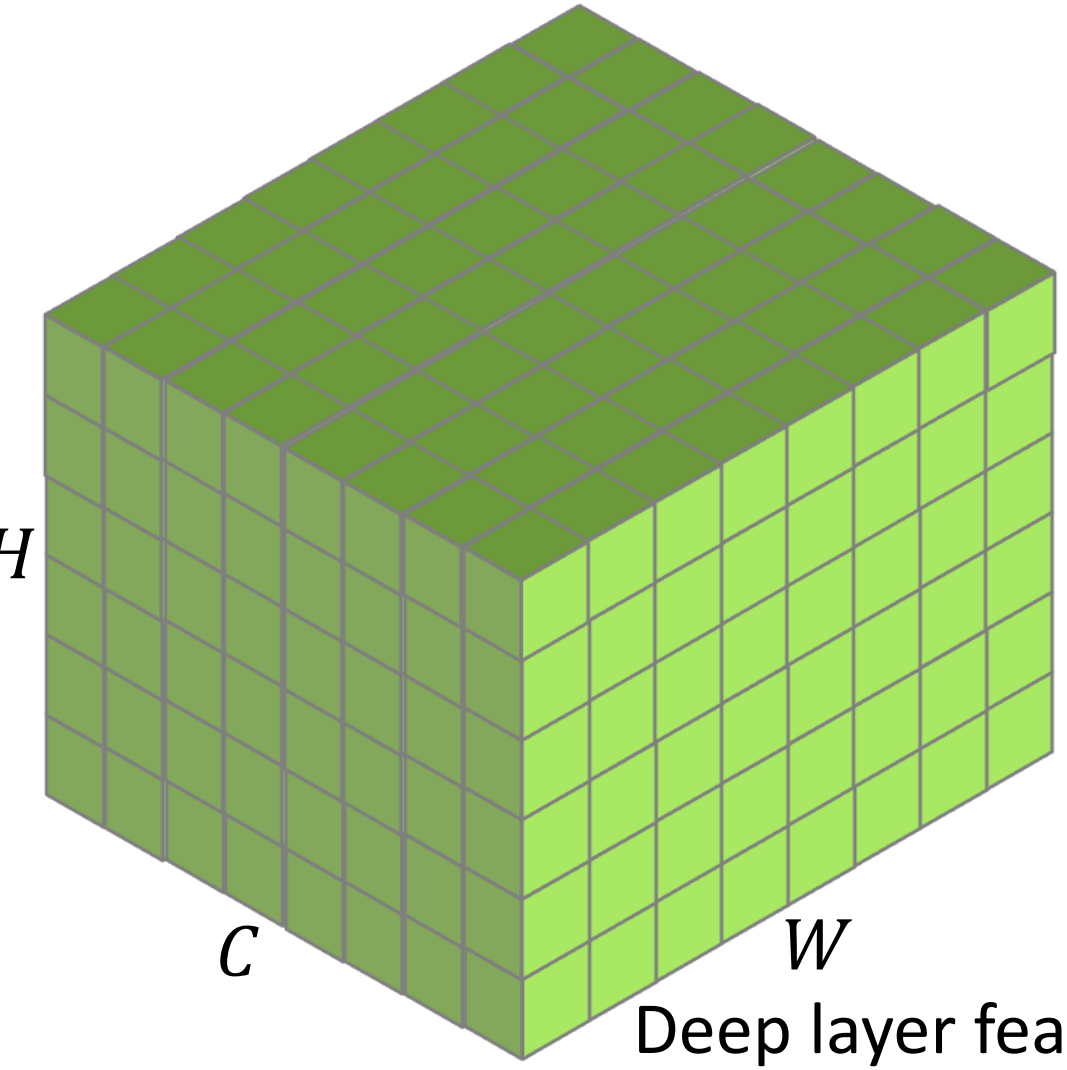




Deep layer feature-map

Two important intuitions about feature maps



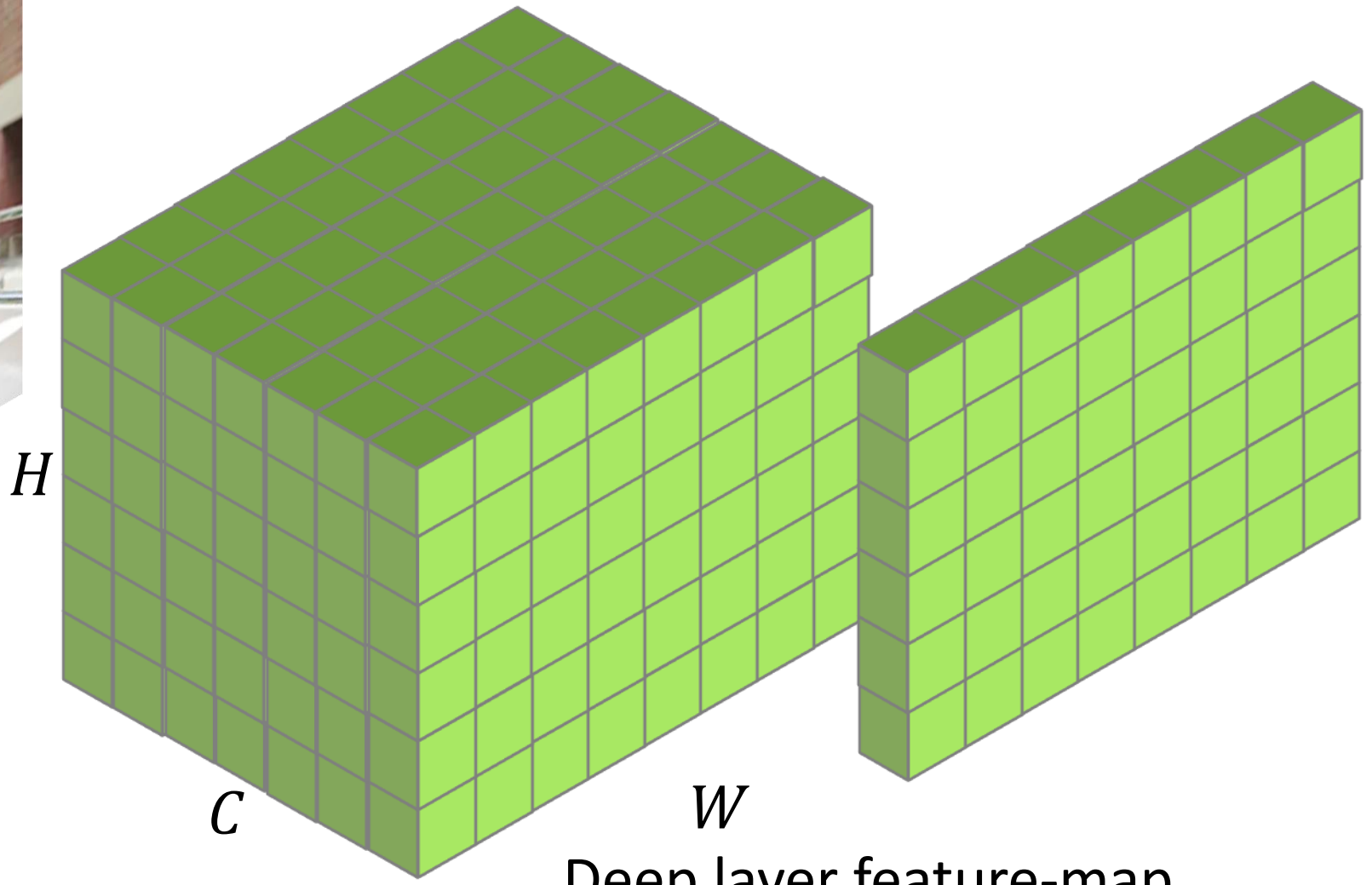


Deep layer feature-map

Two important intuitions about feature maps



Input image

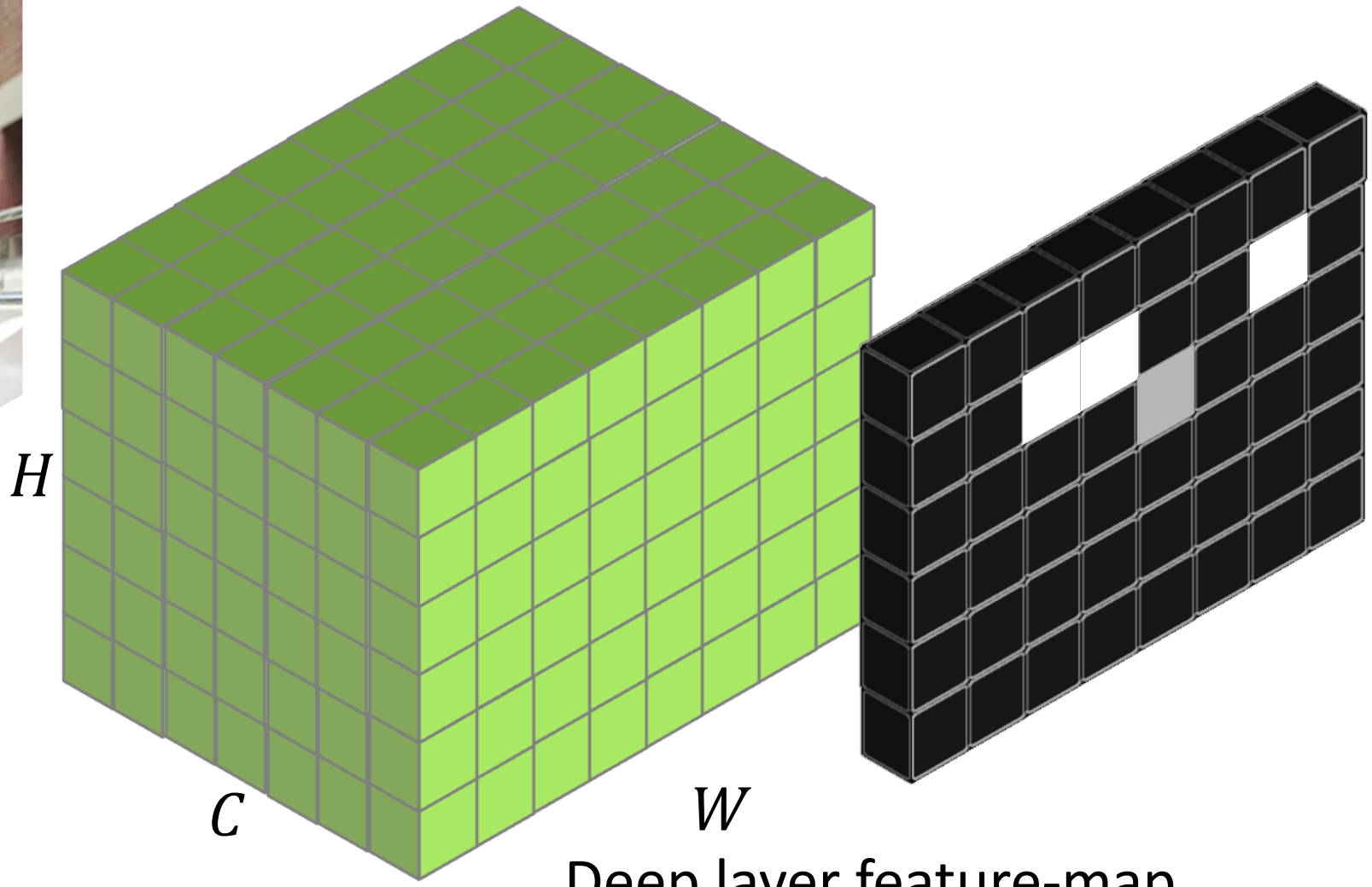


Deep layer feature-map

Two important intuitions about feature maps

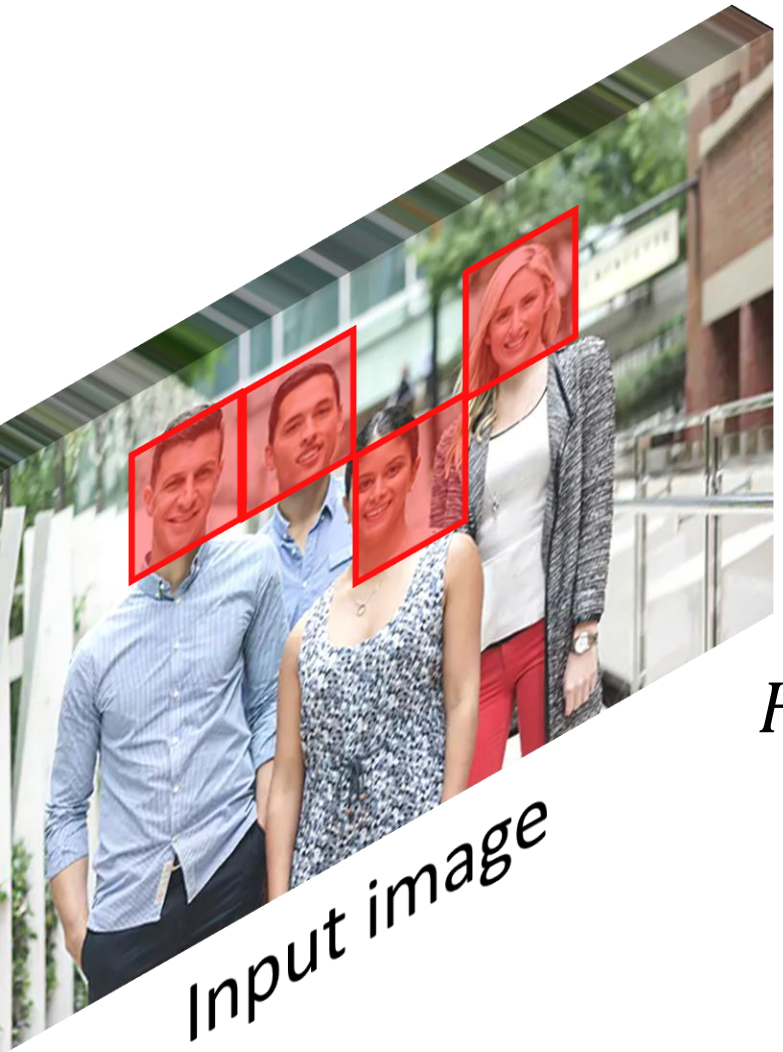


Input image

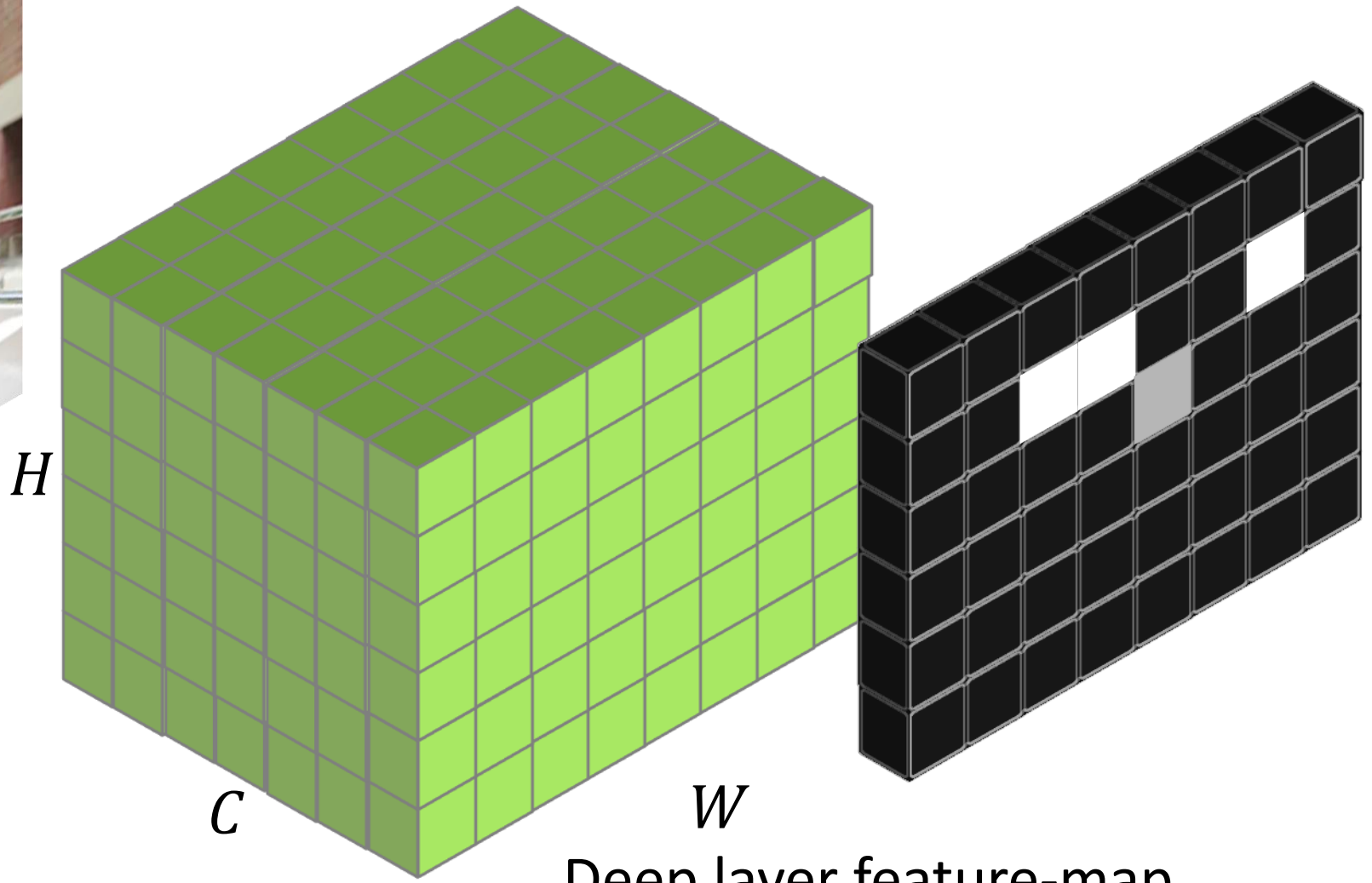


Deep layer feature-map

Two important intuitions about feature maps

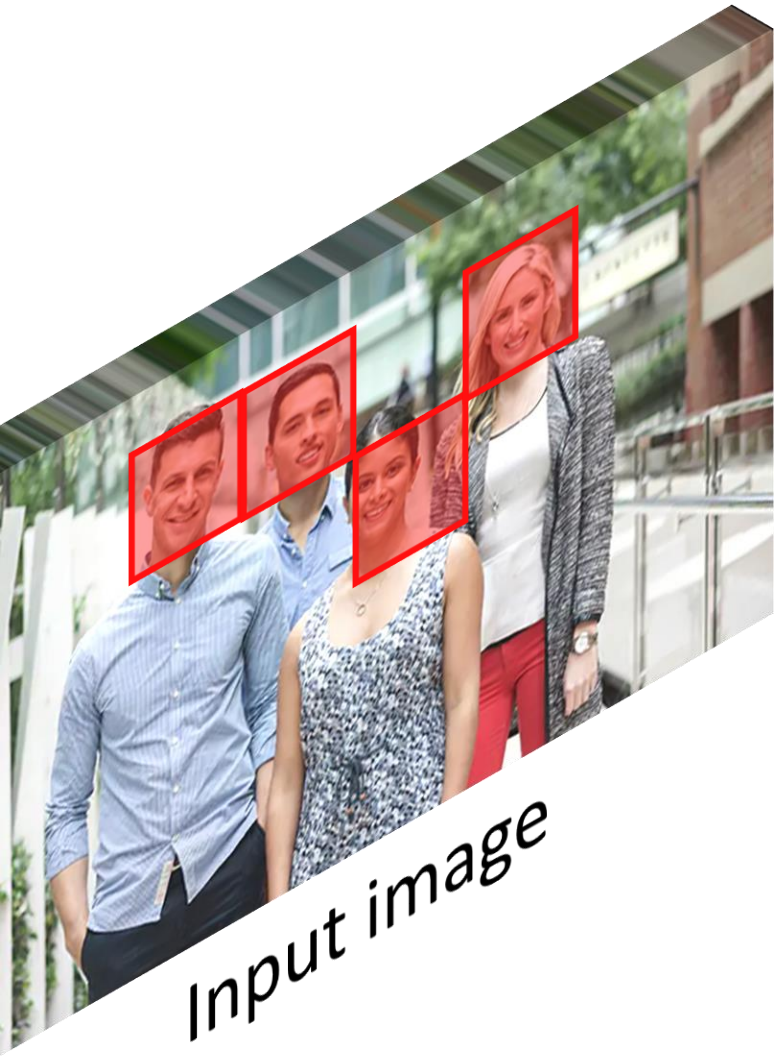


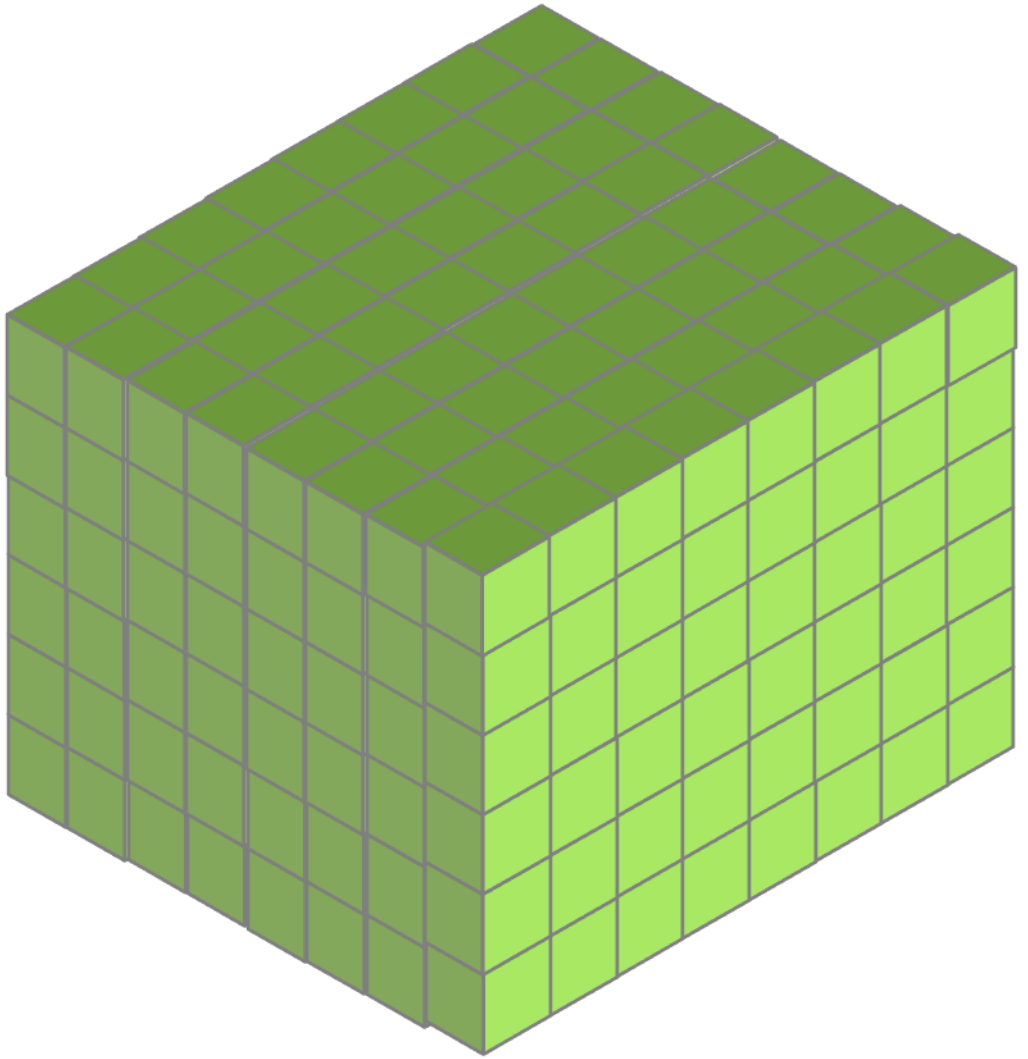
Input image



Deep layer feature-map

Two important intuitions about feature maps

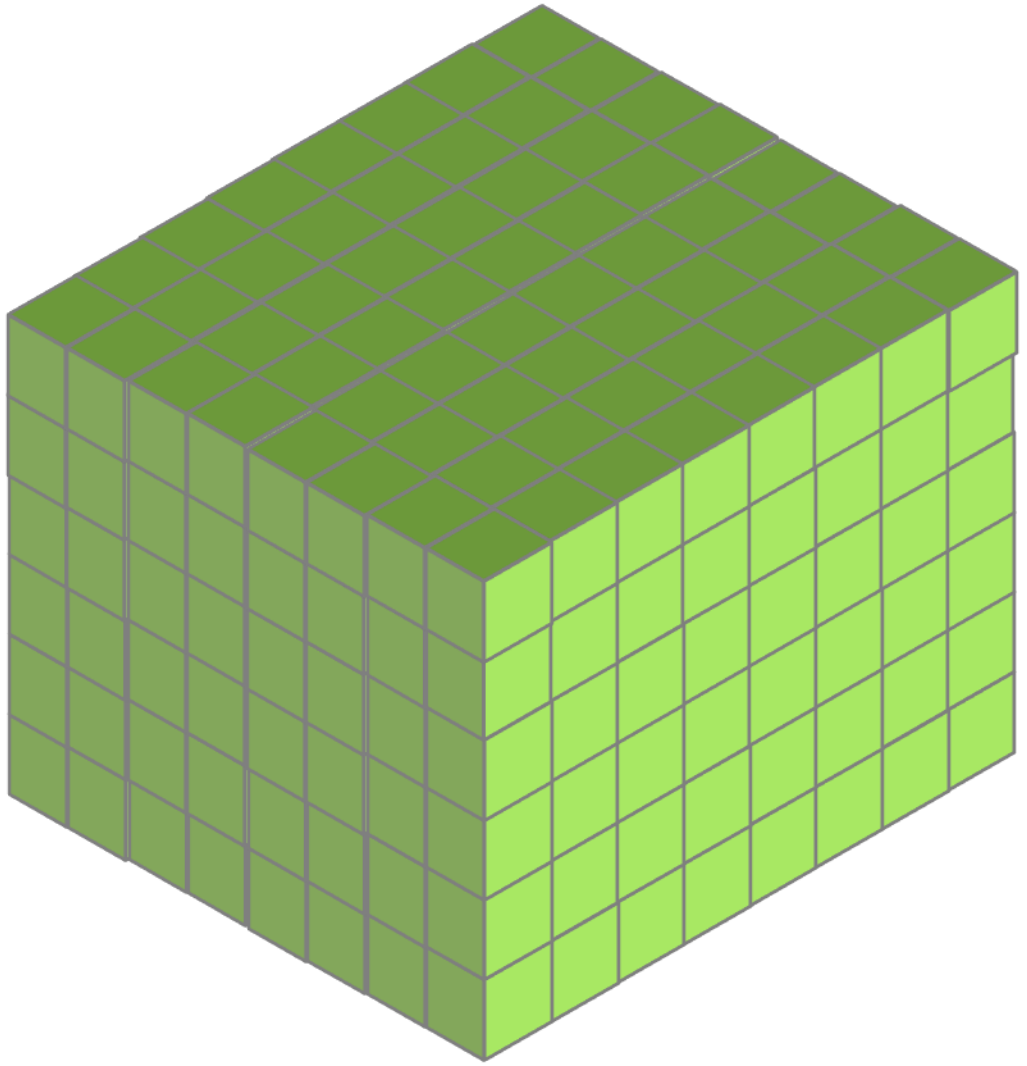




Deep layer feature-map

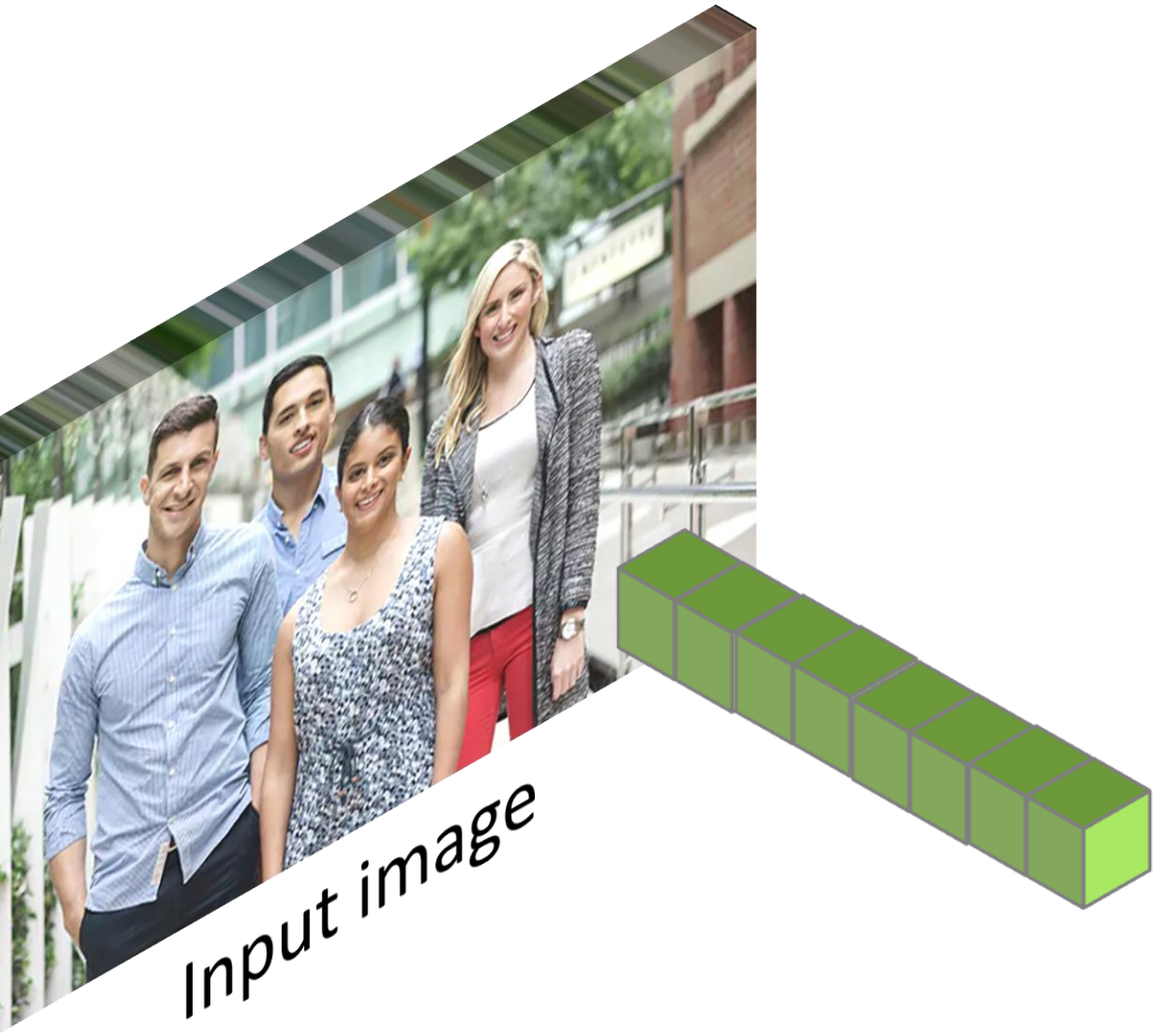
Two important intuitions about feature maps

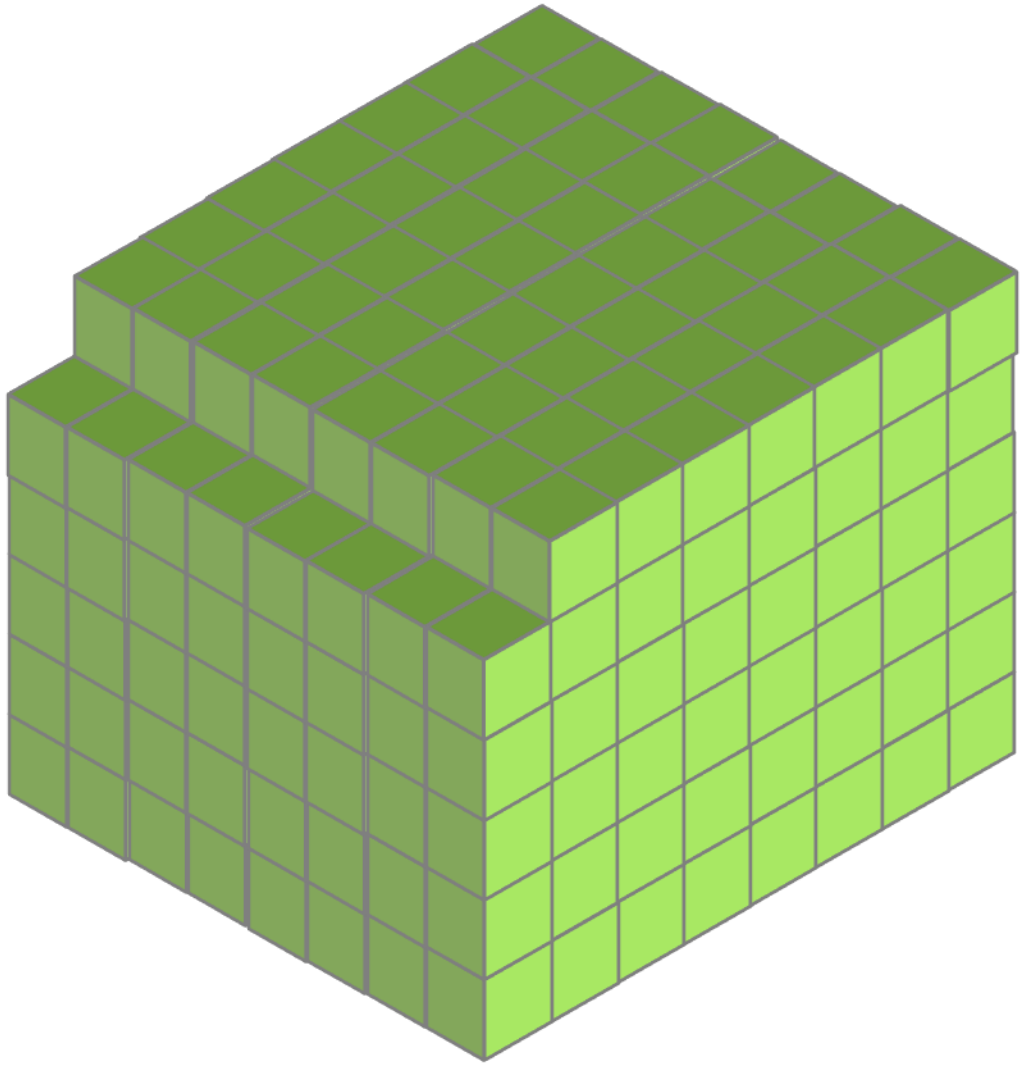




Deep layer feature-map

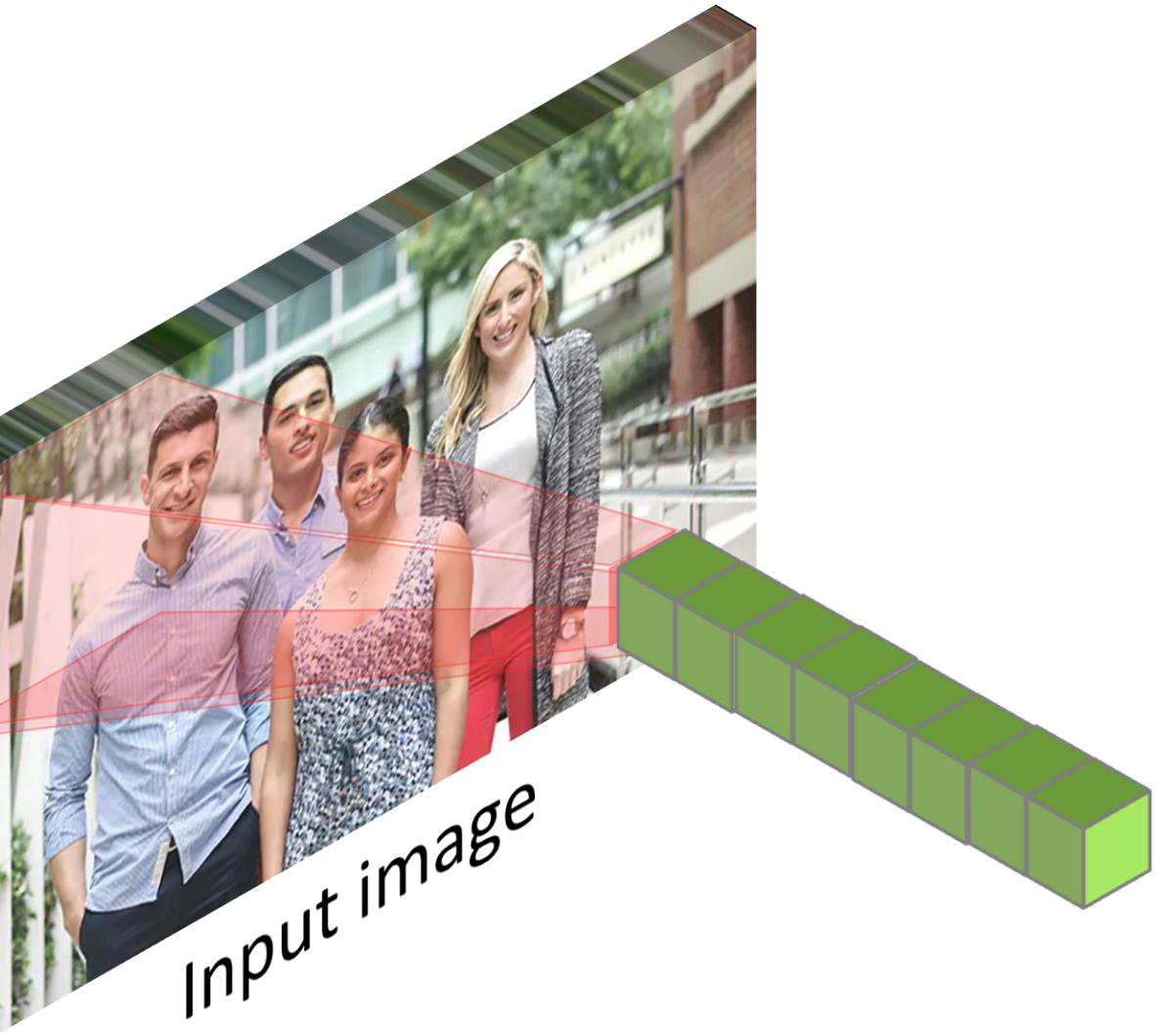
Two important intuitions about feature maps

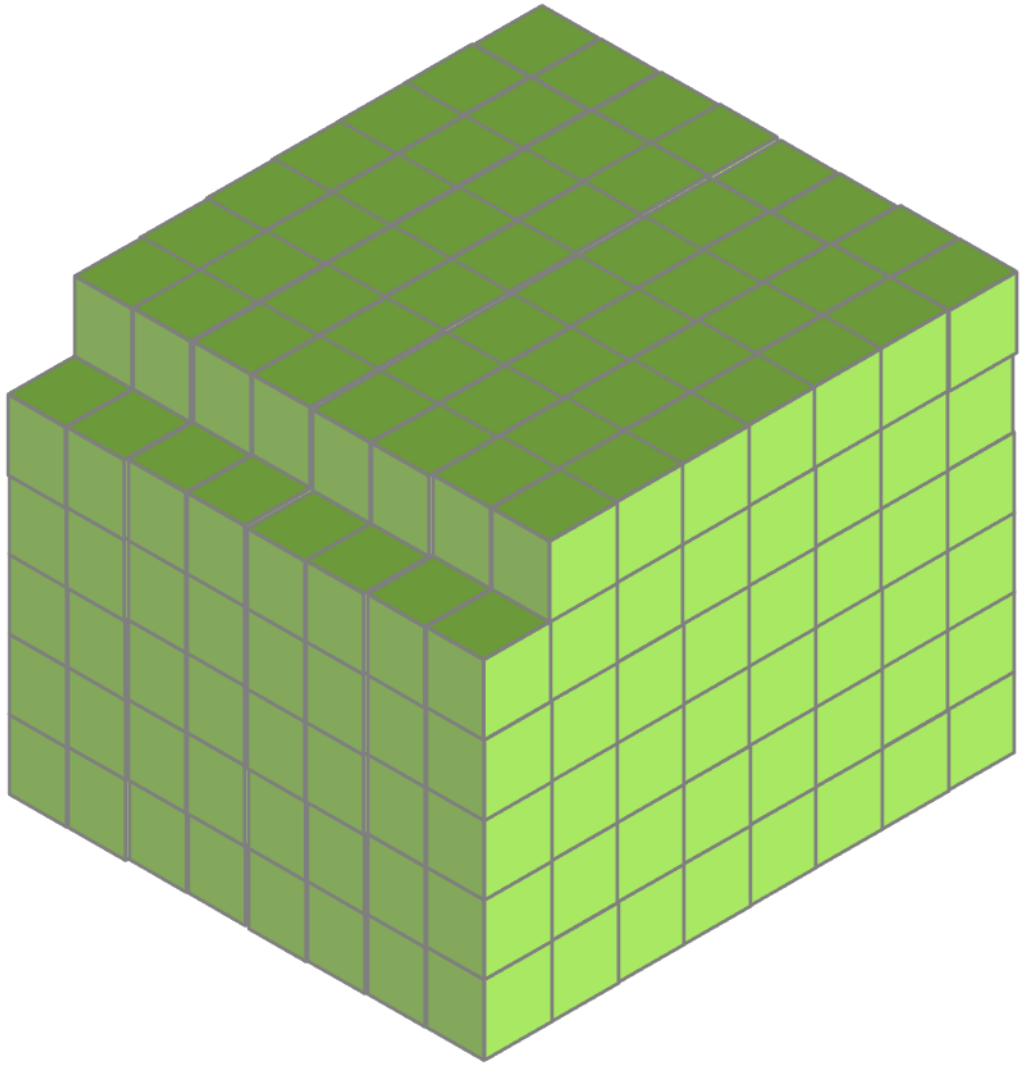




Deep layer feature-map

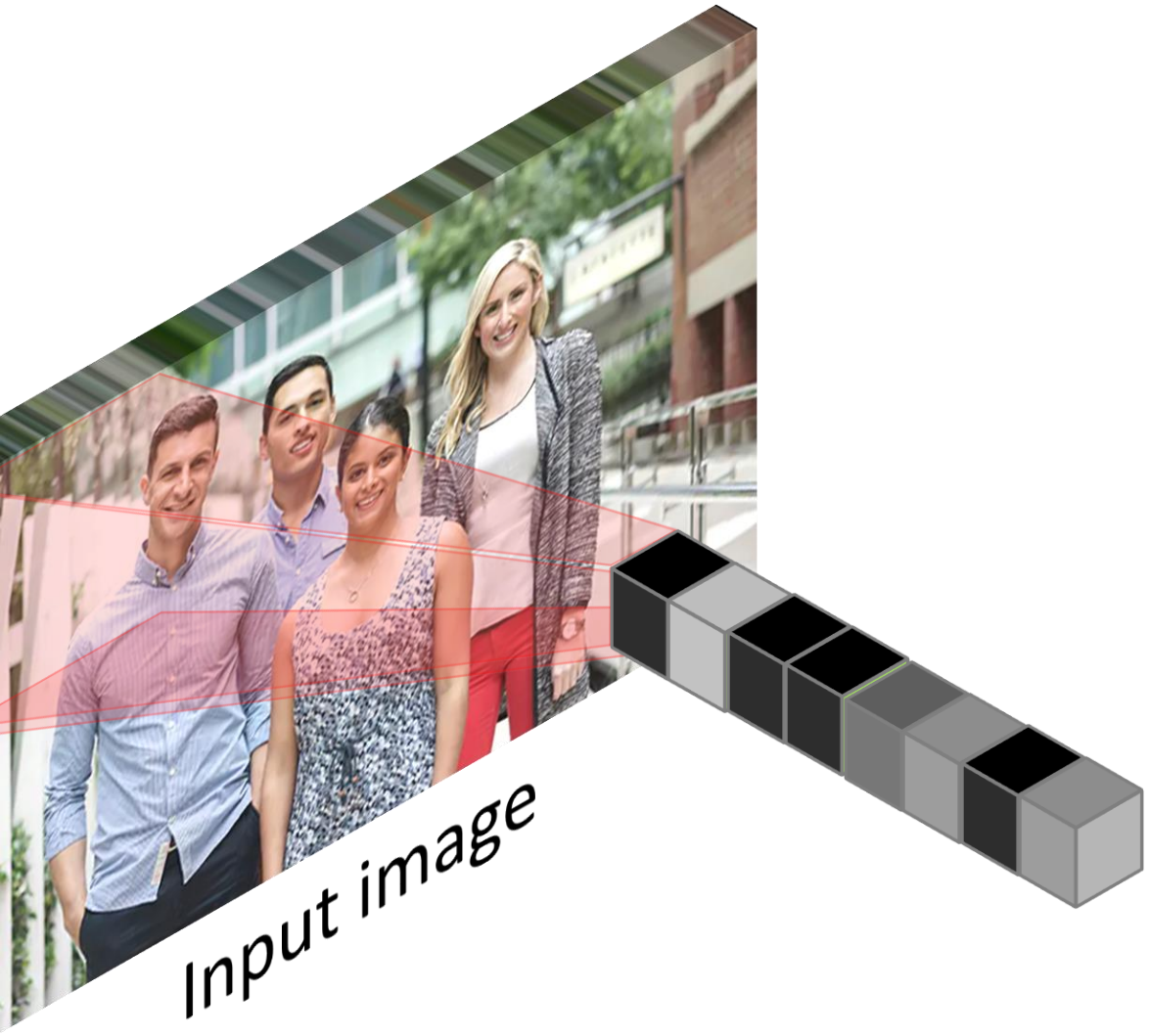
Two important intuitions about feature maps

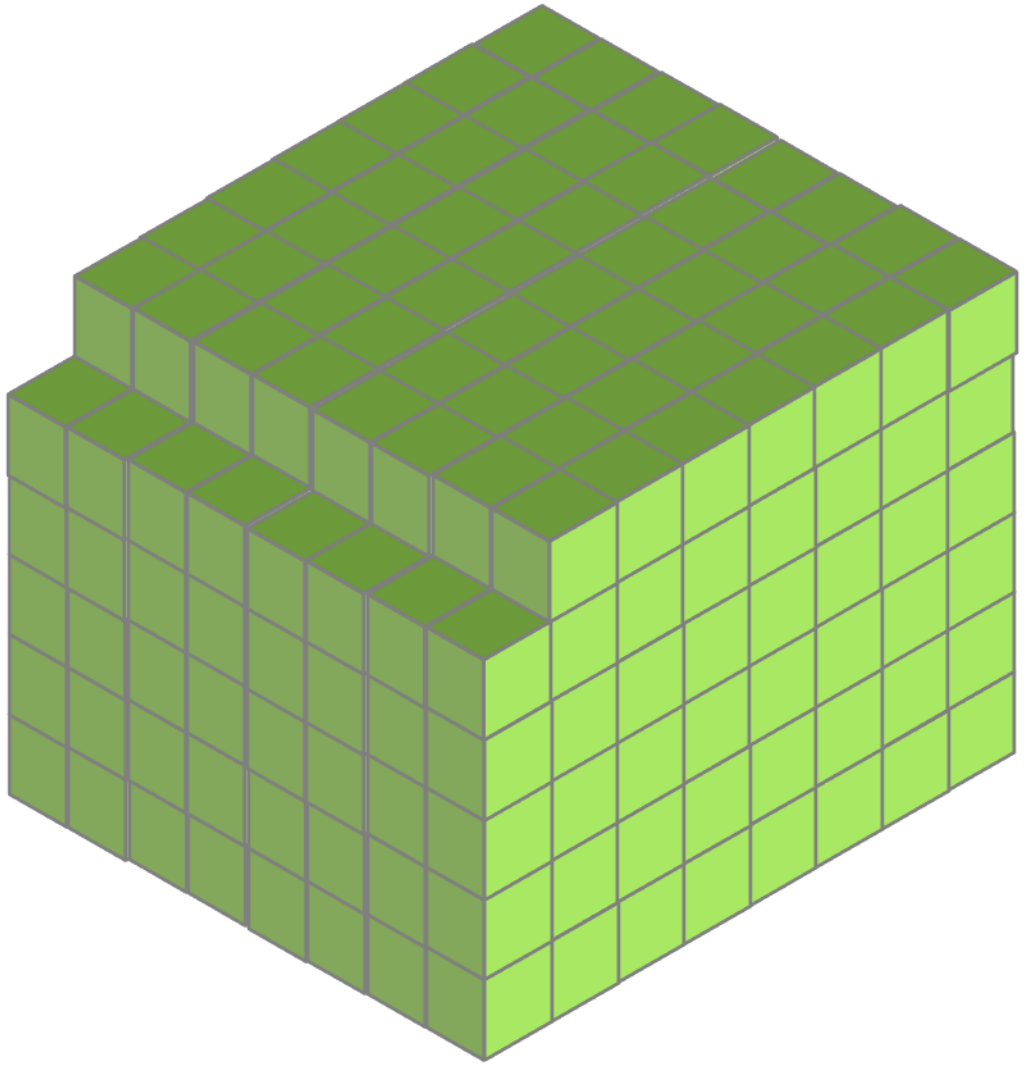




Deep layer feature-map

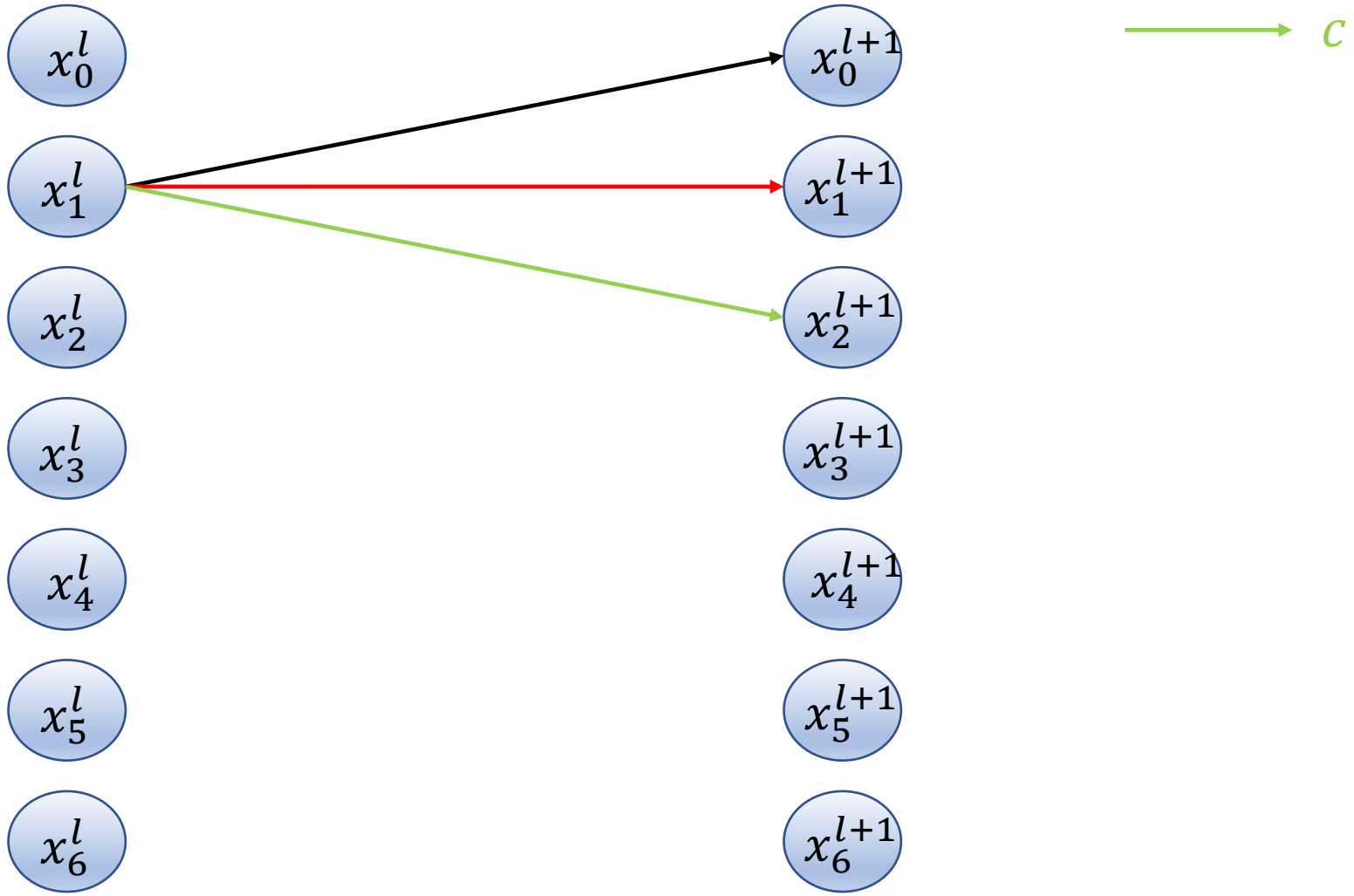
Two important intuitions about feature maps



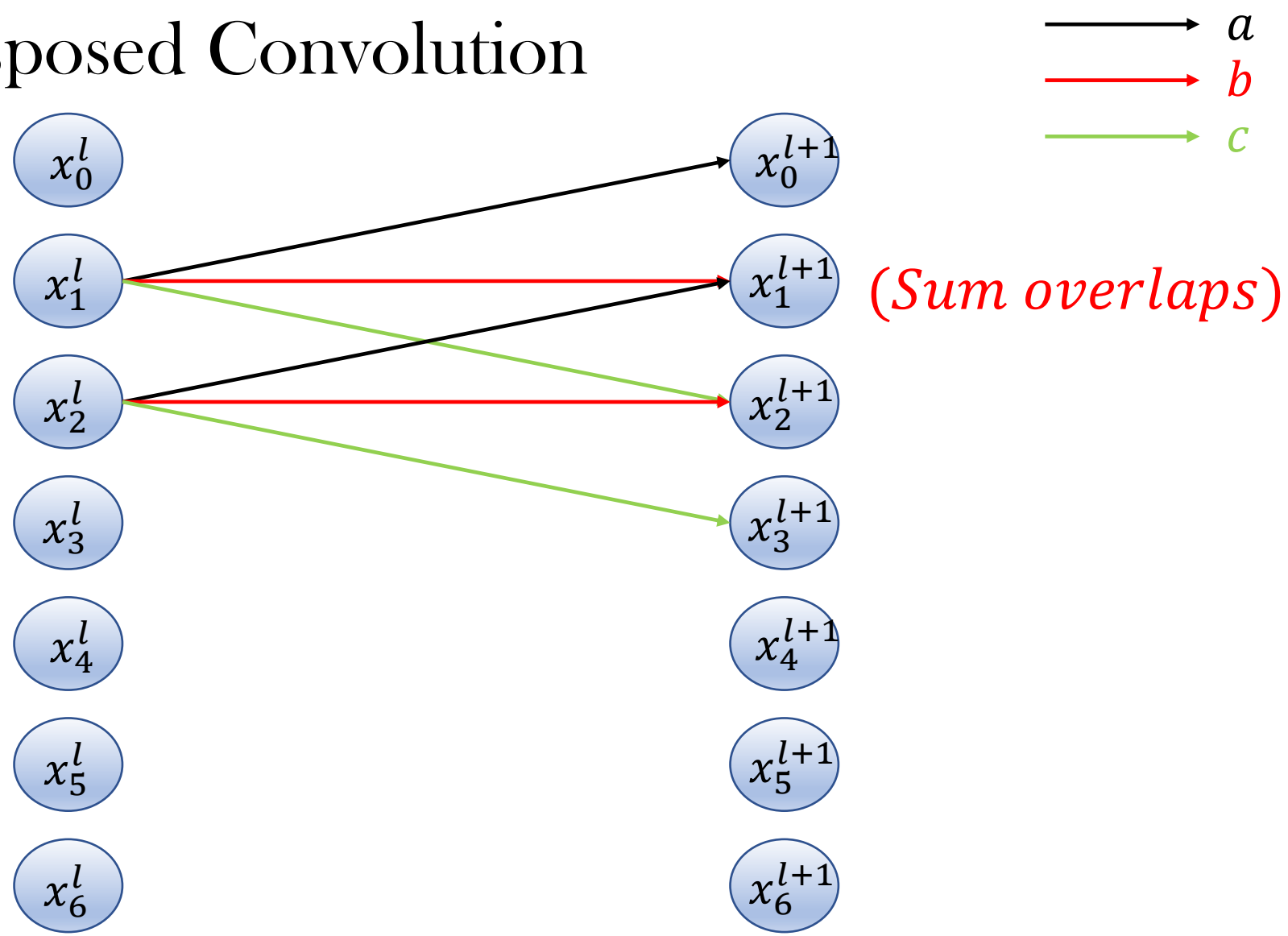


Deep layer feature-map

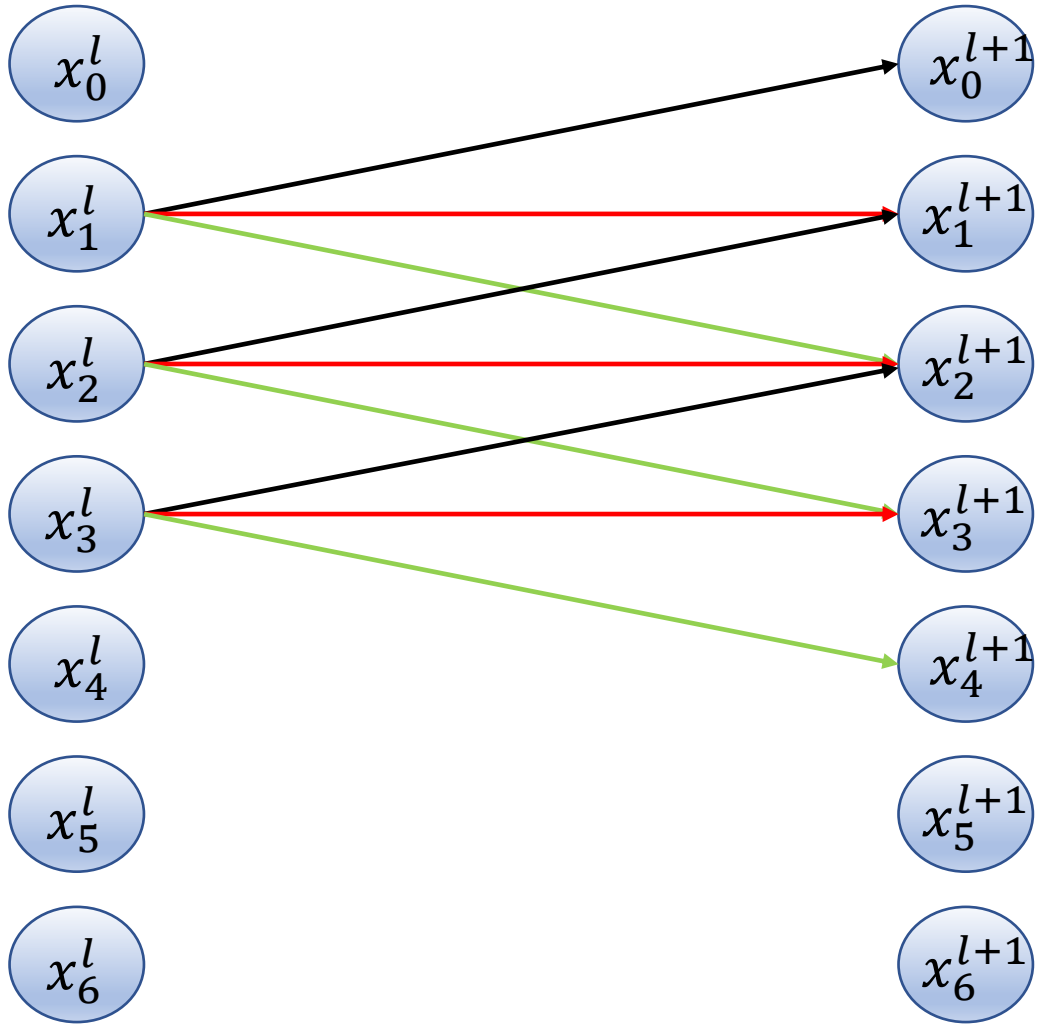
Transposed Convolution

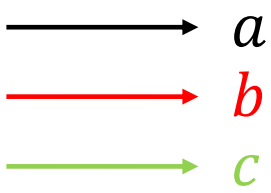


Transposed Convolution

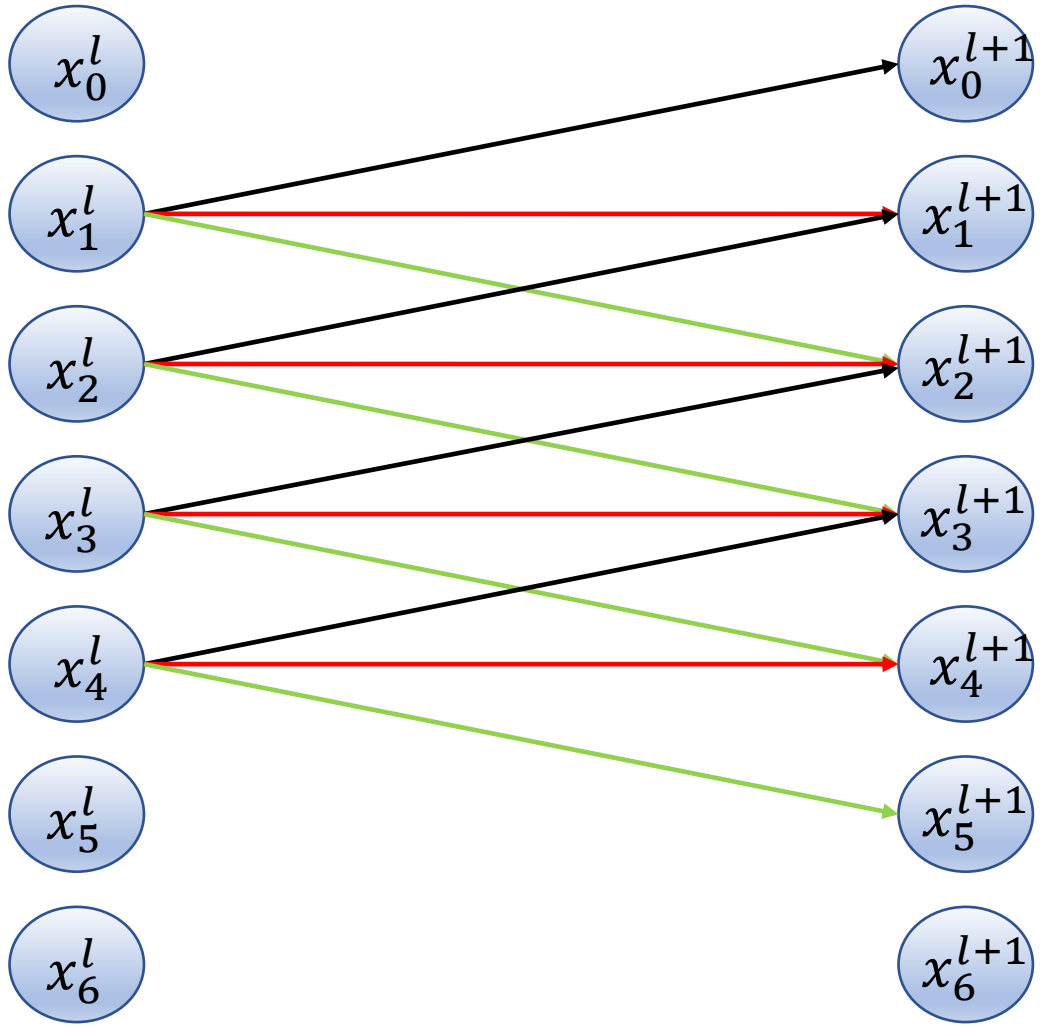


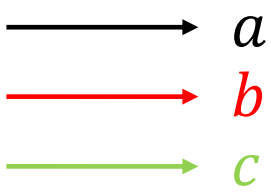
Transposed Convolution



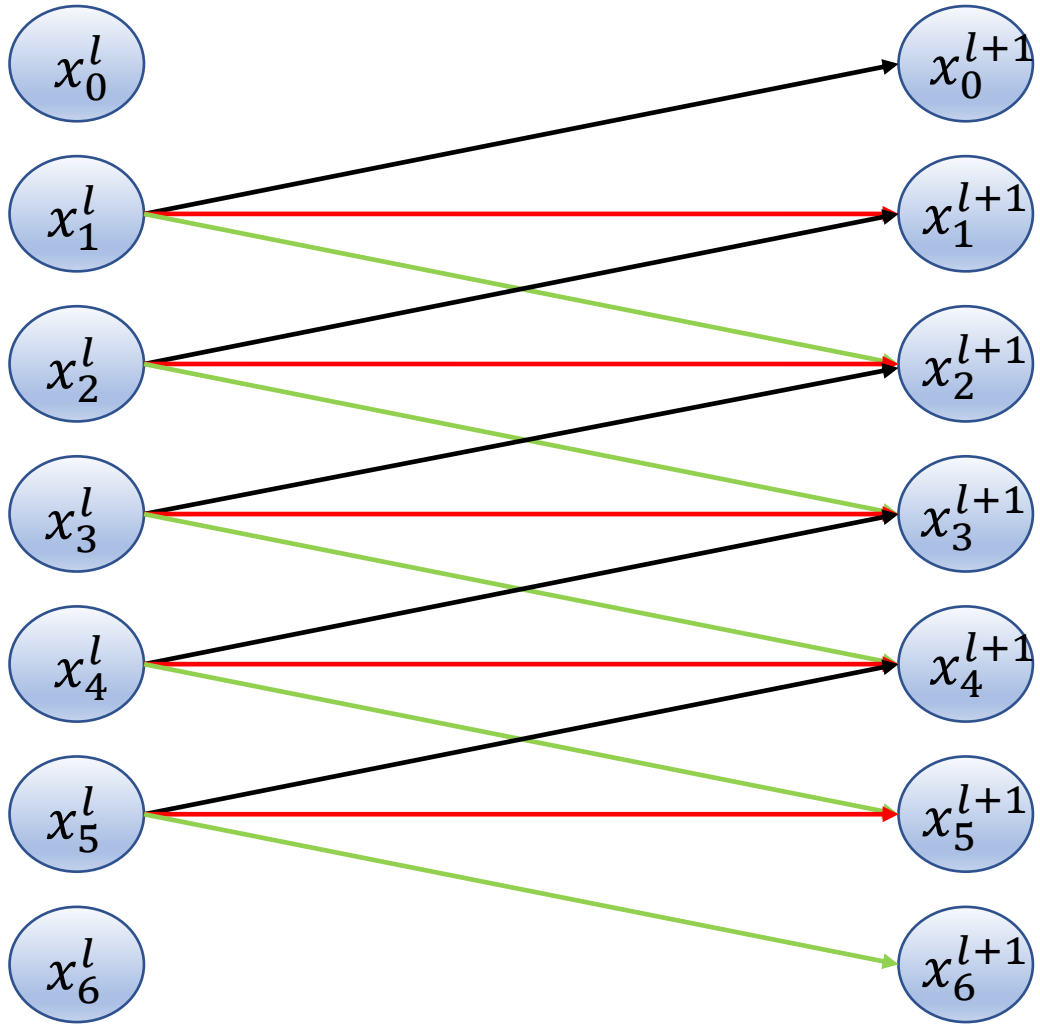


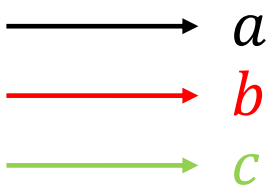
Transposed Convolution



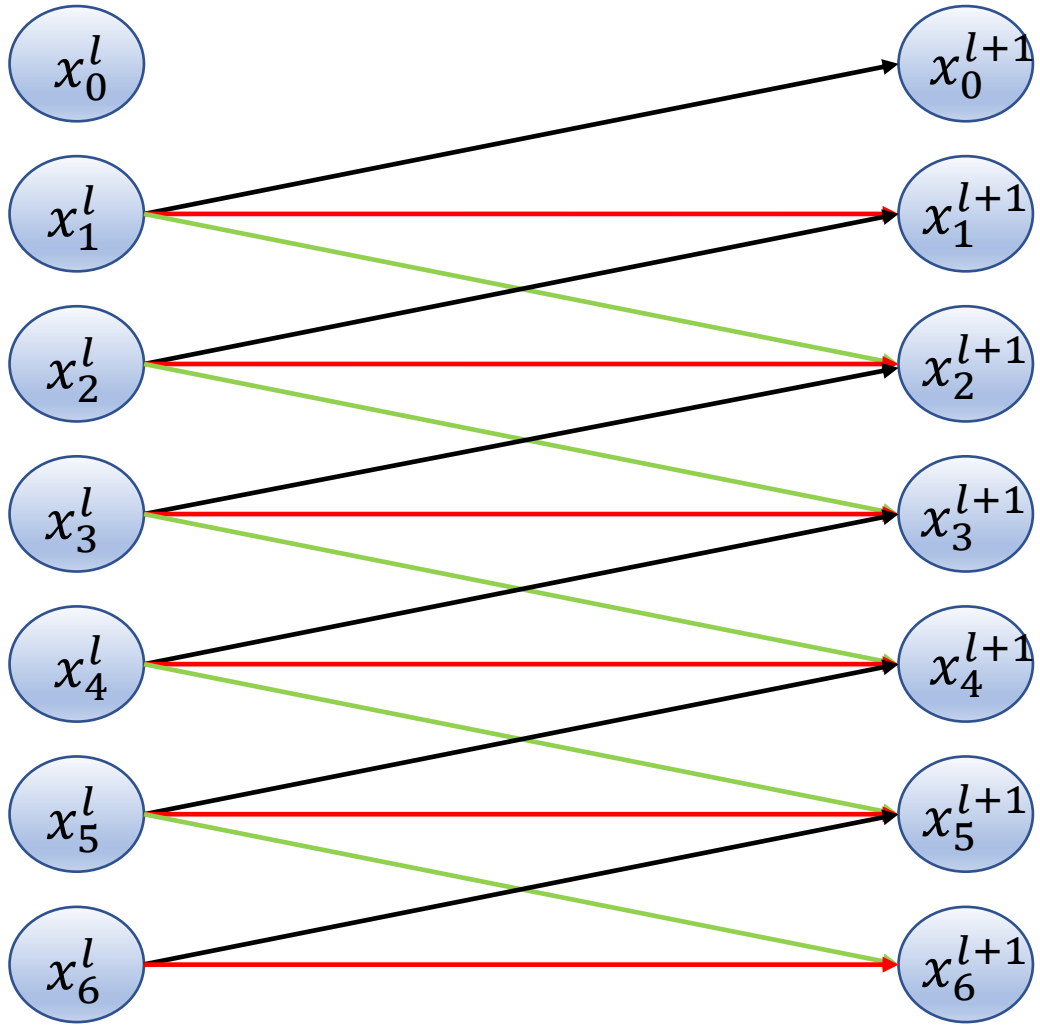


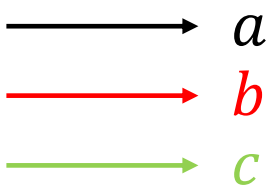
Transposed Convolution



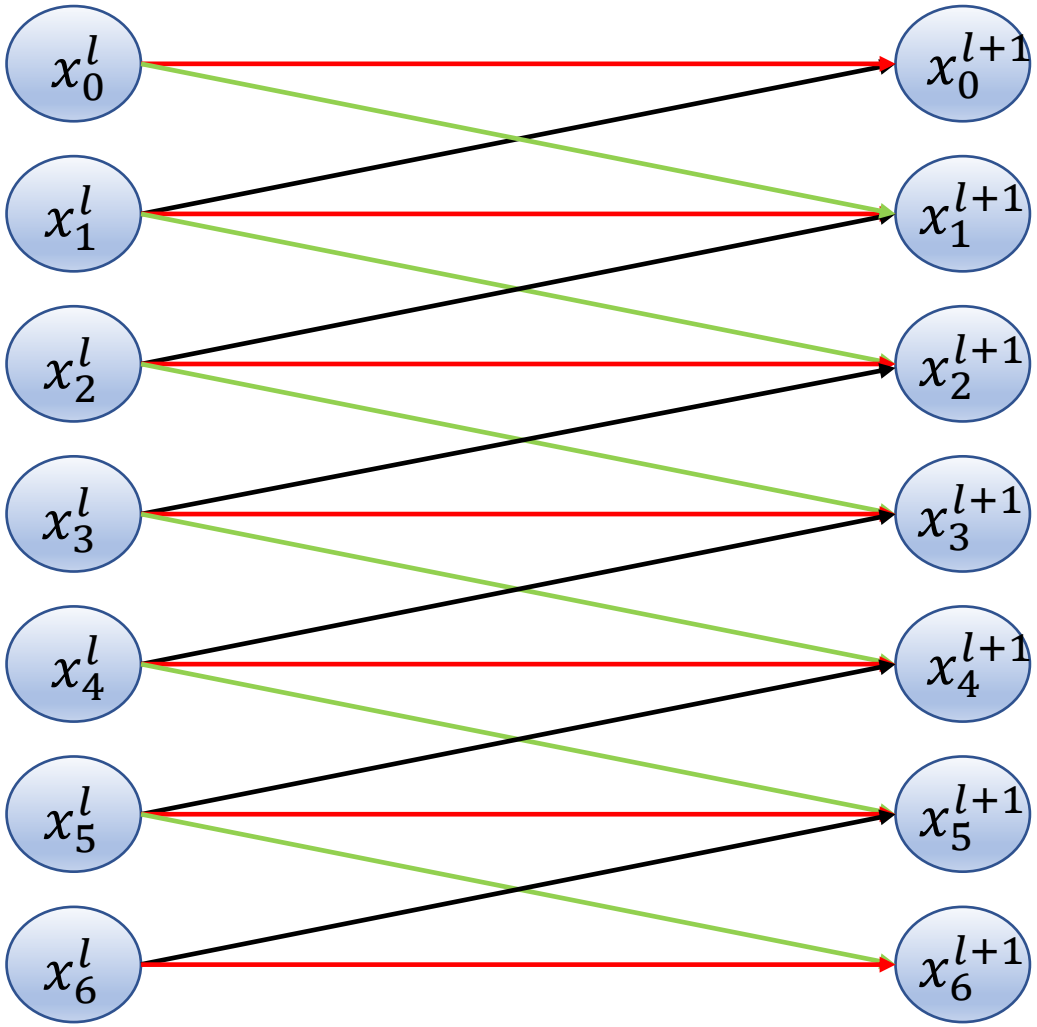


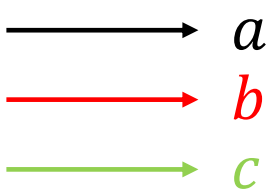
Transposed Convolution





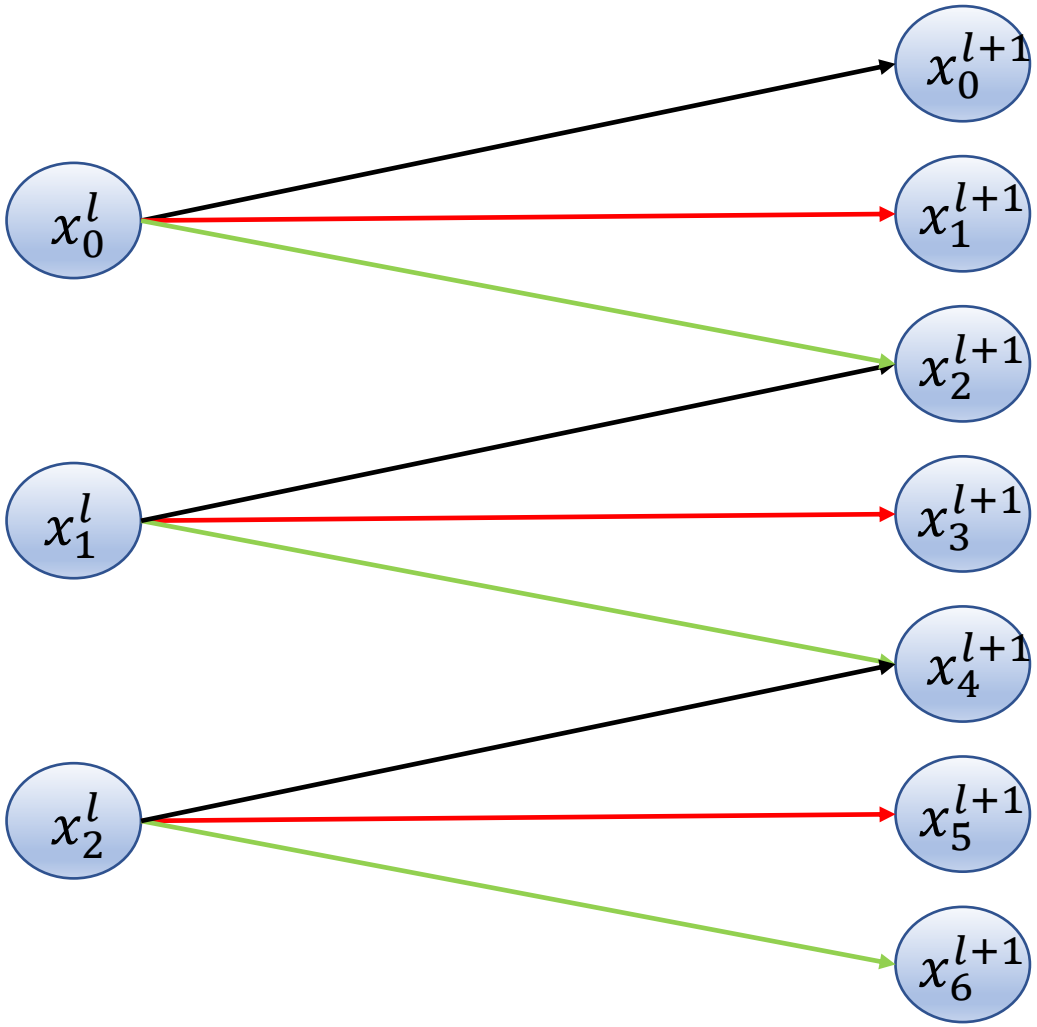
Transposed Convolution

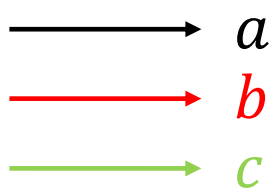




$$\begin{bmatrix} b & a & 0 & 0 & 0 & 0 & 0 \\ c & b & a & 0 & 0 & 0 & 0 \\ 0 & c & b & a & 0 & 0 & 0 \\ 0 & a & c & b & a & 0 & 0 \\ 0 & 0 & a & c & b & a & 0 \\ 0 & 0 & 0 & 0 & c & b & a \\ 0 & 0 & 0 & 0 & 0 & c & b \end{bmatrix}$$

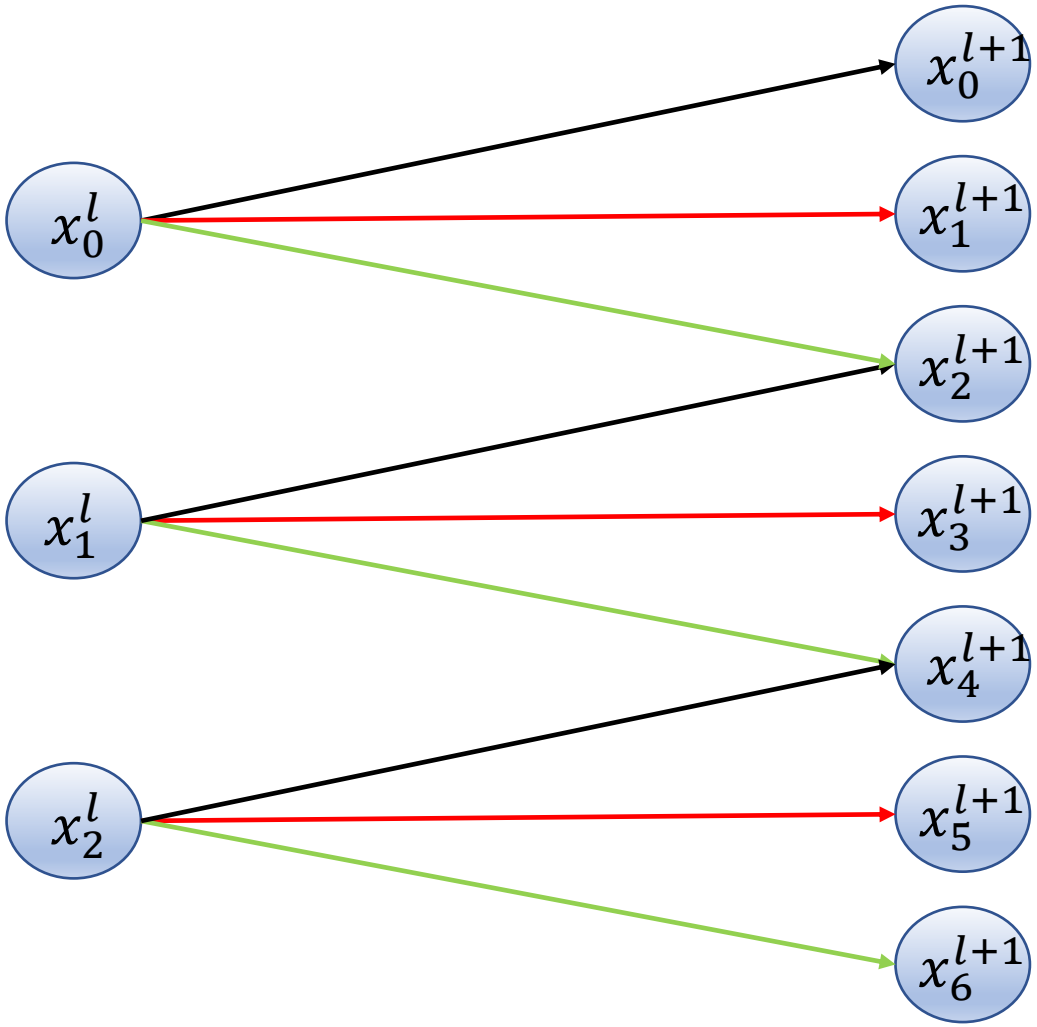
Transposed Convolution with stride

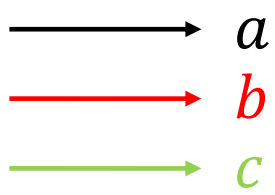




$$\begin{bmatrix} b & a & 0 & 0 & 0 & 0 & 0 \\ c & b & a & 0 & 0 & 0 & 0 \\ 0 & c & b & a & 0 & 0 & 0 \\ 0 & a & c & b & a & 0 & 0 \\ 0 & 0 & a & c & b & a & 0 \\ 0 & 0 & 0 & 0 & c & b & a \\ 0 & 0 & 0 & 0 & 0 & c & b \end{bmatrix}$$

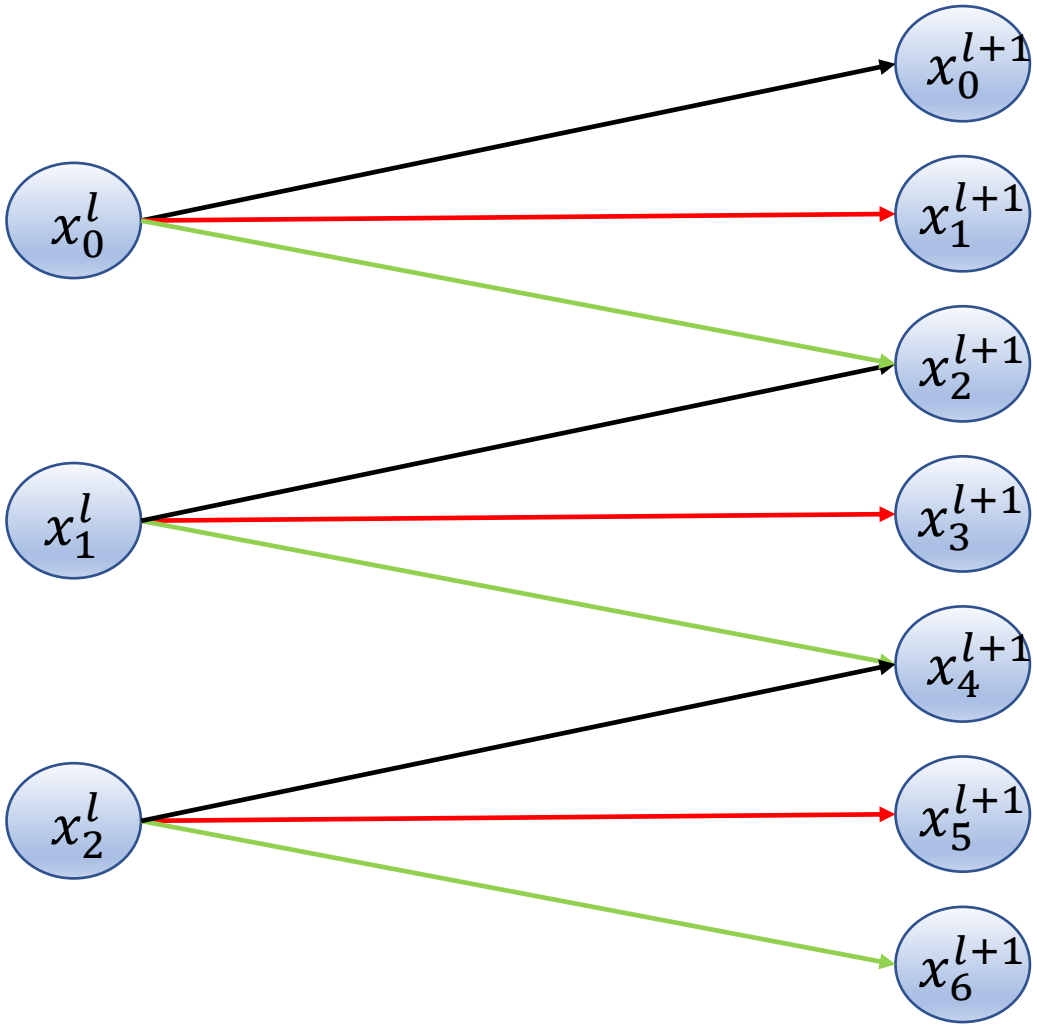
Transposed Convolution with stride





$$\begin{bmatrix} b & a & 0 & 0 & 0 & 0 & 0 \\ c & b & a & 0 & 0 & 0 & 0 \\ 0 & c & b & a & 0 & 0 & 0 \\ 0 & a & c & b & a & 0 & 0 \\ 0 & 0 & a & c & b & a & 0 \\ 0 & 0 & 0 & 0 & c & b & a \\ 0 & 0 & 0 & 0 & 0 & c & b \end{bmatrix}$$

Transposed Convolution with stride



a
b
c

$$\begin{bmatrix} a & 0 & 0 \\ b & 0 & 0 \\ c & a & 0 \\ a & b & 0 \\ 0 & c & a \\ 0 & 0 & b \\ 0 & 0 & c \end{bmatrix}$$

Recall- stride 2 conv:

$$\begin{bmatrix} a & b & c & 0 & 0 & 0 & 0 \\ 0 & 0 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & 0 & a & b & c \end{bmatrix}$$

Transposed Convolution with stride



a
b
c

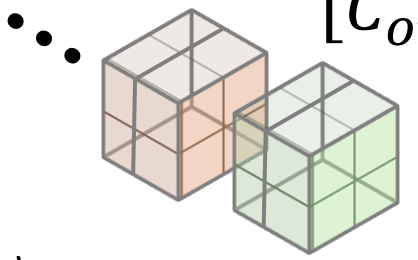
$$\begin{bmatrix} a & 0 & 0 \\ b & 0 & 0 \\ c & a & 0 \\ a & b & 0 \\ 0 & c & a \\ 0 & 0 & b \\ 0 & 0 & c \end{bmatrix}$$

Recall- stride 2 conv:

$$\begin{bmatrix} a & b & c & 0 & 0 & 0 & 0 \\ 0 & 0 & a & b & c & 0 & 0 \\ 0 & 0 & 0 & 0 & a & b & c \end{bmatrix}$$

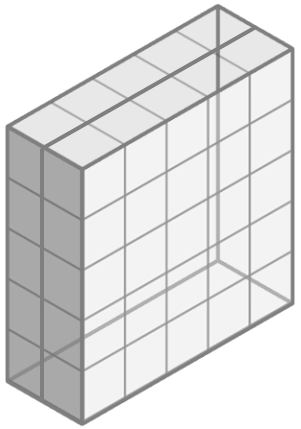
A note about the implementation of conv

$[C_{out}, C_{in}, h, w]$



$Unfold$

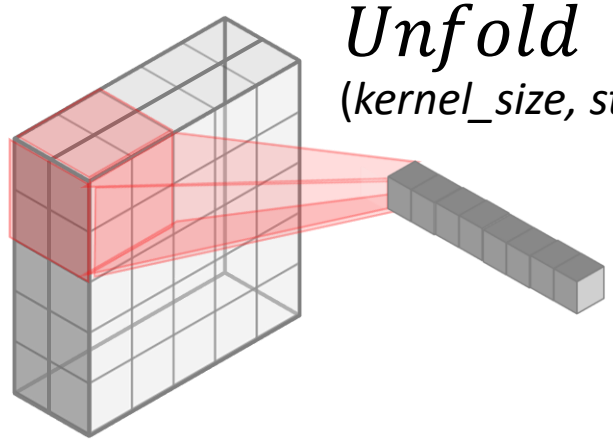
(kernel_size, stride)



$[N, C_{in}, H, W]$

A note about the implementation of conv

$[C_{out}, C_{in}, h, w]$

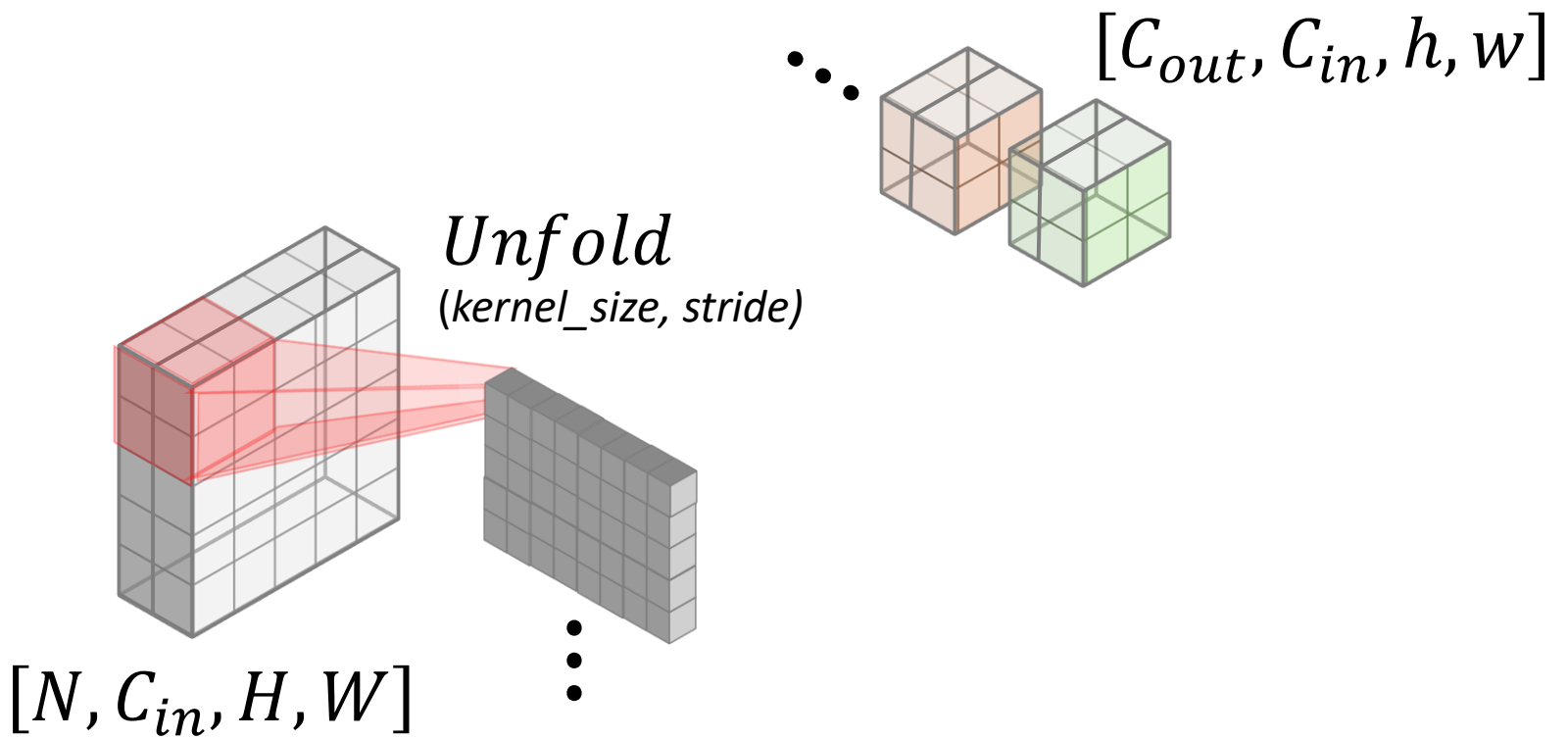


$Unfold$

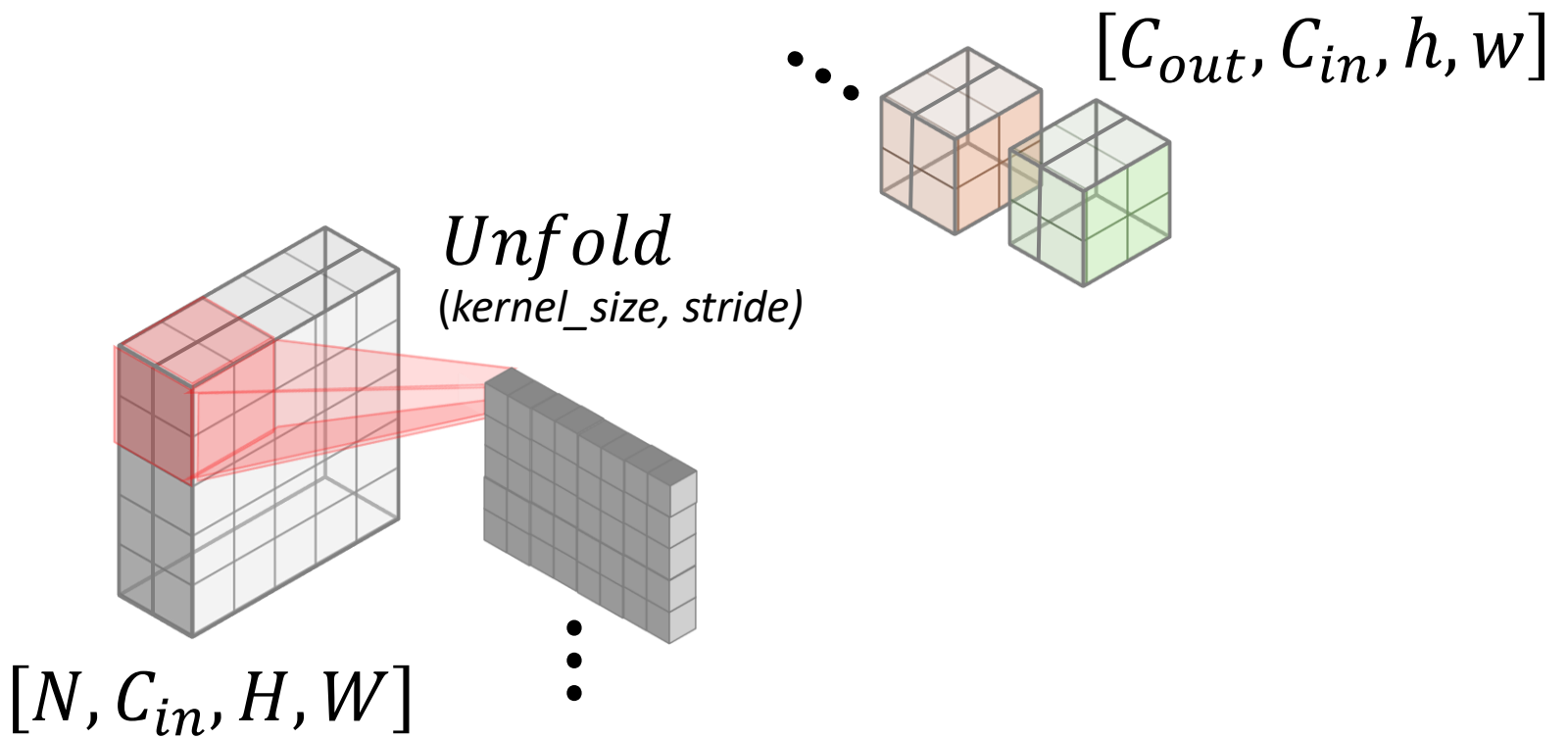
(kernel_size, stride)

$[N, C_{in}, H, W]$

A note about the implementation of conv

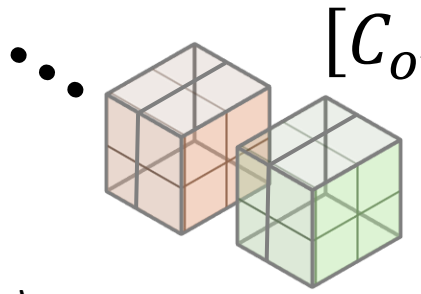


A note about the implementation of conv



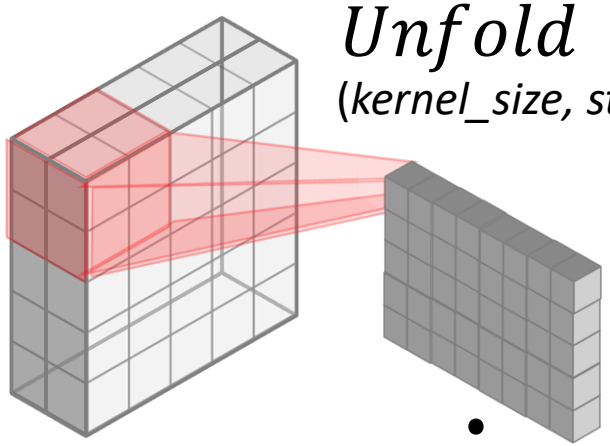
A list of flattened patches!

A note about the implementation of conv



$[C_{out}, C_{in}, h, w]$

$Unfold$
(kernel_size, stride)

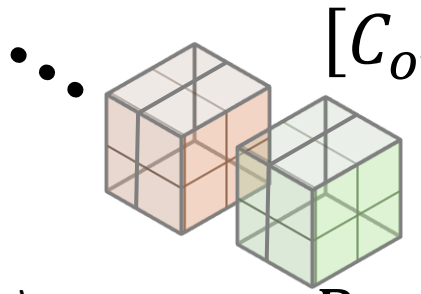


$[N, C_{in}, H, W]$

$$\left[(N \cdot \widetilde{H} \cdot \widetilde{W}), (C_{in} \cdot h \cdot w)\right]$$

A list of flattened patches!

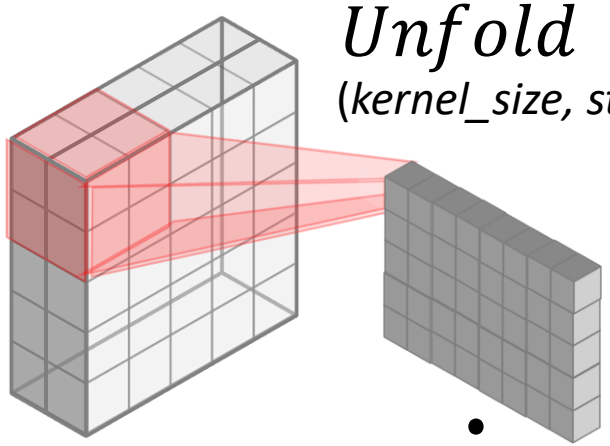
A note about the implementation of conv



$[C_{out}, C_{in}, h, w]$

$Unfold$
(kernel_size, stride)

$\downarrow Reshape$

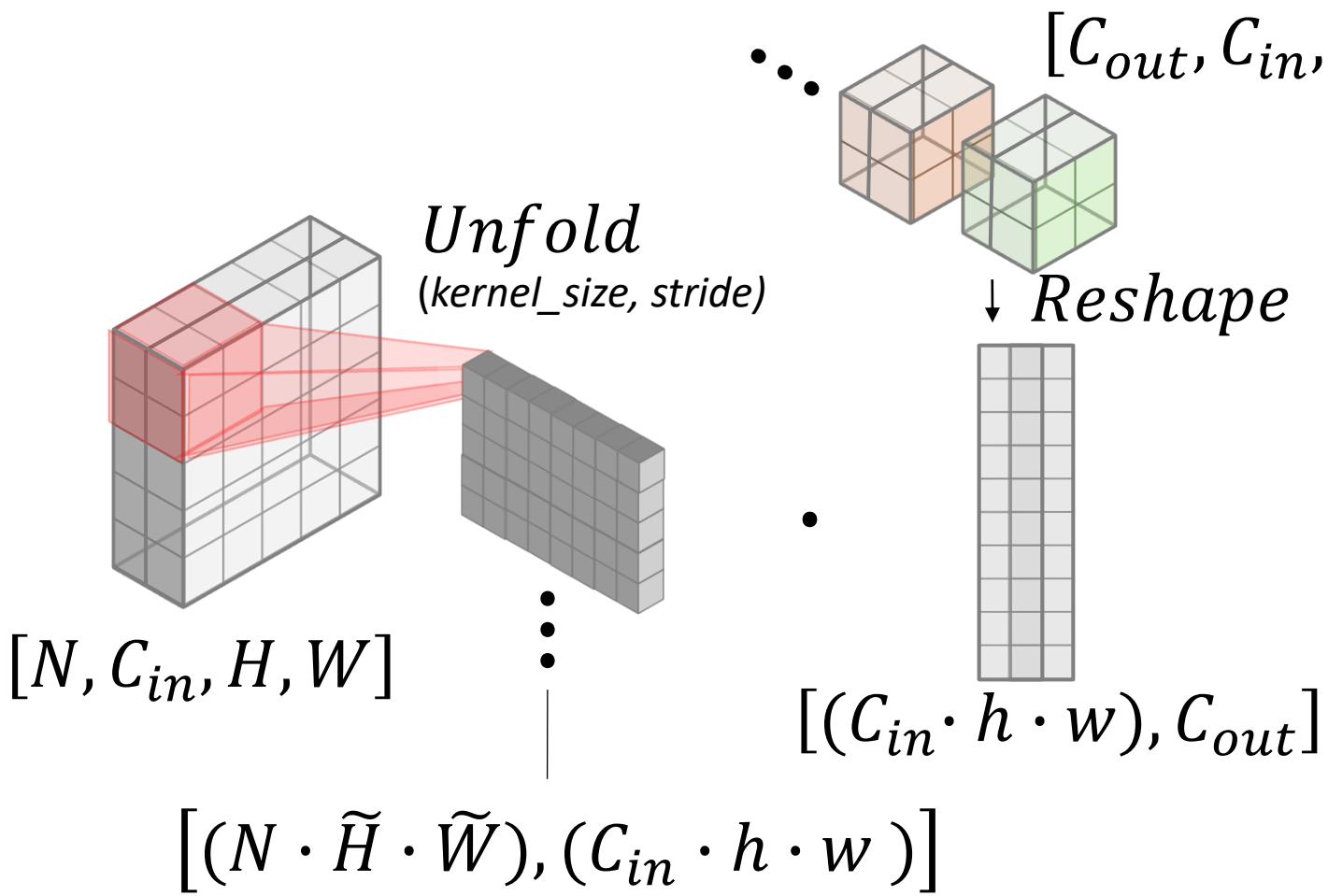


$[N, C_{in}, H, W]$

$\left[(N \cdot \widetilde{H} \cdot \widetilde{W}), (C_{in} \cdot h \cdot w)\right]$

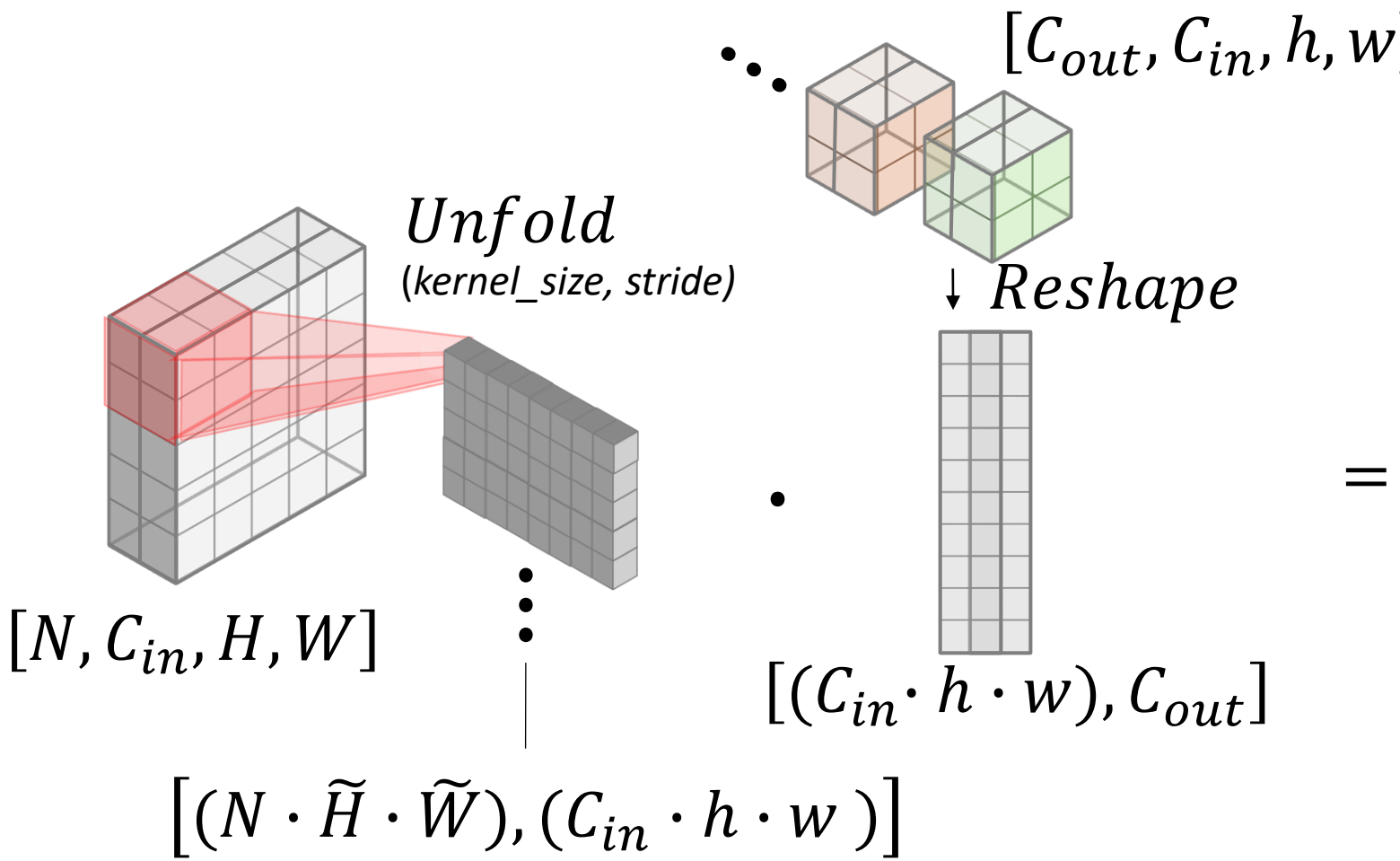
A list of flattened patches!

A note about the implementation of conv



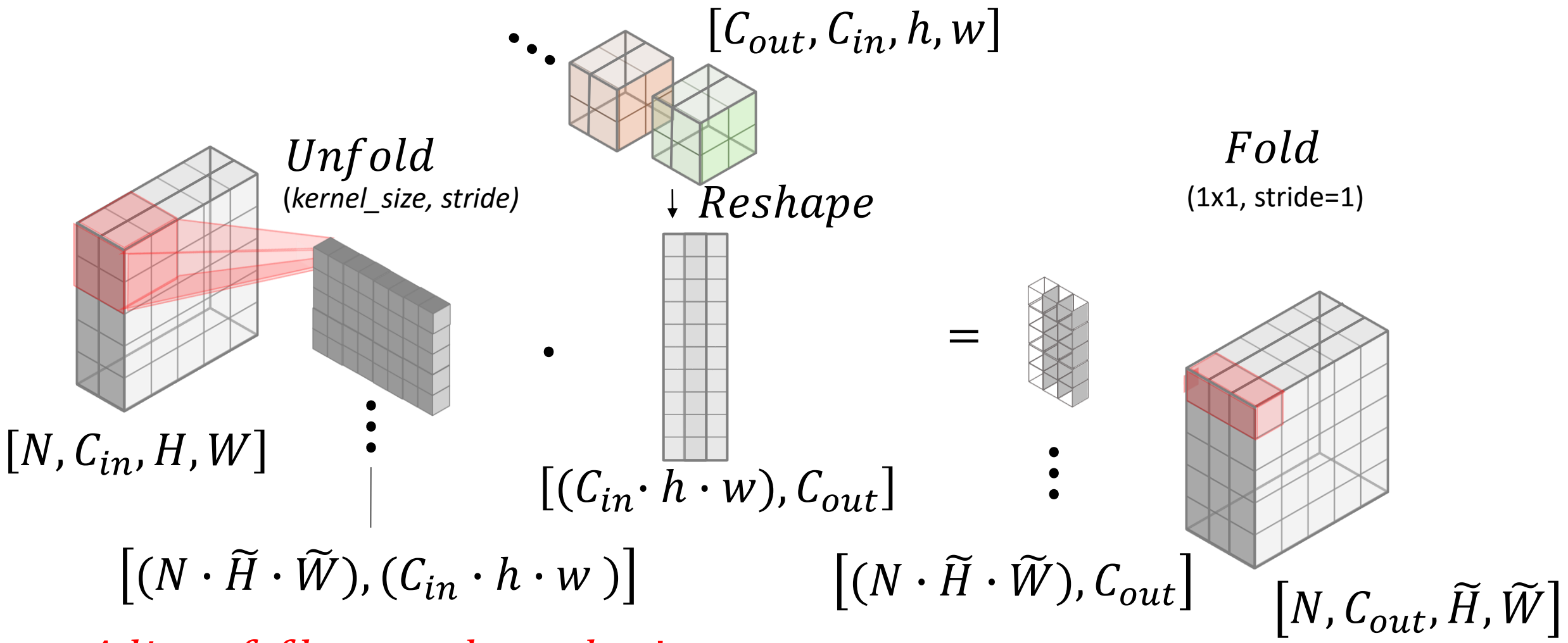
A list of flattened patches!

A note about the implementation of conv



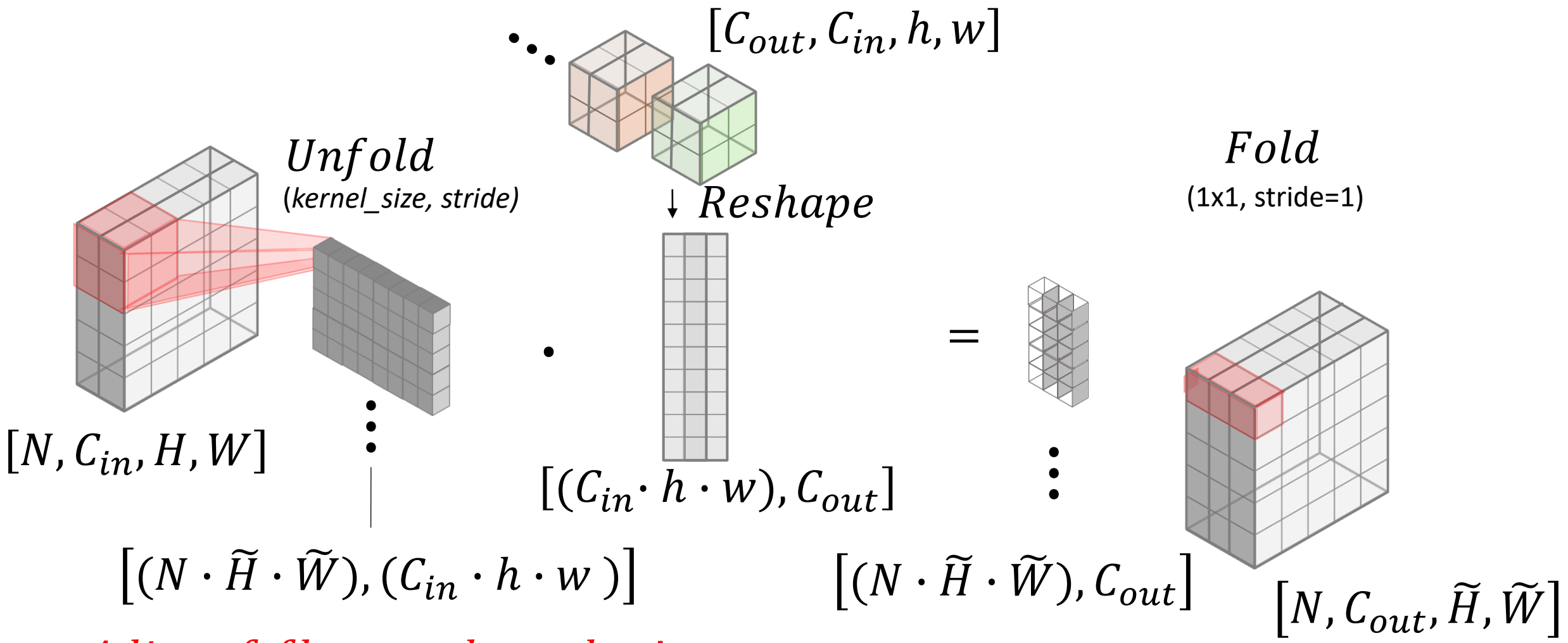
A list of flattened patches!

A note about the implementation of conv



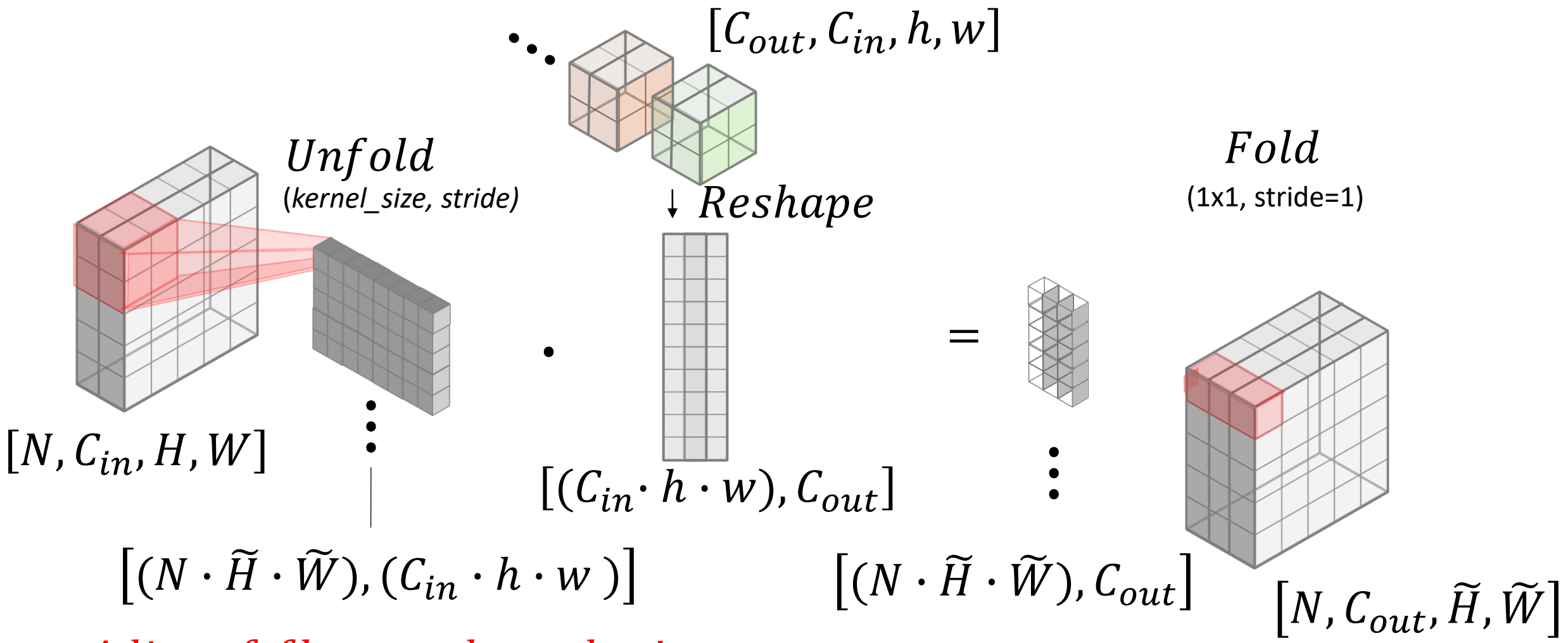
A list of flattened patches!

A note about the implementation of conv



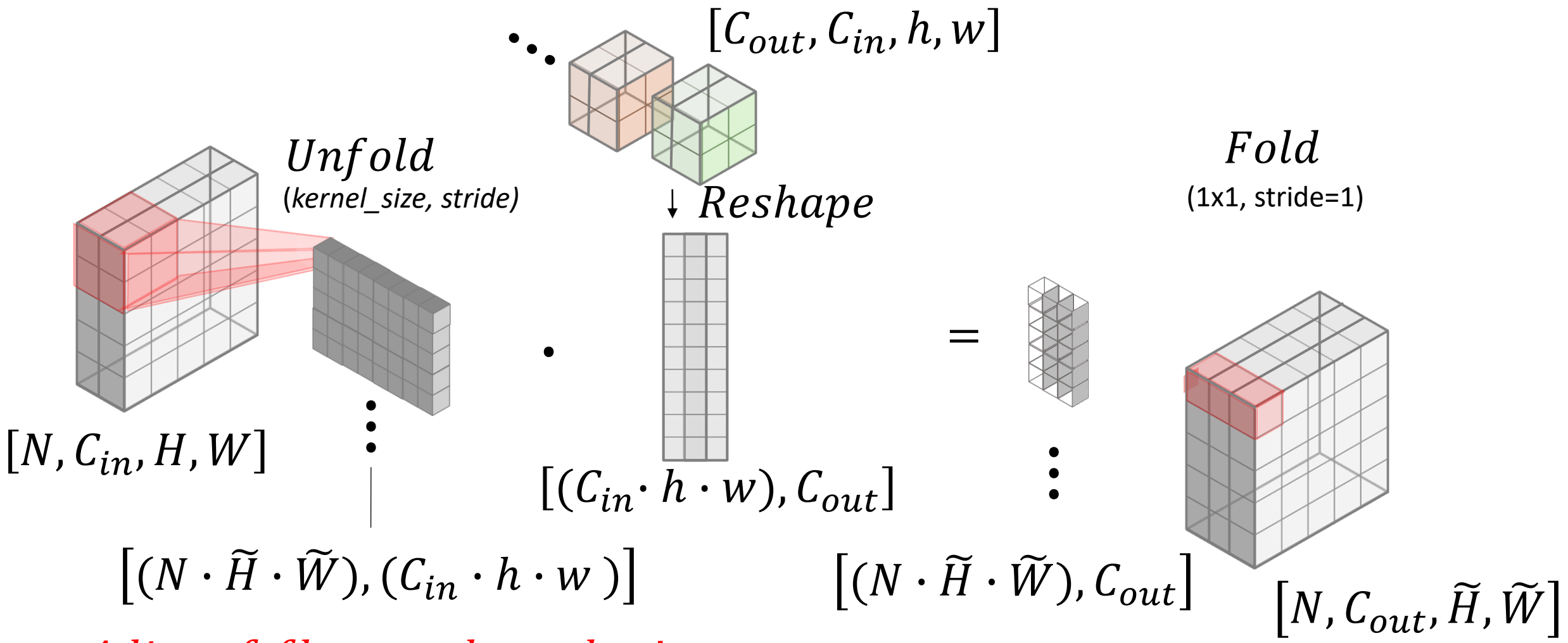
A list of flattened patches!

A note about the implementation of conv



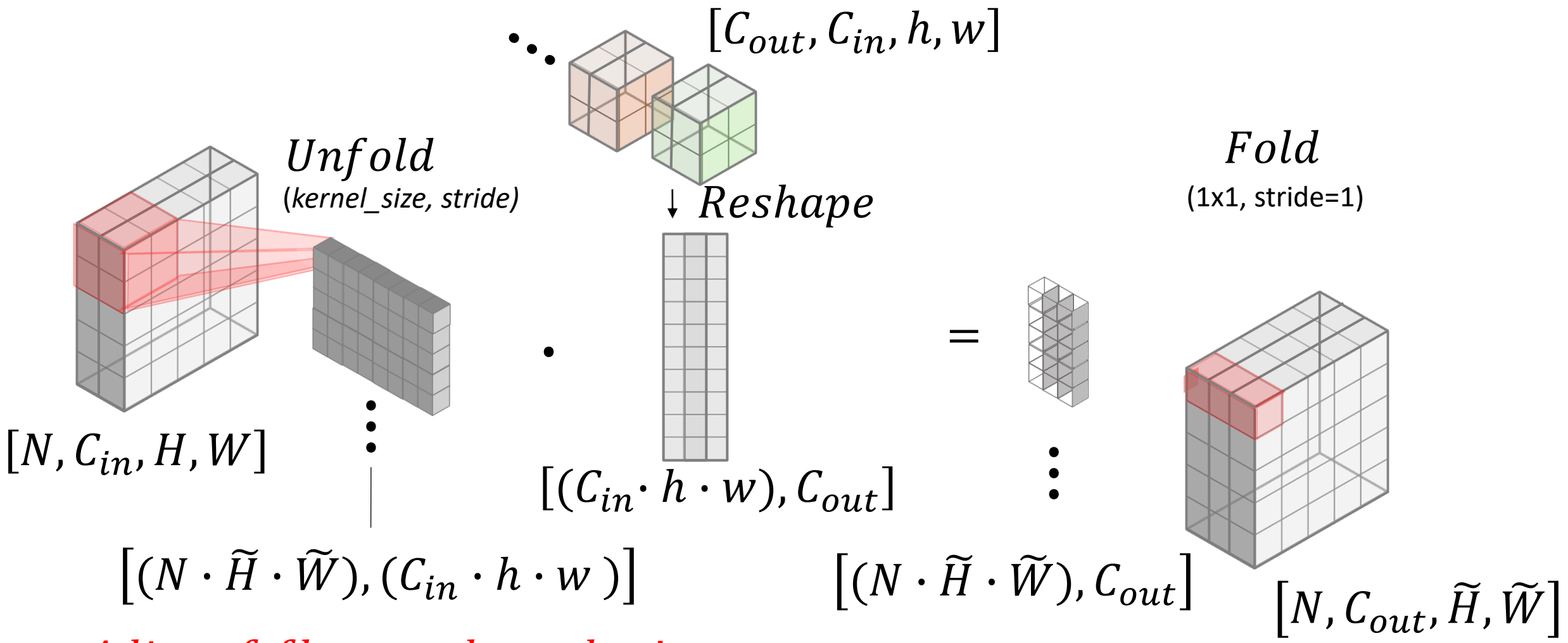
A list of flattened patches!

A note about the implementation of conv



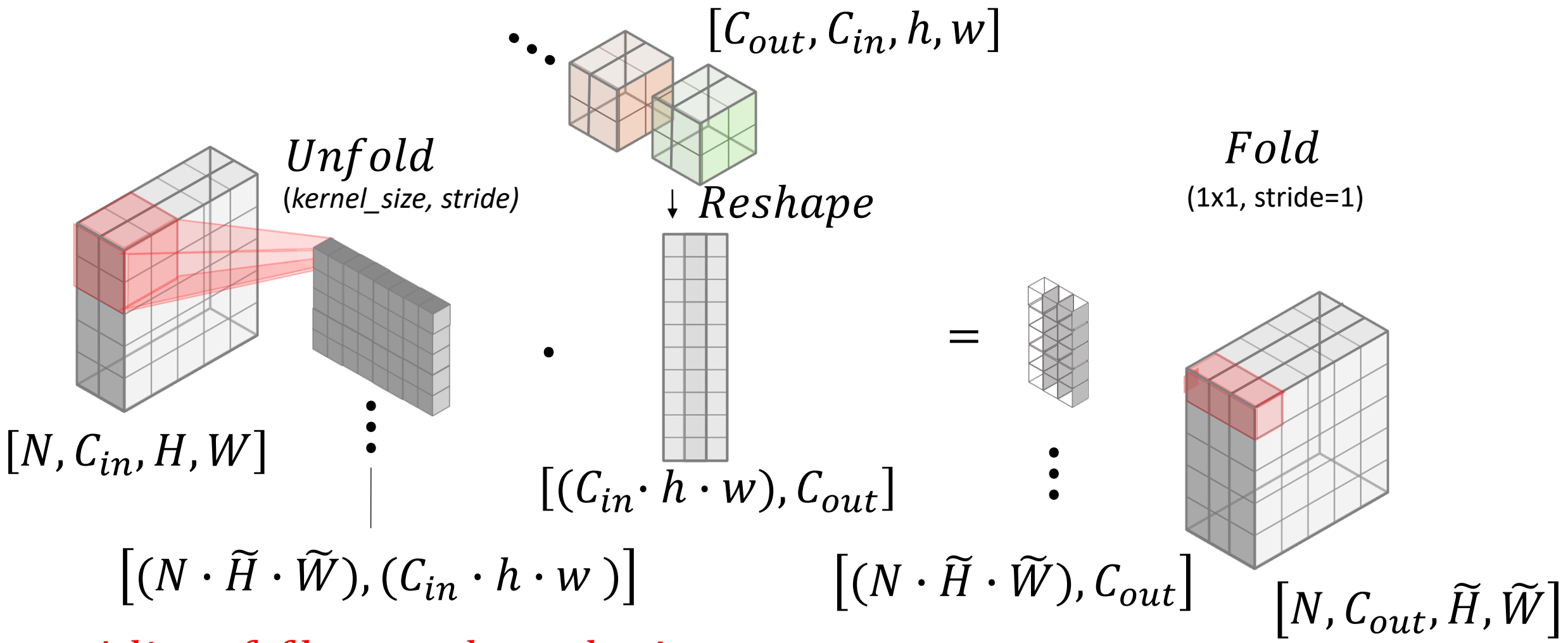
A list of flattened patches!

A note about the implementation of conv



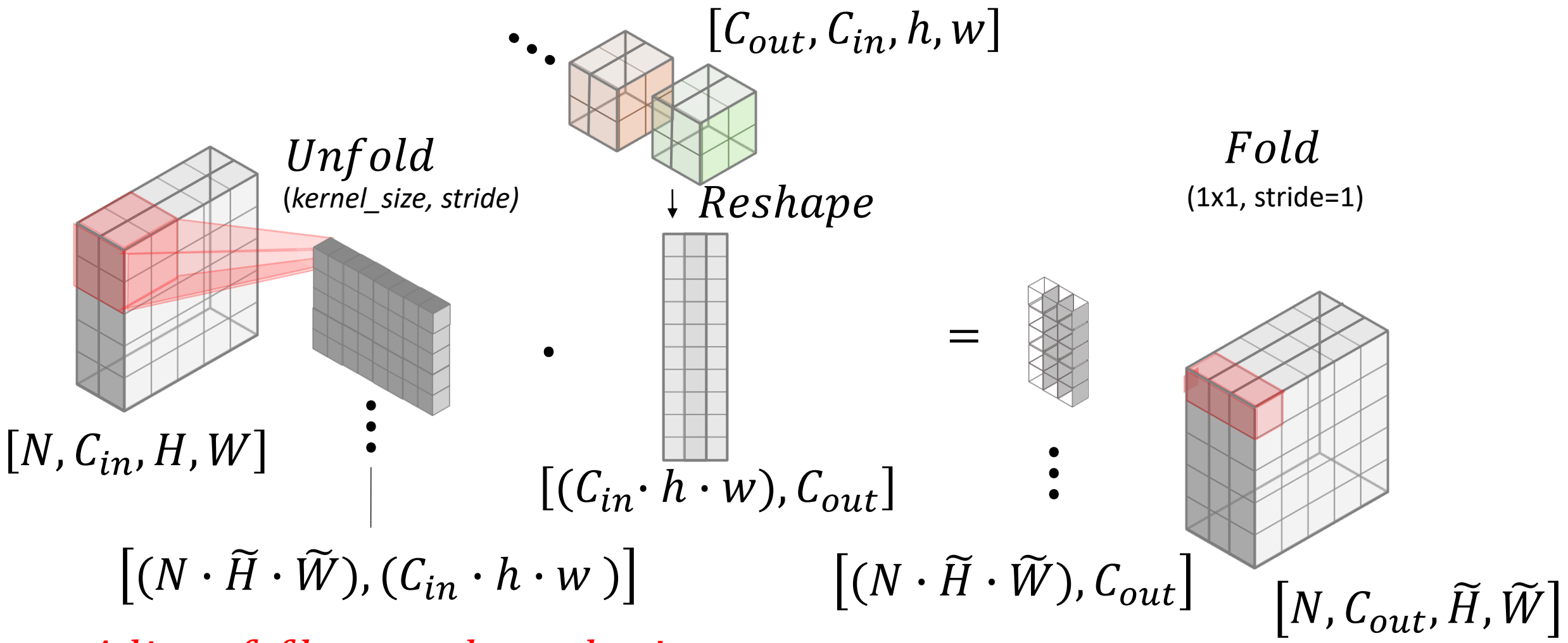
A list of flattened patches!

A note about the implementation of conv



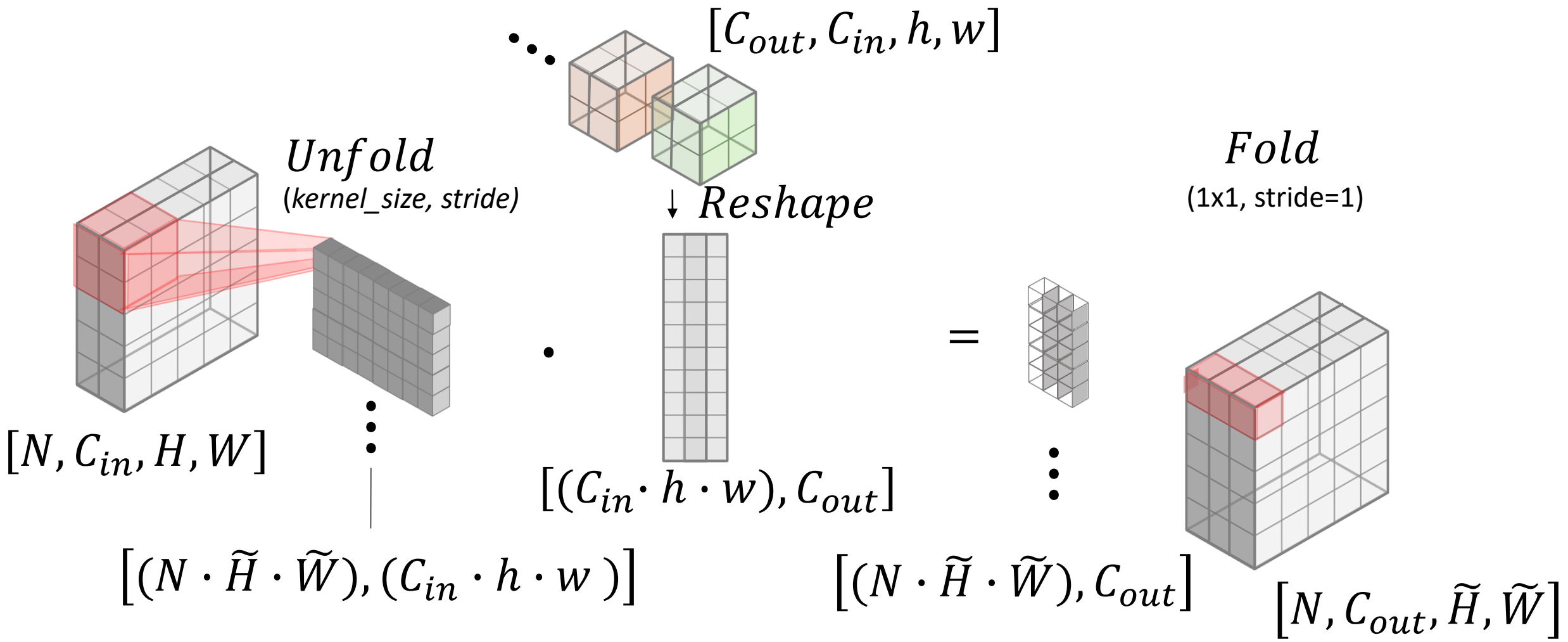
A list of flattened patches!

A note about the implementation of conv

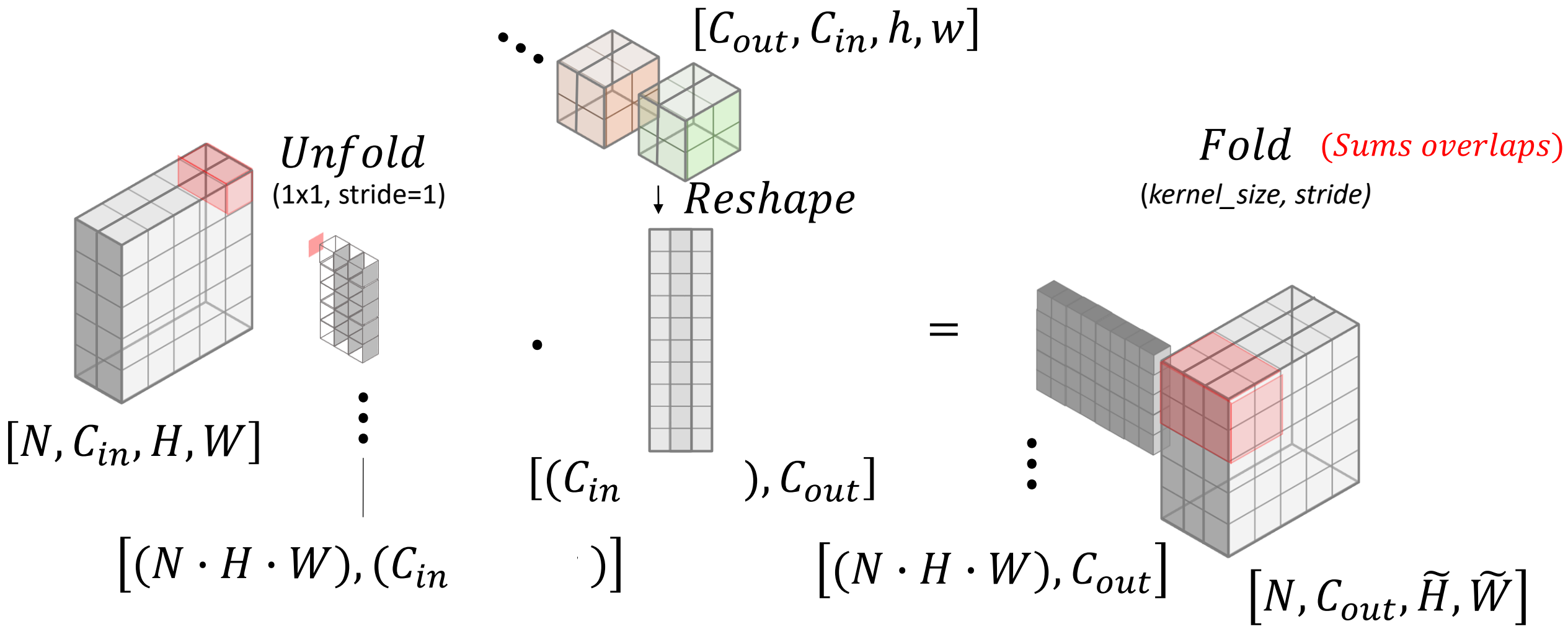


A list of flattened patches!

A note about the implementation of conv *TRANSPOSED*



A note about the implementation of conv *TRANSPOSED*



Q: How do you backprop a Conv? (1D, single channel)

For derivative of loss w.r.t all neurons:
Transposed Conv with the same filter!

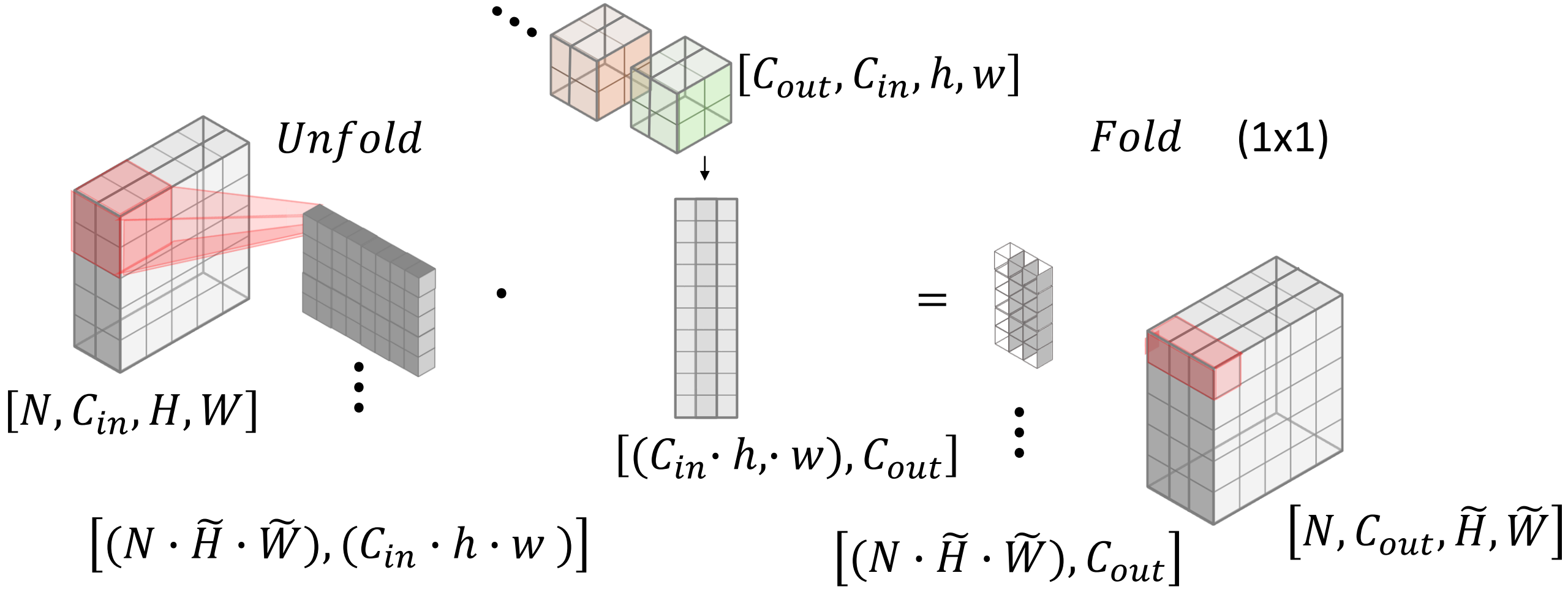
(Conv is a special case of linear)

Q: How about Conv2D?

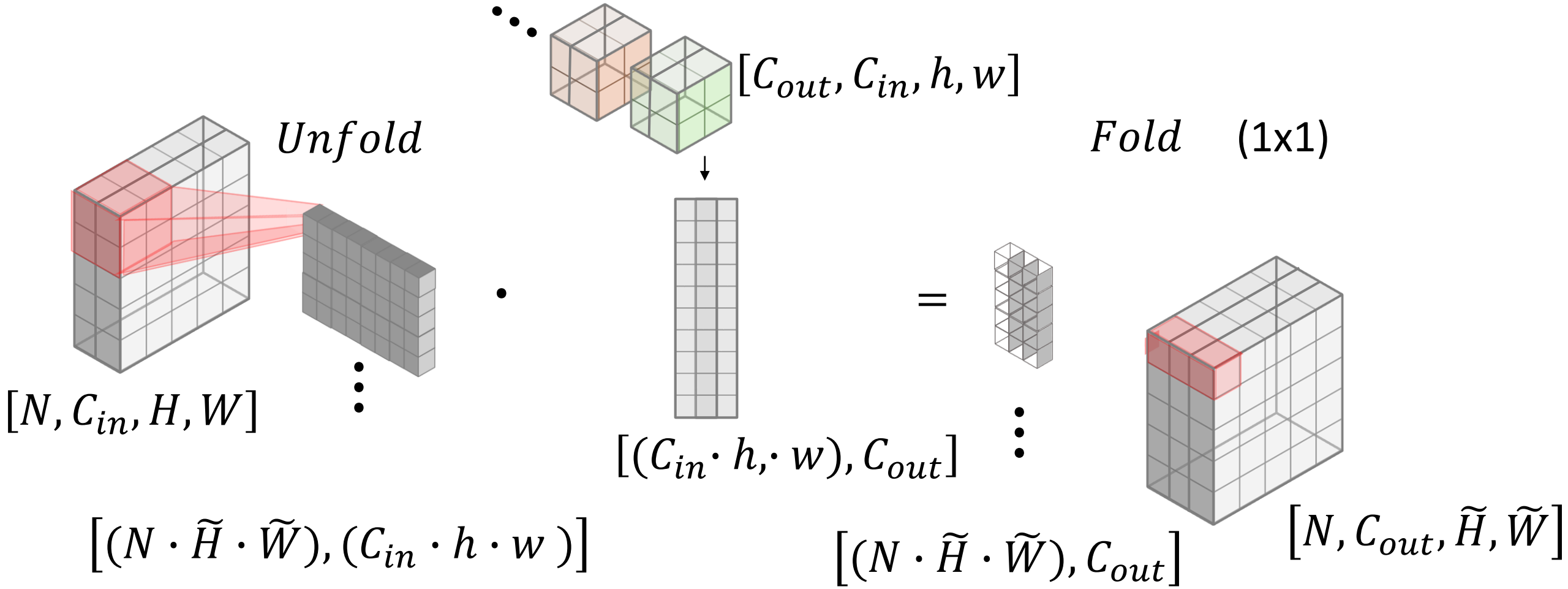
For derivative of loss w.r.t all neurons:
1. Transpose dims of filter c_{in} , c_{out} (as in FC)
2. Transposed Conv2D with the modified filter!

(Think of a conv as a linear layer applied to each patch separately, going from c_{in} to c_{out})

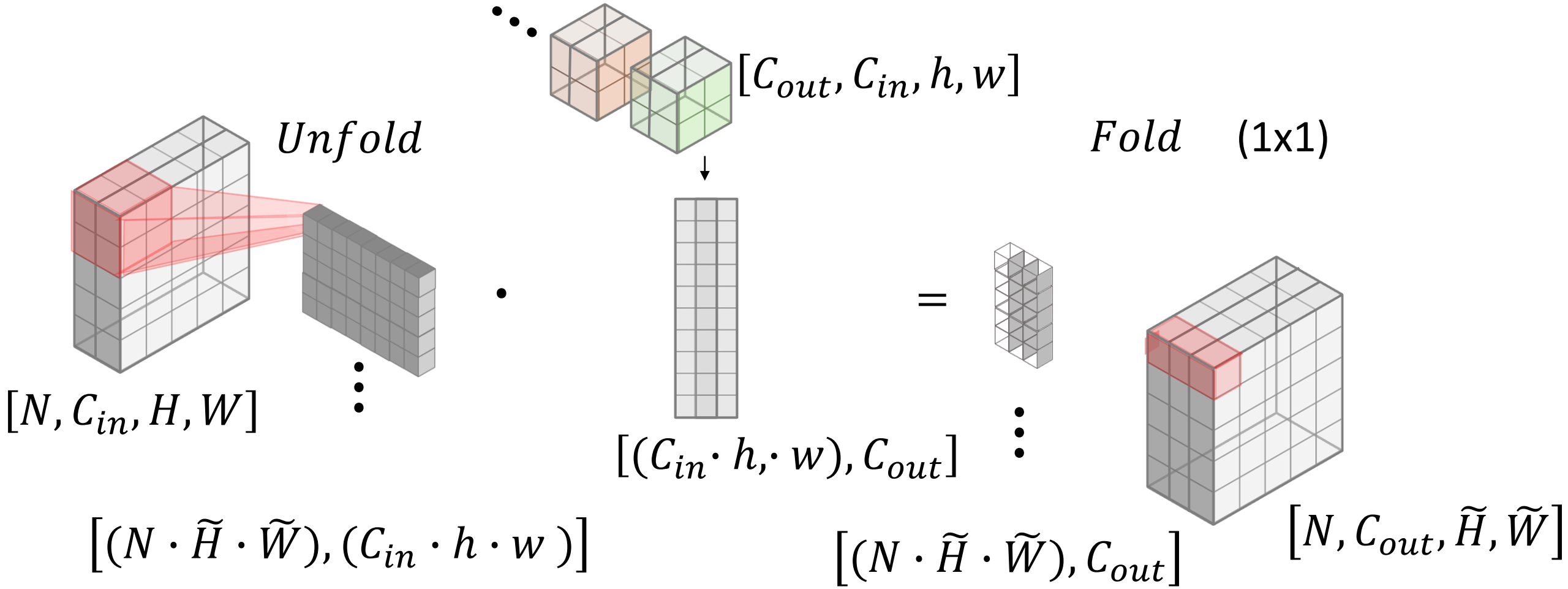
A note about the implementation of conv - BACKWARD



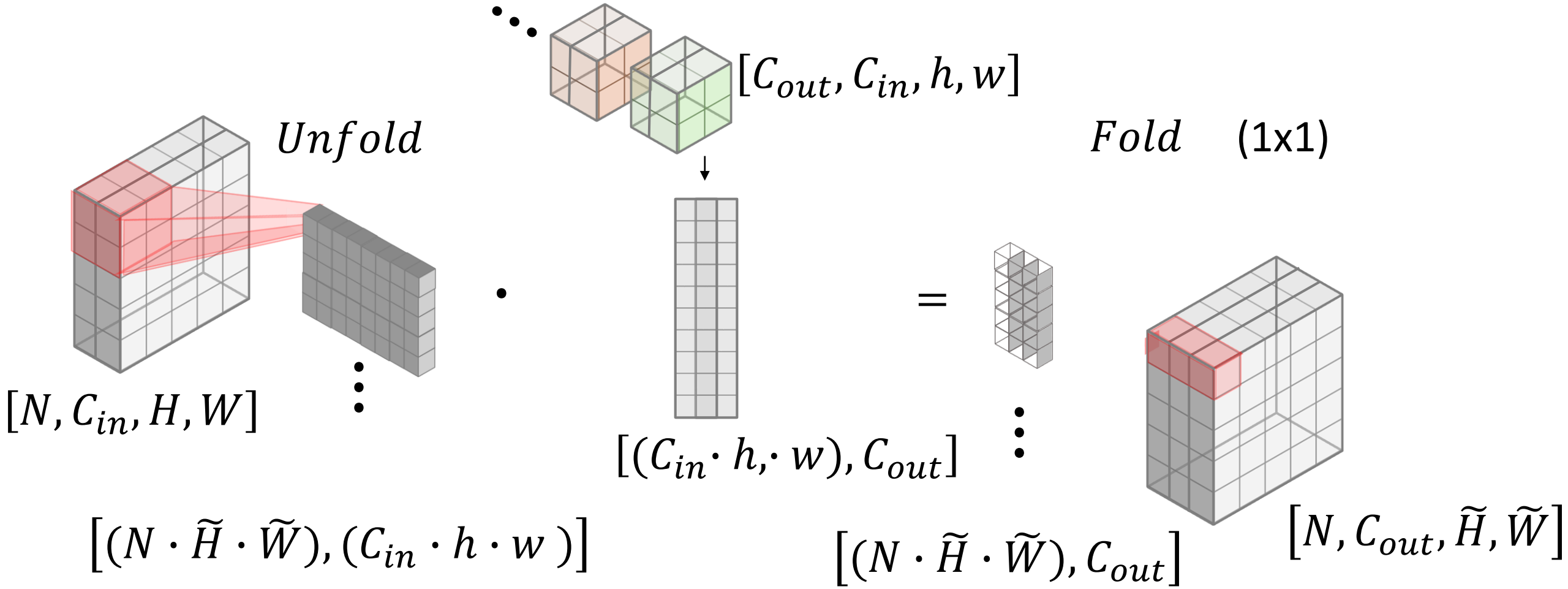
A note about the implementation of conv - BACKWARD



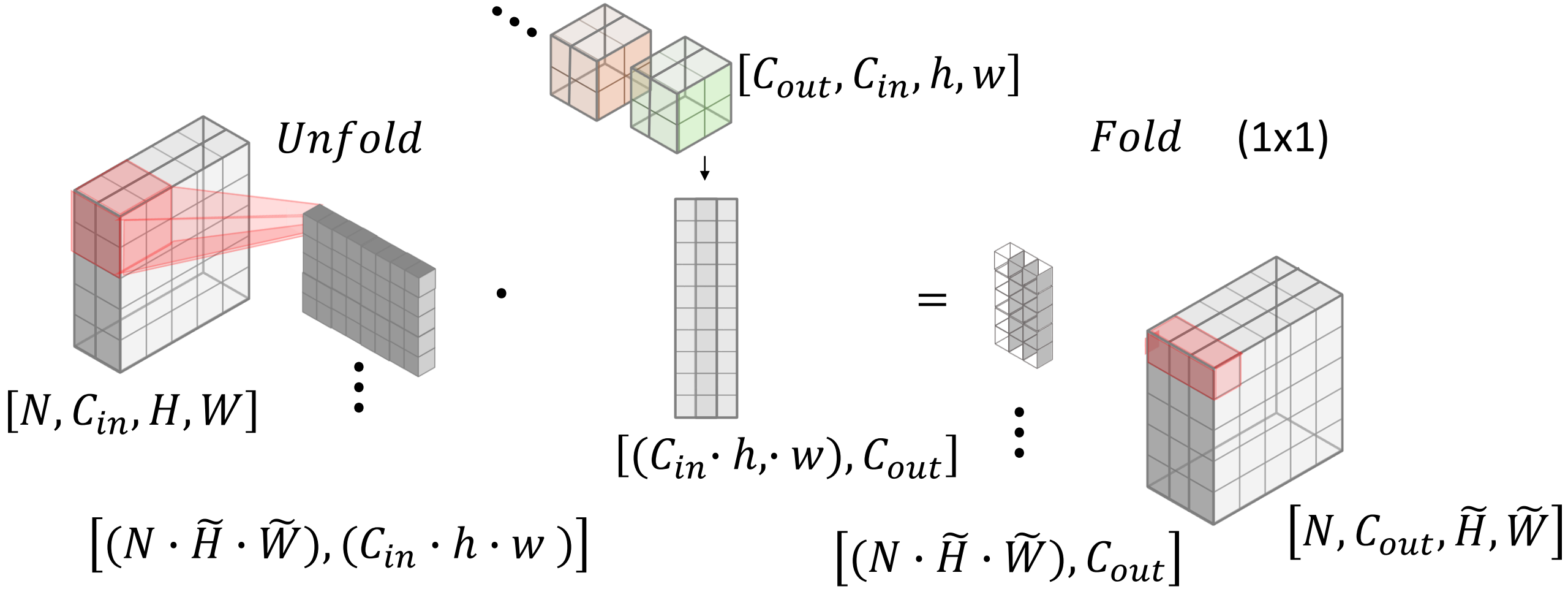
A note about the implementation of conv - BACKWARD



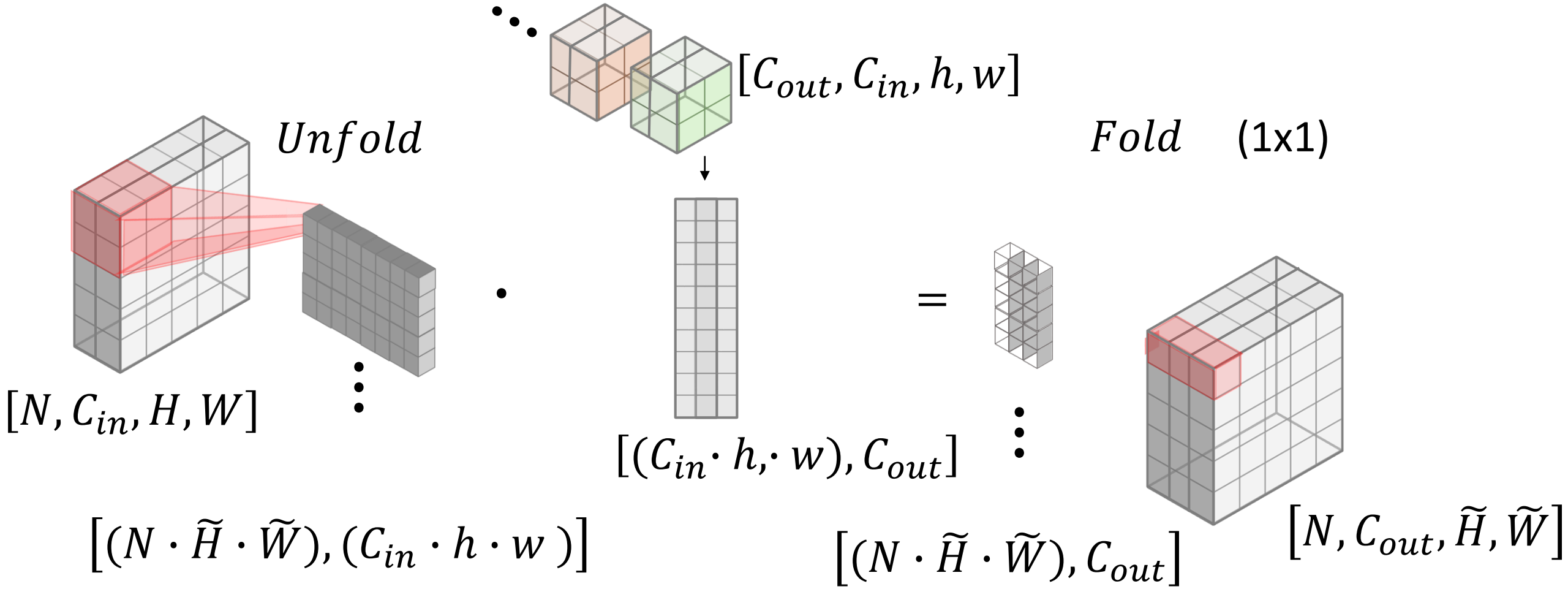
A note about the implementation of conv - BACKWARD



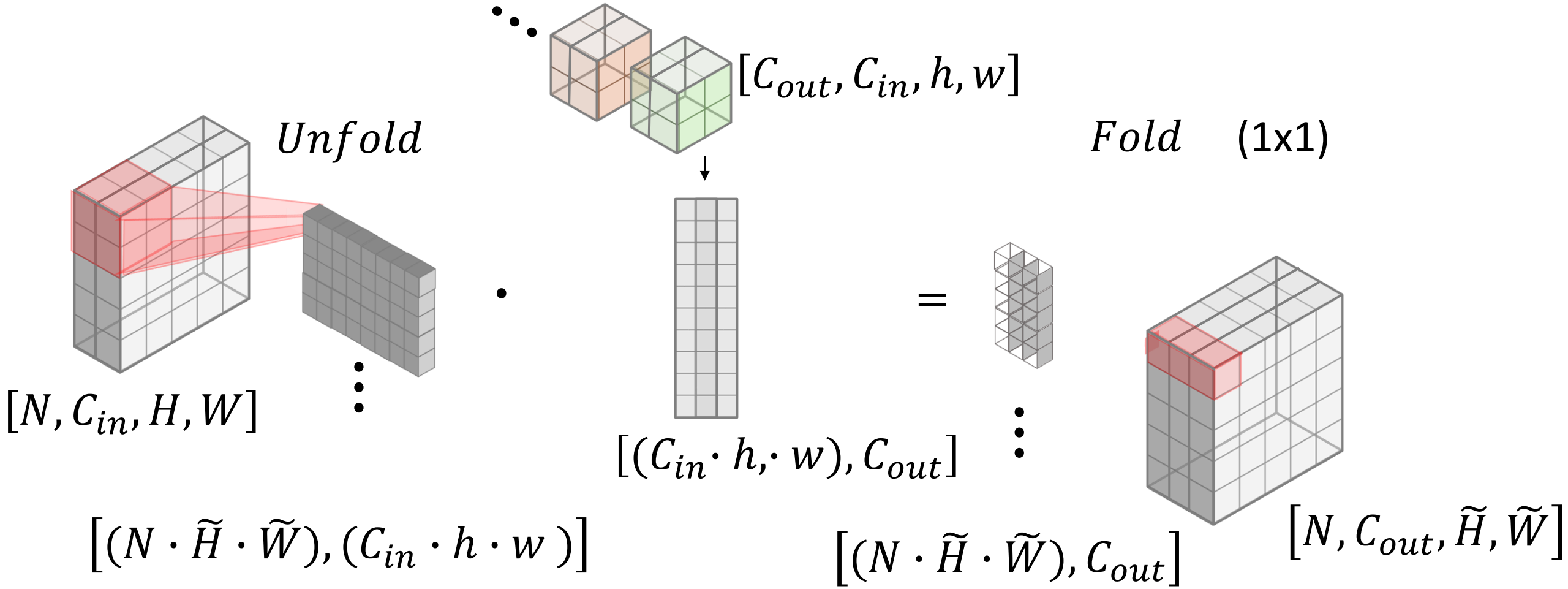
A note about the implementation of conv - BACKWARD



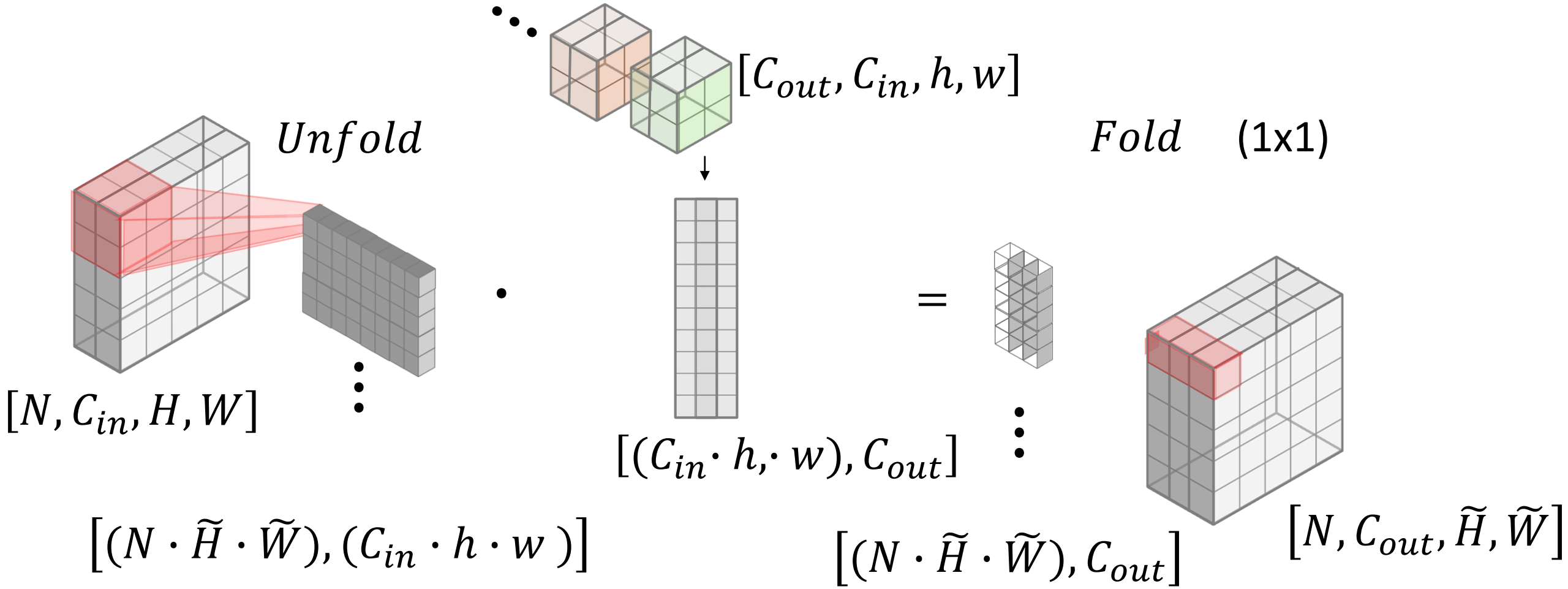
A note about the implementation of conv - BACKWARD



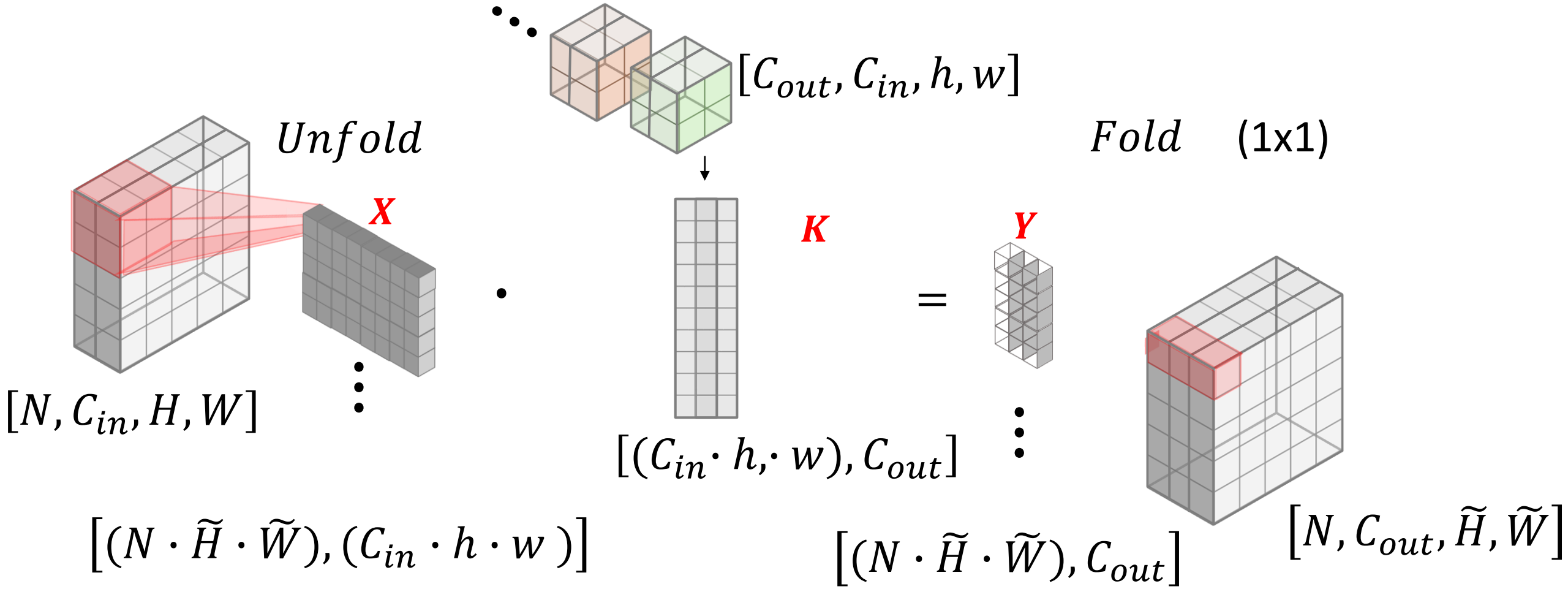
A note about the implementation of conv - BACKWARD



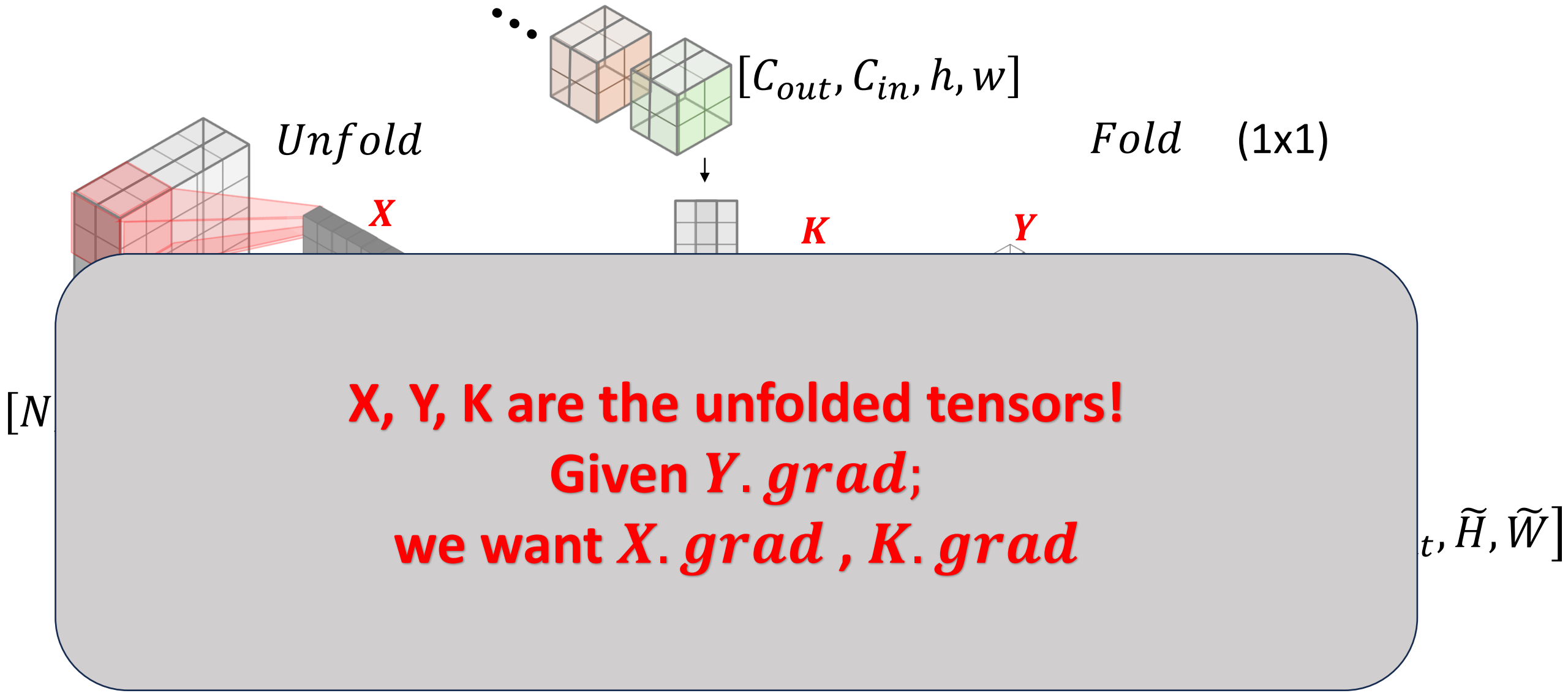
A note about the implementation of conv - BACKWARD



A note about the implementation of conv - BACKWARD



A note about the implementation of conv - BACKWARD

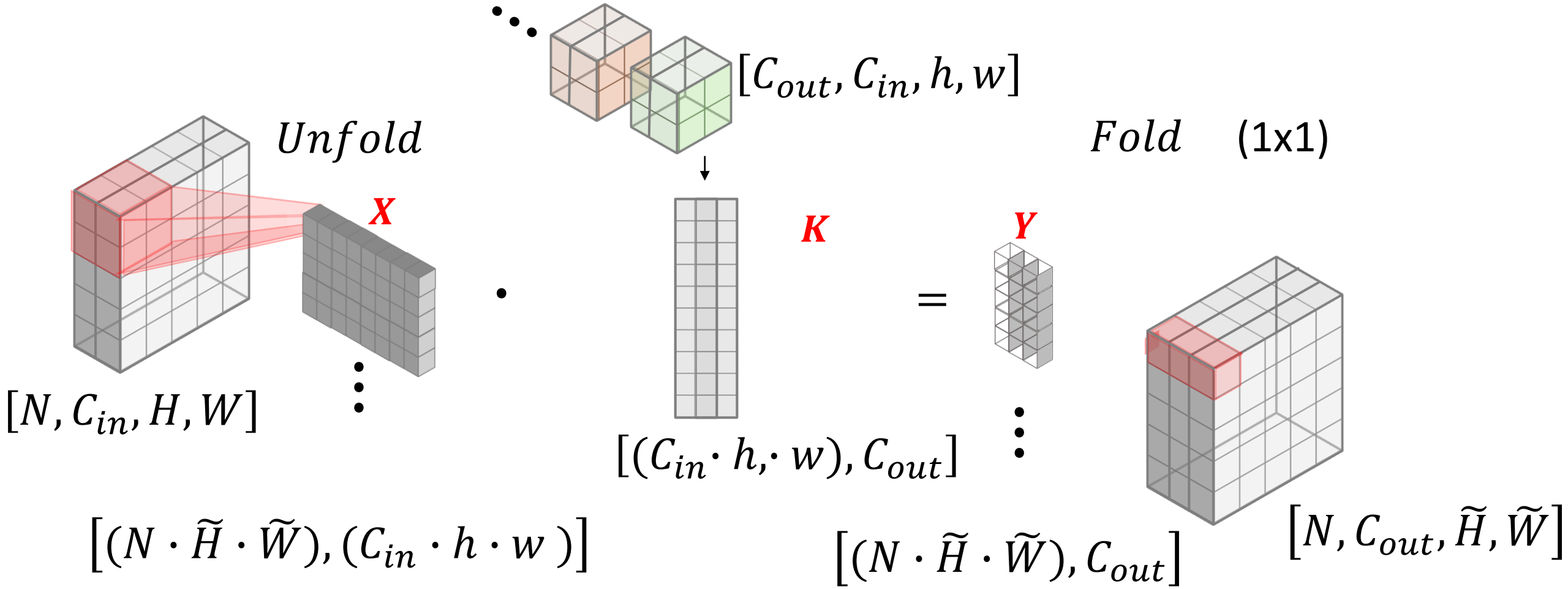


X, Y, K are the unfolded tensors!

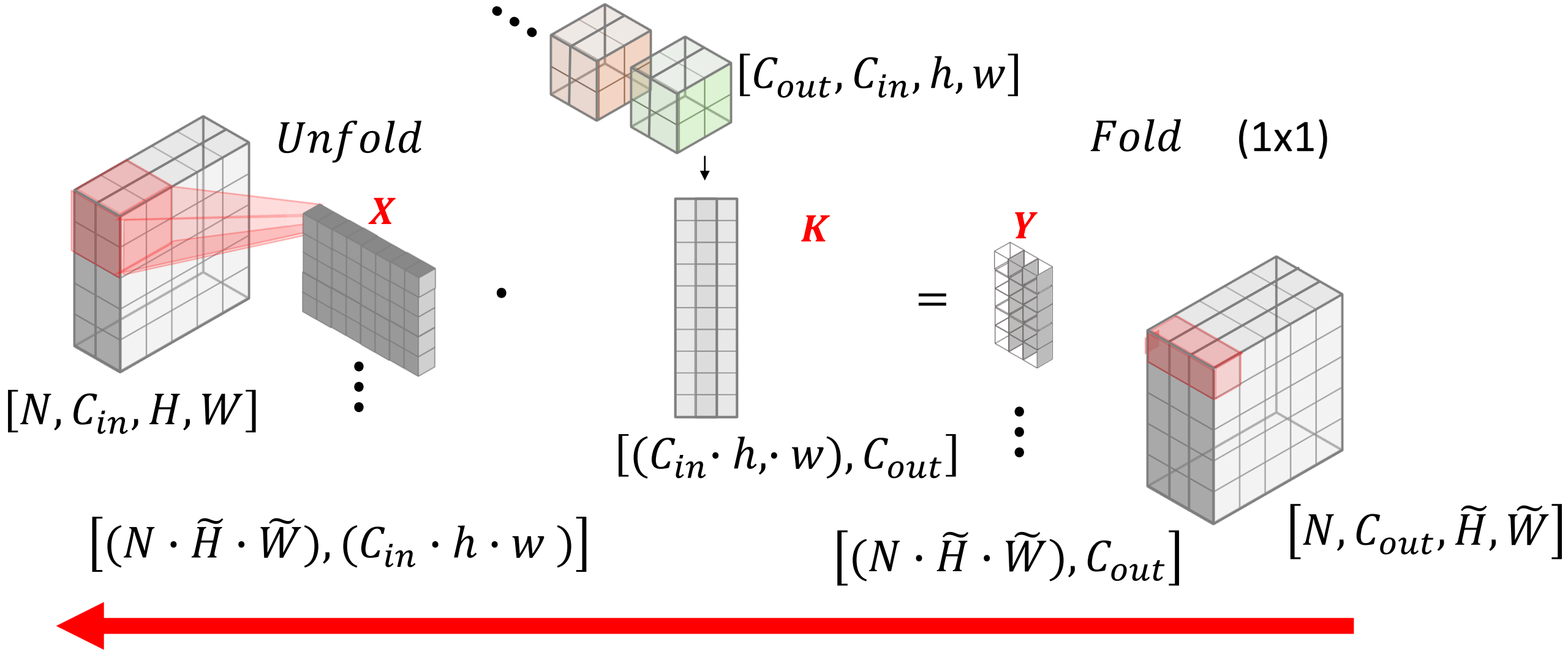
Given $Y.grad$;

we want $X.grad$, $K.grad$

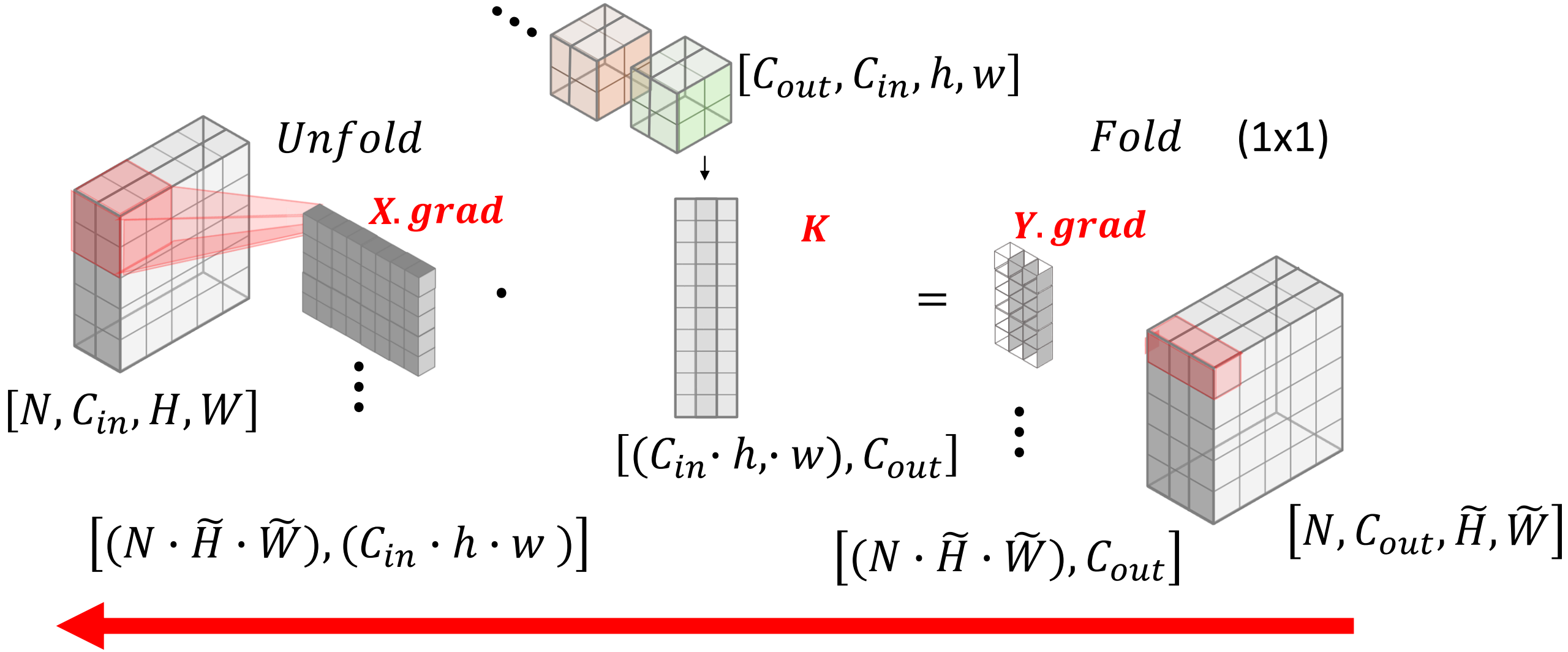
A note about the implementation of conv - BACKWARD



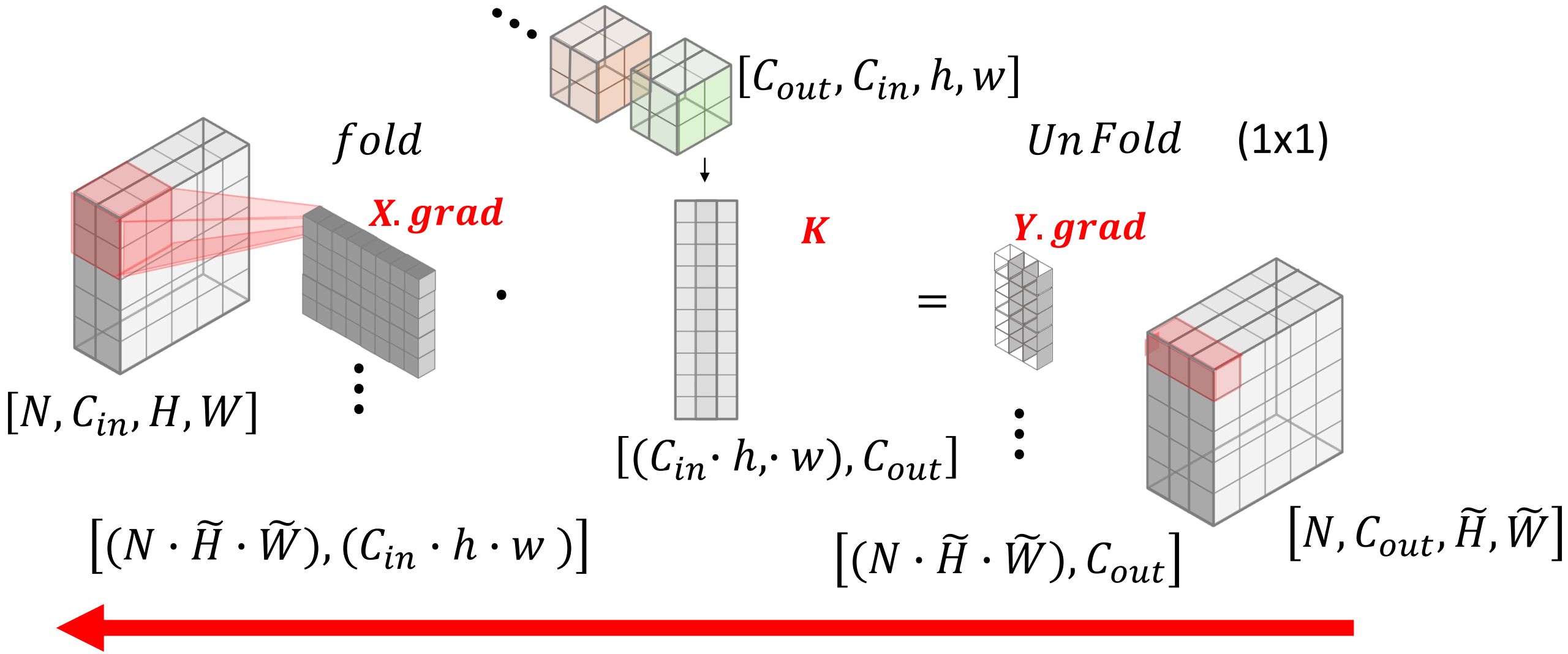
A note about the implementation of conv - BACKWARD



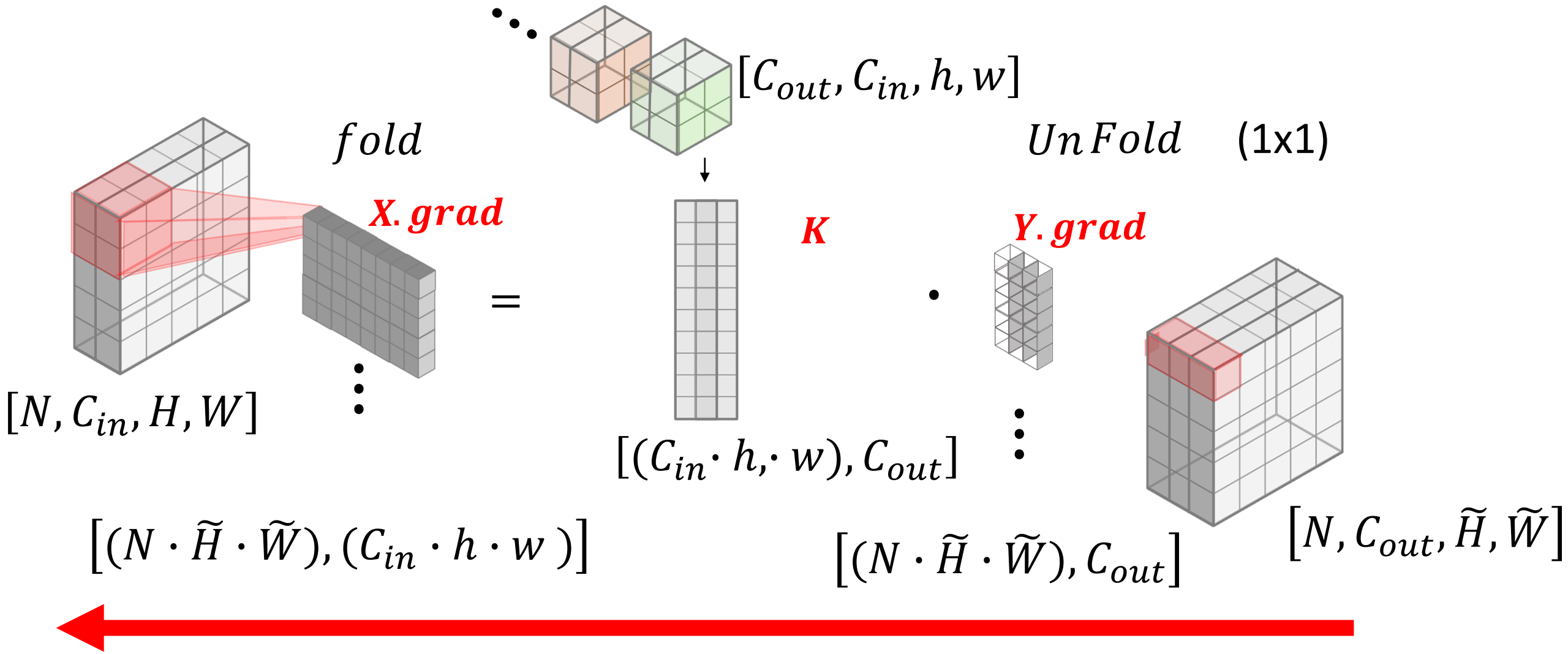
A note about the implementation of conv - BACKWARD



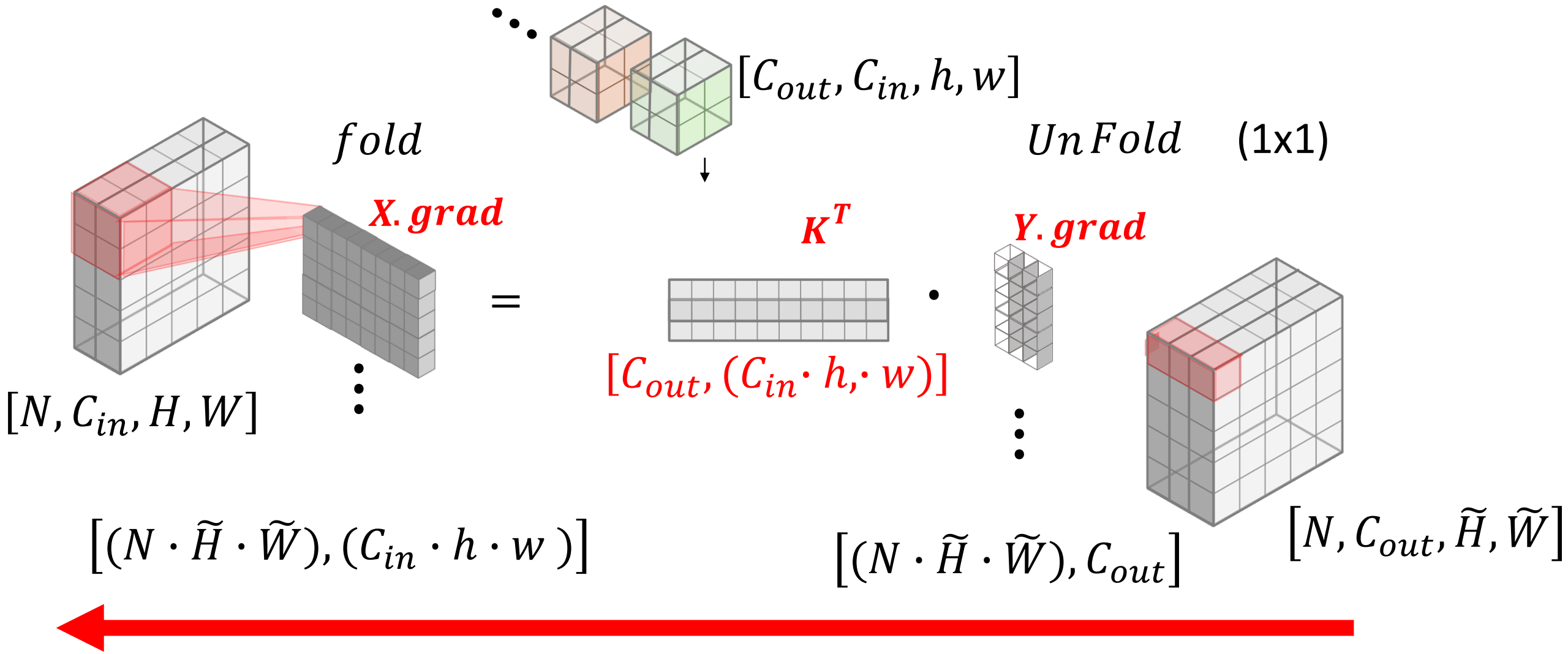
A note about the implementation of conv - BACKWARD



A note about the implementation of conv - BACKWARD

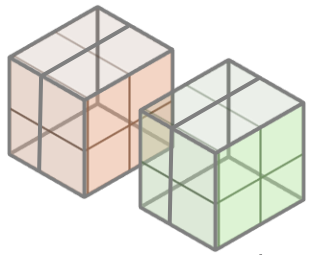


A note about the implementation of conv - BACKWARD



A note about the implementation of conv - BACKWARD

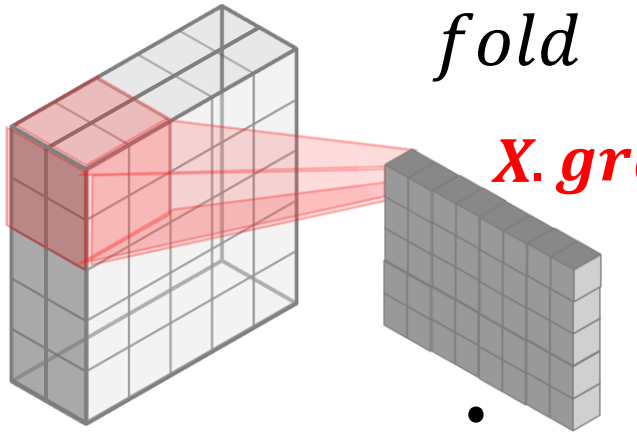
$$X.grad = (Y.grad)K^T$$
(Then Fold!)



$[C_{out}, C_{in}, h, w]$

$fold$

$UnFold$ (1x1)

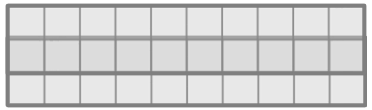


$X.grad$

K^T

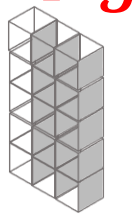
$Y.grad$

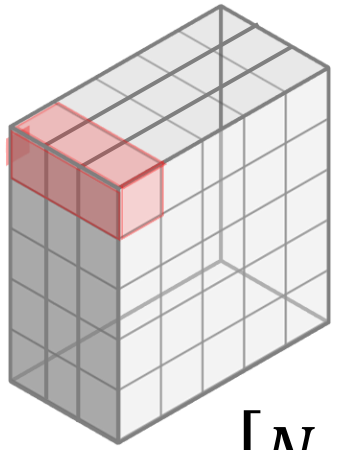
$=$



$[C_{out}, (C_{in} \cdot h, \cdot w)]$

$\cdot$





$[N, C_{in}, H, W]$

$[(N \cdot \tilde{H} \cdot \tilde{W}), (C_{in} \cdot h \cdot w)]$

$[(N \cdot \tilde{H} \cdot \tilde{W}), C_{out}]$

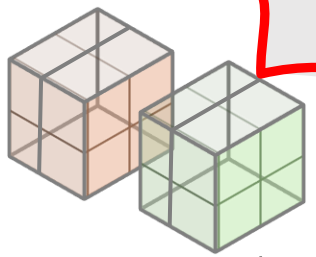
$[N, C_{out}, \tilde{H}, \tilde{W}]$



A note about the implementation of conv - BACKWARD

$$X.grad = (Y.grad)K^T$$
(Then Fold!)

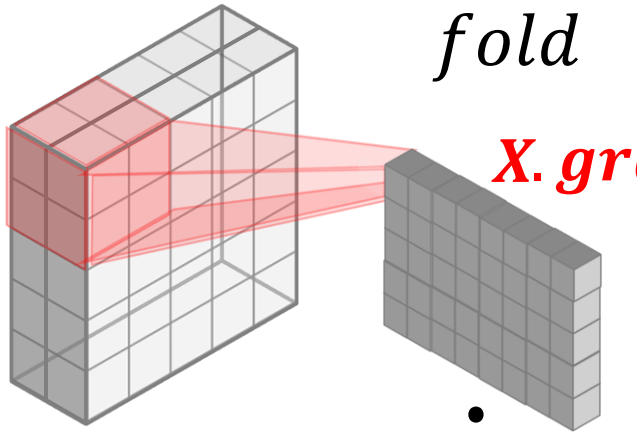
$$K.grad = (Y.grad)^T X$$
(Then Rehsape!)



$[C_{out}, C_{in}, h, w]$

$fold$

$UnFold$ (1x1)

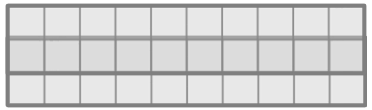


$X.grad$

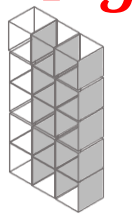
K^T

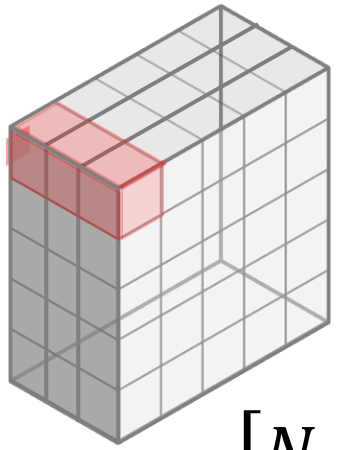
$Y.grad$

$=$



$[C_{out}, (C_{in} \cdot h, \cdot w)]$





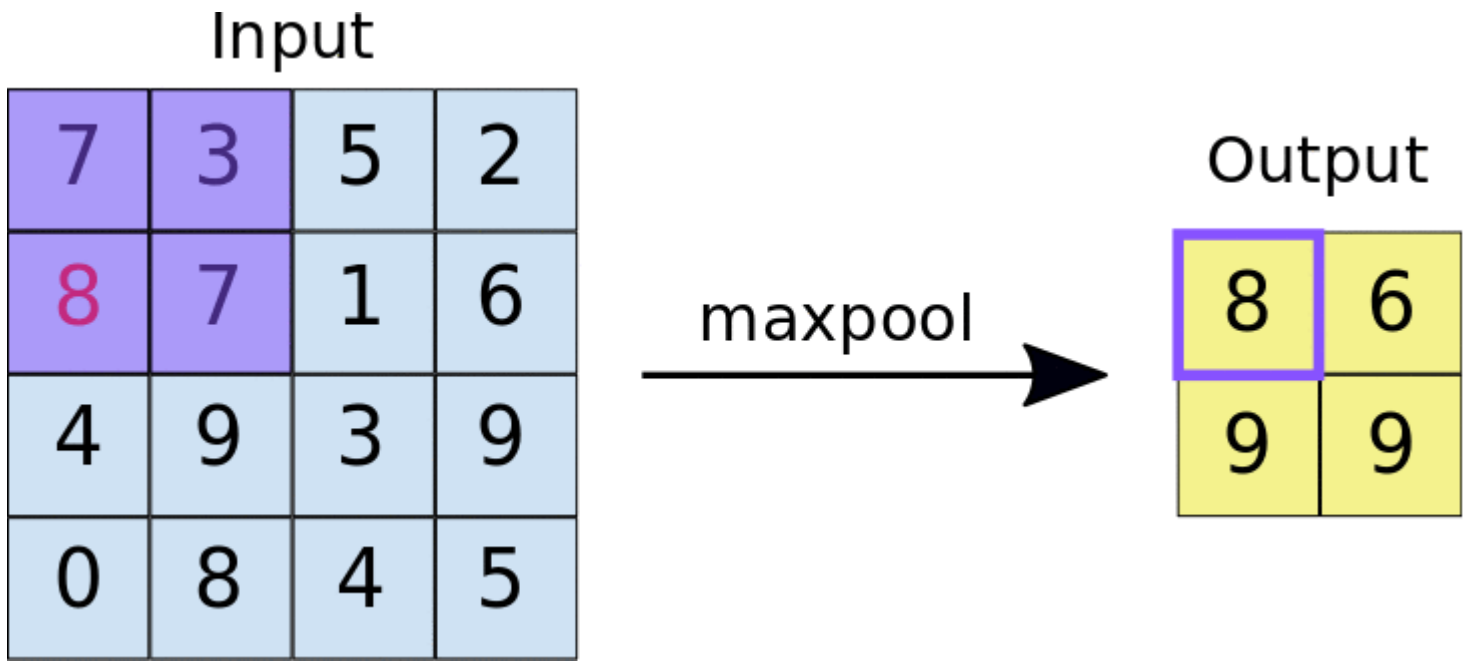
$[N, C_{in}, H, W]$

$[(N \cdot \tilde{H} \cdot \tilde{W}), (C_{in} \cdot h \cdot w)]$

$[(N \cdot \tilde{H} \cdot \tilde{W}), C_{out}]$

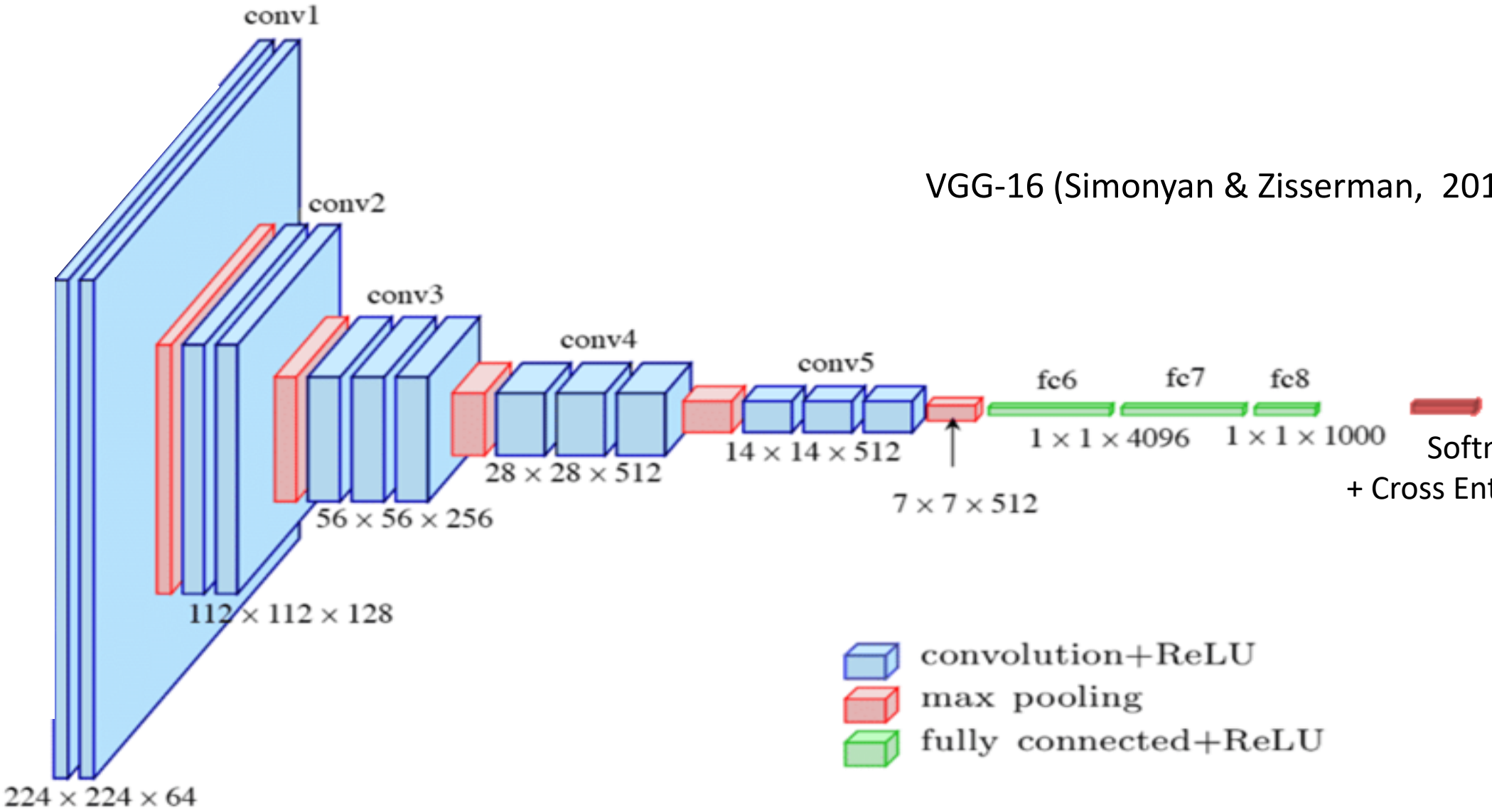
$[N, C_{out}, \tilde{H}, \tilde{W}]$

Max Pooling



- Usually stride=win-size, but not always.
- Each channel separately.

ConvNet Example

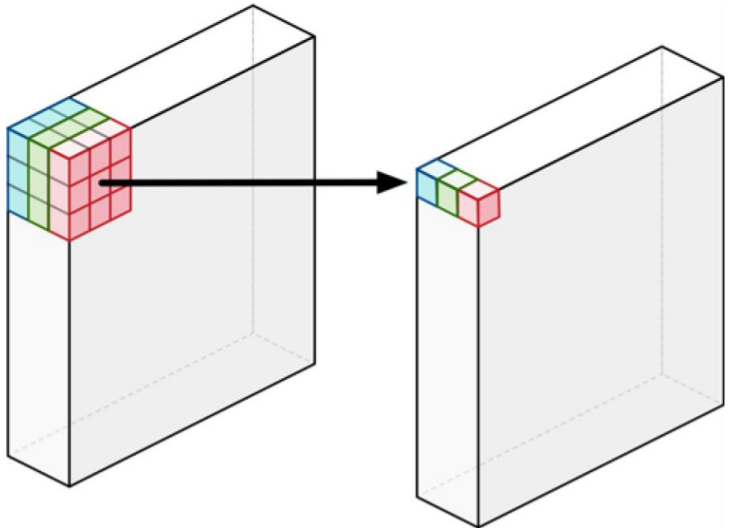


VGG-16 (Simonyan & Zisserman, 2014)

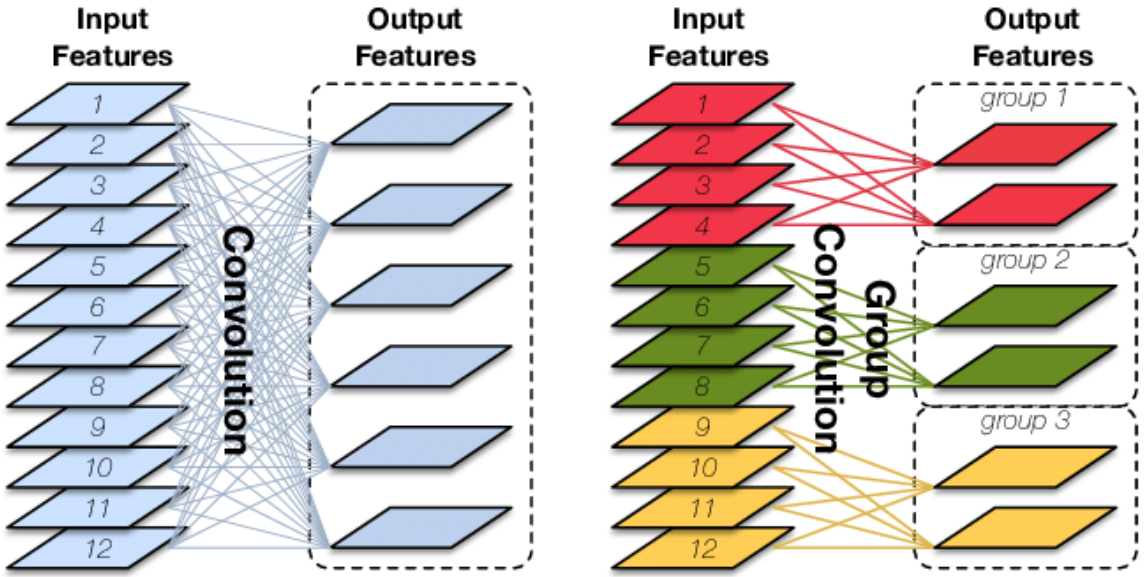
Softmax
+ Cross Entropy loss

More special Convs!

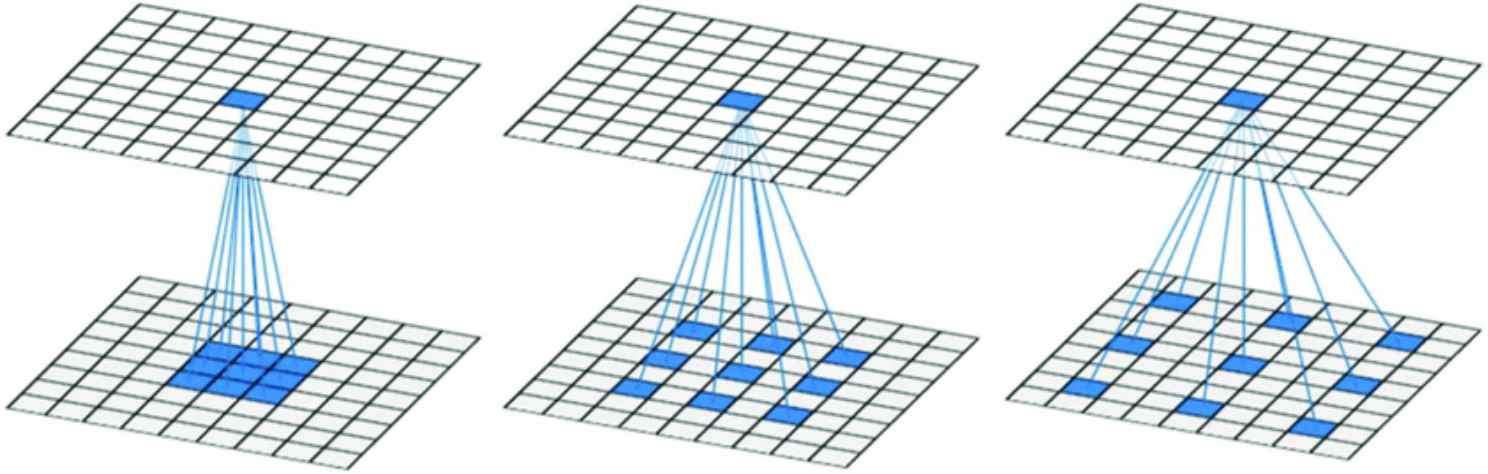
Depthwise Conv



Group Conv (Krihzevsky 2012)



Dilated Conv (Yu&Koltun 2016)



Next week's lecture:



Shai Bagon

Practical Training





Rafail Fridman

Next week's tutorial: CNN Architectures

